RazorCAD User Manual

Parametric parts, assemblies, sheet metal, structures and drawings

Software version 0.21.3-beta
Manual revision 2026-11-02

Click any entry in the Contents or the Index to jump to it. Every page has a **Back to Contents** link in its footer.

Contents

1 Introduction . . . 11

1.1 What RazorCAD is . . . 11

1.2 About this manual . . . 11

1.3 Conventions . . . 11

2 Installing and Starting RazorCAD . . . 13

2.1 Running RazorCAD . . . 13

2.2 RazorCAD in your browser . . . 13

2.3 Installing with the setup program . . . 13

2.4 Updates . . . 13

2.5 The start window . . . 14

2.6 The RazorCAD window . . . 14

2.7 Customising the ribbon . . . 15

2.8 Settings . . . 15

2.9 Language and theme . . . 16

2.10 Getting Started Tour . . . 16

2.11 Search commands . . . 16

2.12 The manual in RazorCAD . . . 17

2.13 Your own keyboard shortcuts . . . 17

2.14 Using the keyboard . . . 17

2.15 Units . . . 17

3 Working with Files . . . 19

3.1 Part files . . . 19

3.2 Properties . . . 19

3.3 Compare With . . . 19

3.4 Tabs and closing . . . 19

3.5 Auto-save and recovery . . . 20

3.6 Opening from Windows and file previews . . . 20

3.7 Point clouds . . . 20

3.8 Importing STEP, IGES, STL, OBJ and 3MF . . . 20

3.9 Exporting . . . 21

3.9.1 Export Face as DXF . . . 22

4 Navigating the 3D View . . . 23

4.1 Mouse and keyboard . . . 23

4.2 Standard views . . . 23
4.3 Up axis: Y or Z . . . 23
4.4 Display styles . . . 23
4.5 Visibility . . . 24
4.6 Section view . . . 24

5 Selecting . . . 25
5.1 Selecting in the part . . . 25
5.2 Selection filters . . . 25
5.3 Quick readout of sizes . . . 25
5.4 Highlighting from the tree . . . 25
5.5 Selecting hidden bodies . . . 25
5.6 Right-click menus . . . 26

6 Sketching . . . 27
6.1 Starting and ending a sketch . . . 27
6.2 The sketch tool wheel . . . 28
6.3 Defined and under-defined sketches . . . 28
6.4 Automatic relations . . . 28
6.5 Sketch orientation . . . 28
6.6 Sketch tools . . . 28
6.7 Dimension . . . 30
6.8 Trim . . . 31
6.9 Offset . . . 31
6.10 Project Edges . . . 32
6.10.1 Selecting source sketches for Project and Intersection . . . 32
6.11 Spline . . . 32
6.12 Control-Point Spline . . . 33
6.13 Import DXF . . . 33
6.14 Text . . . 33
6.15 Relations . . . 33
6.16 How profiles are found . . . 34
6.17 Showing and hiding sketches . . . 35
6.18 Change Sketch Plane . . . 35

7 3D Sketches . . . 36
7.1 Starting and ending a 3D sketch . . . 36
7.2 Drawing in space . . . 36
7.3 3D sketch tools . . . 36

7.4 Relations in 3D . . . 37
7.5 Projected edges in 3D . . . 37
7.6 Curves from a 3D sketch . . . 37
7.7 Features that take a 3D sketch . . . 38

8 Gears . . . 39
8.1 Creating a gear train . . . 39
8.2 The Gear Train panel . . . 39
8.3 Rules for a valid train . . . 39

9 Airfoils . . . 40
9.1 Drawing an airfoil . . . 40
9.2 Sections . . . 40

10 Tutorial: a Guided Loft and a Sheet Metal Tray . . . 41
10.1 The bottom profile . . . 41
10.2 A plane, moved and turned . . . 41
10.3 The guide curve . . . 41
10.4 The guided loft . . . 41
10.5 A fillet with a point along its edge . . . 41
10.6 The tray . . . 42

11 Solid Features . . . 43
11.1 Live previews . . . 43
11.2 Extrude and Extrude Cut . . . 43
11.3 Revolve and Revolve Cut . . . 44
11.4 Sweep and Sweep Cut . . . 45
11.5 Loft and Loft Cut . . . 46
11.5.1 Profile correspondence . . . 46
11.5.2 Start and end conditions . . . 47
11.6 Hole . . . 48
11.7 Helix . . . 49
11.8 Thread Cut and Thread Extrude . . . 49
11.9 Cosmetic Thread . . . 49
11.10 Wrap (Emboss / Deboss) . . . 50
11.11 Patterns and Mirror . . . 50
11.12 Fillet . . . 51
11.13 Chamfer . . . 52
11.14 Shell . . . 52

11.15 Draft . . . 53
11.16 Rib . . . 53
11.17 Dome . . . 53
11.18 Plastic part features . . . 53
11.19 Core and Cavity . . . 53
11.20 Move Face . . . 54
11.21 Delete or Resize Fillet, Defeature . . . 54
11.22 Scale . . . 54
11.23 Long operations . . . 54

12 Bodies and Multi-Body Parts . . . 55
12.1 Solid bodies . . . 55
12.2 Body Properties . . . 55
12.3 Materials . . . 55
12.4 Appearances . . . 55
12.5 The Render tab . . . 56
12.5.1 Spotlights {idx:spotlights}{idx:studio lighting} . . . 57
12.6 Exploded views of a part . . . 58
12.7 Combine and Subtract Bodies . . . 58
12.8 Split Body . . . 58
12.9 Move/Copy Bodies . . . 58
12.10 Delete Body . . . 59

13 Surfaces . . . 60
13.1 Making surfaces . . . 60
13.1.1 Guide control for Swept Surface . . . 60
13.1.2 Loft only the portions created by Split Face . . . 61
13.1.3 Start and end conditions for Lofted Surface . . . 61
13.1.4 Guide construction for Lofted Surface . . . 61
13.1.5 Surface Extrude and Revolve selection . . . 62
13.1.6 Planar Surface selection . . . 62
13.1.7 Filled Surface / Patch continuity . . . 63
13.2 Changing surfaces . . . 63
13.2.1 Freeform {idx:Freeform}{idx:freeform face}{idx:push and pull a face} . . . 64
13.3 Curves and Split Face . . . 64
13.4 Flatten Surface . . . 65
13.5 Meshes and closing surfaces . . . 65

14 Reference Geometry . . . 66

14.1 Reference Plane . . . 66
14.1.1 Plane Along Path and point along a curve . . . 67
14.2 Reference Axis . . . 67
14.3 Reference Point . . . 67

15 The Feature Tree and History . . . 69
15.1 Editing features . . . 69
15.2 References that follow edits . . . 69
15.3 Fix References . . . 69
15.4 Rollback bar . . . 70
15.5 Reordering features . . . 70
15.6 Reference arrows . . . 70
15.7 Tree symbols . . . 70

16 Sheet Metal . . . 71
16.1 Base Flange . . . 71
16.2 Gauge tables . . . 71
16.3 Sheet Metal Extrude . . . 71
16.4 Lofted Bend . . . 71
16.5 Transition / Reducer . . . 72
16.6 End Flange . . . 73
16.7 Edge Flange . . . 73
16.8 Miter Flange . . . 74
16.9 Hem . . . 74
16.10 Sketched Bend . . . 74
16.11 Jog . . . 74
16.12 Cross Break . . . 74
16.13 Tab and Slot . . . 74
16.14 Forming Tools . . . 75
16.15 Gusset . . . 75
16.16 Corner Relief . . . 75
16.17 Closed Corner . . . 75
16.18 Unbend and Bend . . . 76
16.19 Cuts and other features on sheet metal . . . 76
16.20 Flatten . . . 76
16.21 Convert to Sheet Metal . . . 76
16.22 Etch Lines . . . 77
16.23 Sheet Metal Check . . . 77

16.24 Sheet metal Body Properties . . . 77
16.25 Exporting flat patterns as DXF . . . 77
16.26 DXF Settings . . . 78

17 Fast Structure . . . 79
17.1 Structural Member . . . 79
17.2 Trim/Extend . . . 79
17.3 End Cap . . . 80
17.4 Gusset . . . 80
17.5 Weld Bead . . . 80
17.6 Cut List . . . 80
17.7 Duct Fitting . . . 80
17.8 Pipe Route and Cable . . . 80

18 Designing for 3D Printing . . . 82
18.1 Print Profile . . . 82
18.2 Place on Bed, Auto-Orient and Show Bed . . . 82
18.3 Check Printability . . . 83
18.4 Overhang Map and Thickness Map . . . 83
18.5 Nut Pocket . . . 83
18.6 Insert Hole . . . 83
18.7 Screw Hole for Plastic . . . 83
18.8 Printable Hole . . . 84
18.9 Bridging Layer . . . 84
18.10 First-Layer Chamfer . . . 84
18.11 Printable Overhangs . . . 84
18.12 Fit Clearance . . . 84
18.13 Print-in-Place Hinge . . . 84
18.14 Split for Printing . . . 84
18.15 Export for Printing and Open in Slicer . . . 84
18.16 Calibrate Fits . . . 85

19 Configurations . . . 86
19.1 The Configurations tab . . . 86
19.2 What a part configuration changes . . . 86
19.3 Design table . . . 86
19.4 Configurations in assemblies . . . 86

20 Assemblies . . . 87

20.1 Starting an assembly . . . 87
20.2 Inserting components . . . 87
20.3 Standard parts . . . 87
20.4 Mates . . . 88
20.4.1 Gear and rack-pinion mates . . . 89
20.5 Moving components . . . 89
20.6 Fasten All Holes . . . 89
20.7 Component patterns and mirrors . . . 90
20.8 Collision detection . . . 90
20.9 The assembly tree . . . 90
20.10 Sub-assemblies . . . 91
20.11 Equations shared with the parts . . . 91
20.12 Assembly holes and cuts . . . 92
20.13 Exploded views . . . 92
20.14 Motion studies . . . 92
20.15 Interference Check . . . 93
20.16 Bill of Materials . . . 93
20.17 STEP assemblies . . . 93
20.18 Saving parts as separate files . . . 93
20.19 Collect Files . . . 94
20.20 Big assemblies . . . 94

21 Drawings . . . 95
21.1 Starting a drawing . . . 95
21.2 The sheet and its template . . . 95
21.3 Drawing templates . . . 95
21.4 Working on the sheet . . . 96
21.5 View Layout tab . . . 96
21.6 View properties . . . 97
21.7 Flat pattern views . . . 97
21.8 Annotation tab . . . 98
21.8.1 Auto Dimension . . . 99
21.8.2 Check Drawing . . . 100
21.8.3 DXF drawings on the sheet . . . 100
21.9 Sheet tab . . . 100
21.10 Views follow the part . . . 100
21.11 Not yet available . . . 100

22 Evaluate and Tools . . . 102

22.1 Measure . . . 102
22.2 DFM Check . . . 102
22.3 3D Annotations . . . 102
22.4 Tolerance Stack-Up . . . 102
22.5 Mass Properties . . . 103
22.6 Interference Check . . . 103
22.7 Surface analysis . . . 103
22.8 Configurations . . . 103
22.9 Equations and global variables . . . 103
22.9.1 Dimension IDs {idx:dimension IDs} . . . 103
22.10 Macros . . . 104
22.11 Edit Material . . . 105

23 Modelling with Claude . . . 106
23.1 Setting up . . . 106
23.2 Ask Claude panel . . . 106
23.3 RazorCAD Part Builder on claude.ai . . . 106

24 Troubleshooting . . . 107
24.1 Crash reports . . . 107
24.2 Reporting a problem . . . 107
24.3 Speed on your PC . . . 107
24.4 Common problems . . . 108
24.5 Settings and data files . . . 108

25 For Developers . . . 109
25.1 Rebuilding the executable . . . 109
25.2 Code signing . . . 109
25.3 Regression tests . . . 109
25.4 Releases . . . 109
25.5 The geometry kernel builds . . . 110
25.6 The website and the browser version . . . 110
25.7 Working on the UI . . . 110
25.8 Project layout . . . 110

Appendix A: Keyboard and Mouse Reference . . . 112
General . . . 112
View . . . 112
Selection filters . . . 112

Sketch 112
3D sketch 113
Assembly 113

Appendix B: Materials 114

Appendix C: Manual Revision History 115

Index 118

1 Introduction

1.1 What RazorCAD is

RazorCAD is a parametric part modeller for Windows with a classic feature-tree interface. You build a part as a **feature history**: sketches, extrudes, revolves, cuts, fillets, patterns and so on, each of which can be edited later. Changing a dimension or an equation rebuilds every feature after it.

Solid modelling runs on the **OpenCascade** B-rep kernel (the engine behind FreeCAD), through the replicad library. Display uses three.js. RazorCAD runs fully offline.

Main capabilities:

- Constraint-driven 2D sketches with relations, dimensions and equations, plus gear and airfoil generators
- 3D sketches for frames, pipe and duct routes and sweep paths
- Solid features: extrude, revolve, sweep, loft, hole, helix, threads, wrap, draft, rib, dome, patterns, mirror, fillet, chamfer, shell
- Multi-body parts with combine, subtract, split, move/copy, per-body materials and appearances
- Surface modelling: extruded, revolved, planar, lofted, swept, filled, boundary and ruled surfaces, trim, extend, knit, thicken and more
- Sheet metal: flanges, hems, miter flanges, sketched bends, jogs, forming tools, tab and slot, lofted bends, transitions, flat patterns and DXF export
- **Fast Structure** (weldments): structural members from standard profiles, end caps, gussets, weld beads and a cut list
- Assemblies with mates (including gear and rack-pinion mates), sub-assemblies, configurations, standard parts and fastener sets, equations shared with the parts, holes and cuts made in the assembly, exploded views, motion studies, collision detection, an interference check, a bill of materials with mass and cost, and STEP export
- 2D drawings of parts and assemblies, with projected, section, detail, auxiliary, broken, exploded and flat pattern views, dimensions (also made automatically), GD&T, cosmetic threads, balloons, bill-of-materials tables, a drawing check, DXF drawings placed on the sheet, and PDF and DXF export
- Millimetres or inches, chosen for each document
- STEP, IGES, STL, OBJ and 3MF import; STEP, STL, OBJ, 3MF and glTF export; DXF import into sketches and drawings, and DXF export of faces and flat patterns
- Macros, recorded or written, that drive RazorCAD
- English, German, French or Spanish, a light or dark theme, and keyboard use of the panels
- An **Ask Claude** panel that builds features from a plain-language description
- A browser version at razorcad.com/app, the same program without installing anything

1.2 About this manual

This manual describes RazorCAD as it is in the version shown on the cover. It is checked against the software every day and updated when features change; Appendix C lists what changed in each revision.

The **Contents** and the **Index** are clickable. The footer of every page has a **Back to Contents** link. Your PDF viewer's bookmarks pane shows the same outline.

1.3 Conventions

Convention	Meaning
File ▸ Save	A menu command: the File menu, then Save
`Ctrl+S`	A key or key combination
Property panel	The panel on the left that holds a feature's settings while you edit it
Feature tree	The feature history list on the left
Command ribbon	The tabbed toolbar across the top (Features, Sketch, Surfaces …)
`width/2`	Text you type, such as a value or an equation

> **Note:** Lengths are in **millimetres** or **inches**, chosen for each document (see Units). Angles are in **degrees**.

2 Installing and Starting RazorCAD

2.1 Running RazorCAD

Double-click **RazorCAD.exe**. It is one standalone program: it needs no browser, Python or installation, and it runs offline. It uses the Microsoft Edge WebView2 runtime, which comes with Windows 10 and 11.

2.2 RazorCAD in your browser

razorcad.com/app runs the same RazorCAD in a web browser (Microsoft Edge, Chrome and other current browsers), with nothing to install: parts, sketches, assemblies, drawings and macros, on the same geometry kernel.

- **Open** reads a file from your PC and **Save** downloads it. Preferences stay in that browser.
- There is no auto-save, no Windows fonts (the fonts that come with RazorCAD are used) and no Ask Claude sign-in. A notice at the top says so and links to the Windows app; ✕ closes it for good.
- Parts that take a while to build are kept in the browser's own storage (up to 400 MB), so they open quickly the next time.
- Closing the browser tab with unsaved work asks first.

2.3 Installing with the setup program

Testers can be given **RazorCAD-Setup.exe**, a single setup program with RazorCAD inside it.

- Install for the current user (`%LOCALAPPDATA%\Programs\RazorCAD`, no administrator rights), for all users (Program Files), or into a folder you choose.
- Setup installs the Microsoft Edge WebView2 Runtime and .NET Framework 4.8 from Microsoft if they are missing.
- It creates Start-menu and desktop shortcuts and registers RazorCAD under Windows **Installed apps** with an uninstaller.
- The uninstaller asks before removing settings and crash reports. Part files are never touched.
- Running setup again updates an existing installation.
- Until RazorCAD is code-signed, Windows may show "Windows protected your PC" for the setup: click **More info**, then **Run anyway**. The *Read me first* file that comes with the setup says what to expect for that version.

Silent install and uninstall:

Command	Effect
`RazorCAD-Setup.exe /S`	Silent install for the current user
`/D=folder`	Install into this folder
`/ALLUSERS`	Install for all users
`/NODESKTOP`, `/NOSTARTMENU`	Skip the desktop or Start-menu shortcut
`uninstall.exe /S`	Silent uninstall

2.4 Updates

A little after it starts (at most once a day), RazorCAD asks GitHub for the newest RazorCAD release. When there is a newer one, a notice in the corner offers **Download** (the release page), **Later** or **Skip this version**. Nothing is shown when you are up to date or offline. Only GitHub is asked; nothing about you or your parts is sent.

Help ▸ Check for Updates... checks at once and says what it found. Its window also turns the check at start on or off. **Help ▸ About RazorCAD** shows the version you have.

Update Now (in the notice and in Check for Updates) updates in one click: RazorCAD downloads the new setup from its GitHub release, checks it against the release (size and SHA-256), installs it into the same folder (asking for administrator rights only when it was installed for all users), closes and starts again as the new version. Unsaved work is saved first (**Save All and Update**), or the update waits. The helper waits for this installation's app and launcher processes to close, checks the installer result, verifies that the executable was replaced and that the installed version matches the download, then restarts. A failed installation shows an error and its log path instead of silently reopening the old version. A copy of RazorCAD that wasn't installed with the setup offers **Download** instead.

If updating fails, inspect `%TEMP%\RazorCAD-update\update.log` and run the downloaded setup in that folder manually after closing RazorCAD. The helper script is saved beside it as `install-update.ps1`. Fixes to the updater itself take effect once a build containing them has been installed; an older installation may need the newer setup run manually once.

2.5 The start window

At launch RazorCAD shows a start window with:

- **New to RazorCAD? Take the one-minute tour** (until you have taken it; see Getting Started Tour)
- **New Part**, **New Assembly** and **New Drawing**
- **Open...**
- **Sample Part**, a ready-made part to explore
- **Recent Parts**: name, folder and date. Missing files are greyed out and can be removed with ×. Parts that have an auto-save waiting are marked.
- **Recover Previous Session**, offered once when unsaved work from an earlier version is found
- A notice when new crash reports have been written

Reopen it any time with **File ▸ Recent Parts / Start...**.

2.6 The RazorCAD window

Area	What it does
Menu bar	File, Edit, View, Insert, Tools, Help. Next to the menus are quick buttons for **Save**, **Save As**, **Rebuild**, **Settings** (the gear) and **Search commands** (the magnifier).
Document tabs	Each open part or drawing has its own tab. × closes a tab and asks to save.
Command ribbon	Tabs of tool buttons: **Features**, **Sketch**, **Surfaces**, **Sheet Metal**, **Fast Structure**, **Evaluate** and **Claude**. Buttons with a shortcut show the key in their bottom-right corner.
Feature tree	The part's history, with folders for Solid Bodies, Sheet Metal Bodies and Surface Bodies, the material, the planes and every feature. An eye column on the right shows or hides each item.
Configurations tab	Next to the Feature tree and the Property panel at the top of the left panel (see the Configurations chapter)

Area	What it does
Property panel	Opens in place of the tree while you create or edit a feature. ✓ accepts (**Enter**), ✕ cancels (**Esc**). On a new feature, ✓+ (**Shift+Enter**) accepts it and starts another of the same kind straight away (another Extrude, another Fillet…) . While a new feature still needs something picked, the panel says what (in black, not as an error), and the status bar doesn't count it as an error. Starting another feature (or editing another) while one is open asks first: **Apply** it, **Discard** it, or **Keep editing**; it is never thrown away unasked . A feature that fails is not accepted : ✓ shows a dialog saying why, and the panel stays open to put it right; a new one is not added to the tree, and an edit that makes a feature fail is not kept (✕ leaves it as it was).
Heads-up toolbar	Above the 3D view: view orientation, display style, selection filters, visibility (eye button) and section view
Confirm corner	The top-right corner of the view holds ✓ / ✕ for the active feature or sketch
Status line and progress bar	Bottom left. Long operations (opening, rebuilding, importing, meshing, exporting) show progress here. The rebuild time is shown at the right; when features have errors it shows their number in red: click it for Fix References.
Units	Bottom right: **MMGS** (millimetres) or **IPS** (inches). Click it to change the document's units.

2.7 Customising the ribbon

The ribbon shows its buttons in two rows. When the window is too narrow for all of them, the ones that don't fit go under **More** at the end of the ribbon (its menu runs them as the buttons do); widen the window and they come back.

- **Icon size**: Small, Medium or Large, from **View** ▸ **Toolbar Icons** or by right-clicking the ribbon.
- **Rearrange**: drag a button to a new place; the others slide aside.
- **Your own toolbars**: click **+** after the tabs (or right-click a tab ▸ **New Toolbar…**). Add tools by dragging a button onto its tab or with right-click ▸ **Add to "…"**. Drag a button off to remove it.
- Right-click a button on a built-in toolbar to hide it or add a separator after it. **Customize Toolbars…** shows hidden buttons again, and **Reset** puts a toolbar back as it was.
- In **Customize Toolbars…**, choose **Sketch** and set **Sketch toolbar layout** to **Expanded (all tools)** or **Grouped (flyout menus)**. Grouped mode gathers lines, rectangles, arcs, splines and slots under split buttons. Click the main icon to start the last-used variant; click the adjacent arrow to choose another variant. The icon and label follow the saved choice. With a family tool active, its **Type** selector in the Property panel offers the same variants, in both toolbar layouts. Choices made with shortcuts or individual toolbar buttons are remembered too. Each family remembers its own variant between sessions. The layout preference is saved, and changing it preserves the expanded toolbar order and hidden tools. Use Expanded mode to rearrange individual tools.

The layout is kept in `prefs.json` in RazorCAD's data folder.

2.8 Settings

The gear next to Save and Rebuild (or **File** ▸ **Settings…** / **Tools** ▸ **Settings…**, in parts, assemblies and drawings) opens **Settings**: everything you can set in one place, on pages down its left side. A change applies at once.

- **General**: the units of the part, assembly or drawing shown; the **units for new parts** (and assemblies); the up axis (also used for new parts); how often **auto-save** runs (every 20 seconds, every minute, every

5 minutes, or off); and, in the desktop app, whether RazorCAD checks for updates when it starts.

- **Display**: theme, language, the size of the toolbar icons, the Sketch toolbar's layout (all tools, or grouped in flyout menus), and the 3D view's **projection** (orthographic or perspective).
- **Export** : how files are written.
  - **STEP** : the **format**: **AP242** (the default: the newest, read by most programs, and the one that carries semantic PMI), **AP214** or **AP203** for older programs. The **units in the file** (millimetres, inches or metres; the part is the same size, only the file's unit changes). Whether the bodies' **colours** are written. The **author** and **organisation** put in the file's header. For parts, single bodies and assemblies.
  - **STL**: **binary** (smaller) or **ASCII** (text), and the **quality**: how finely curved faces are cut into triangles (coarse, normal, fine).
  - **DXF**: **DXF Settings...** (layers, colours, line types and options for flat patterns and faces).
- **Sheet Metal**: the gauges you saved (from a sheet metal feature's **Gauge** row, **Save...**); delete the ones you no longer want.
- **Customize**: Customize Shortcuts, Customize Toolbars, Claude Settings and the Getting Started Tour.

A window opened from Settings (DXF Settings, the shortcut and toolbar editors, Claude Settings) has a ‹ **Settings** button at its top left that goes back to the page you came from; its own buttons (Save, Cancel, Close) go back there too, and ✕ closes it.

2.9 Language and theme

- **View ▸ Language**: English, Deutsch, Français or Español, in parts, assemblies and drawings. The menus, ribbons, Property panels, dialog titles and buttons, the tree's fixed items and the start window are translated; longer help texts and messages are still in English. The command search finds a command by its translated name or its English one. Macros, recordings and your shortcuts work in every language, and files are the same in all of them.
- **View ▸ Theme**: Light, Dark, or Follow Windows (switches when Windows does). In the dark theme the window, the 3D view and sketches are dark; drawing sheets stay white paper.

Both are kept with your preferences.

2.10 Getting Started Tour

A one-minute tour of the window in eight short steps, each pointing at its part of the window: the ribbon, the feature tree, the view and its mouse buttons, how to make a first part, the command search and the units. It ends with **Open the Sample Part**. Use **Next** / **Back** (or → / ← and **Enter**); **Esc** or **Skip the tour** ends it.

The first time RazorCAD starts, the start window offers it. If you close the start window on an empty part, it starts by itself, once. Afterwards it is in **Help ▸ Getting Started Tour**.

2.11 Search commands

Press **Ctrl+K** (or **/**), or click the magnifier next to Save, and type part of a command's name: "fil" finds Fillet, "circ pat" Circular Pattern. Pick a result with ↑ / ↓ and **Enter**, or click it.

- It searches every command of the menus and the ribbon of the document in front: part, sketch, 3D sketch, assembly or drawing.
- Each result shows where the command lives (its menu or ribbon tab) and its shortcut.
- With nothing typed, the commands you used lately come first.

- Sections of this manual about what you typed are listed after the commands (**Manual ·** and their chapter): pick one to read it in the manual panel.

2.12 The manual in RazorCAD

This manual is built into RazorCAD. **Help ▸ User Manual**, the **?** at the top right of the window or **F1** opens it in a panel at the right of the window, beside the model; with a Property panel open, **F1** or the panel's **?** opens it at that feature.

- **Search**: type in the box at its top what you are looking for (a feature, a command, a word): the sections that have all the words are listed, the best first, with the words marked. Click one (or **Enter** for the first) to read it; the words are marked in it too. The arrows ‹ › go back and forward through what you read; with nothing typed it lists the chapters.
- **Wider or narrower**: drag its left edge. RazorCAD keeps the width for next time.
- **A window of its own**: ⧉ puts the manual in a window of its own, at the same place. Move it anywhere, onto another monitor too, and size it as you like; help you open while it is open shows there. ⇤ in that window puts it back in the panel; closing that window puts help back in the panel next time.
- **Close**: ✕ closes the panel. Your parts, assemblies and drawings stay open as they are; nothing is saved or closed with it.

2.13 Your own keyboard shortcuts

Help ▸ Customize Shortcuts... (or **Customize...** in Help ▸ Keyboard Shortcuts) gives commands keys of your own. Find a command, click **Set** and press the keys, for example **Ctrl+Shift+F**, **G** or **F6**.

- A shortcut works wherever its command is, in the document in front.
- It takes the place of a key RazorCAD uses itself; the window says when it does (for example **F**, Zoom to Fit). Giving a key to another command moves it there.
- **Esc**, **Enter**, **Tab** and **Backspace** can't be used.
- ✕ removes a shortcut; **Remove All** removes them all. Your shortcuts are listed at the top of Help ▸ Keyboard Shortcuts and shown in the command search. They are kept in `prefs.json`.

2.14 Using the keyboard

- **The trees**: click the feature tree, assembly tree or drawing tree, then ↑ / ↓ move from row to row (selecting as a click does), ← / → close and open folders, **Enter** opens the item as a double-click does, and **Home** / **End** go to the first and last row. While the tree has the keyboard, the arrow keys don't turn the view; click the view to give them back.
- **Property panels**: **Tab** goes through a panel's fields and buttons. The group headings are buttons: **Enter** or **Space** opens or closes them. The small ✕ / ▲ buttons in lists can be pressed the same way. **Enter** on a focused button presses it; elsewhere in the panel it accepts the feature, as before.
- **F6** / **Shift+F6** move the keyboard focus round the window: the Property panel (or the tree), the ribbon, the view.
- **Dialogs** take the focus when they open (their first field), keep **Tab** inside, and give the focus back when they close.
- **Screen readers**: every panel field is named by its label and unit (for example "Depth (mm)"), fields with an invalid value are marked so, and icon-only buttons are named by their tooltips. The status bar and panel messages are read out as they change, and errors at once.

2.15 Units

Each part, assembly and drawing is in **millimetres** or **inches**. Click **MMGS / IPS** at the right of the status bar, or use **Tools ▸ Units: Millimetres / Units: Inches**. Changing the units keeps every size exactly as it is; only how sizes are shown and typed changes.

In an inch document:

- Feature fields show and take inches. The arrow keys step 0.1 in (**Shift**: 1 in).
- Sketch dimensions, the Modify box and the sizes shown while drawing are in inches.
- Measure shows in, in^2 and in^3; Mass Properties also gives the mass in pounds.
- New features start at round sizes, to the nearest 1/8 in.
- Equation variables used in a length count in inches: with `w = 3`, a depth of `w/2` is 1.5 in.

Any value can name its unit: `2in`, `30 mm`, `1.5cm`, `2ft`, `(w/2)*in`. Values typed before a document became inch keep their size; an expression typed in millimetres shows as `(…)*mm`.

Drawings have units of their own (see the Drawings chapter). A drawing made from an inch part starts in inches.

3 Working with Files

3.1 Part files

Parts are saved as **.rcad** files, assemblies as **.rasm**, drawings as **.rdrw** and drawing templates as **.rdrt** files (see the Assemblies and Drawings chapters).

Command	Shortcut	Notes
File ▸ New Part	`Ctrl+N`	Opens in a new tab
File ▸ New Assembly		Opens in a new tab
File ▸ Open...	`Ctrl+O`	Opens parts, assemblies, drawings, templates, and STEP, IGES, STL, OBJ and 3MF files
File ▸ Open Sample Part		The built-in sample, in a new tab
File ▸ Save	`Ctrl+S`	Uses the normal Windows save dialog the first time
File ▸ Save As...	`Ctrl+Shift+S`	Saves a copy under a new name
File ▸ Where Used...		The assemblies and drawings that use this part or assembly: those open now, and (once it is saved) those in its folder and the folders under it. **Open** opens one
File ▸ Revisions...		The revision history of the part or assembly: **New Revision** records the next revision (A, B ... Z, AA; or the next number) with its date, description and who made it, sets the **Revision** property, and keeps a copy of the document as it is; **Open** opens that copy again as a new document
File ▸ Properties...		The part's or assembly's custom properties (see below)
File ▸ Compare With...		Compares the part with another version of it (see below)

3.2 Properties

File ▸ Properties... in a part or assembly holds custom properties such as Part Number, Description, Vendor, Finish or Cost. Common names are suggested; any name works. Undo takes a change back. A drawing of the part takes its drawing number from the Part Number, bills of materials can show any property as a column, and the **Cost** property gives the cost columns (see Bill of Materials).

3.3 Compare With

File ▸ Compare With... compares the part in front with its last saved version, another part file, or another open part.

- **Features and settings**: features added or removed, features changed (with which settings: a depth, sketch points, an end condition, a new name), features moved in the tree, and the part's own settings (material, equations, units, configurations, properties).
- **Shapes**: both versions' volumes, and the material this version **adds** (shown yellow on the model) and **removes** (red), with their volumes.

3.4 Tabs and closing

Several parts and drawings can be open at once, each in its own tab. New, Open and Sample all open a new tab. Closing a tab with unsaved changes asks whether to save it, and closing RazorCAD asks to save every unsaved part.

3.5 Auto-save and recovery

Auto-save starts once a part has been saved for the first time. While the part has unsaved changes it is written next to the part file, at most every 20 seconds (**Settings ▸ General ▸ Auto-save** sets how often, or turns it off), as `Name.autosave.rcad`. The title bar shows the time of the last auto-save. Saving the part removes the auto-save file.

If RazorCAD closes with unsaved changes, opening the part again offers to **recover** the auto-saved version. It is also marked in the recent list.

> **Warning:** A part that was never saved is not auto-saved. RazorCAD warns before closing it. Save new parts early.

The recent-files list is kept in `%APPDATA%\RazorCAD\recent.json`.

3.6 Opening from Windows and file previews

Double-clicking a .rcad, .rasm or .rdrw file opens it in RazorCAD; if RazorCAD is already running, the file opens in a new tab there. RazorCAD sets the file types up itself when it starts. **Help ▸ Open Files With RazorCAD...** shows whether they are registered and can register them again.

Saving keeps a small picture of the view in the file. Windows Explorer, the Open and Save dialogs and the start window show it as the file's thumbnail.

A part that takes a while to build is stored, once built, in `%LOCALAPPDATA%\RazorCAD\modelcache` (kept under 1 GB). With it the kernel keeps the geometry after each feature. Opening the unchanged file again shows the model at once, ready to edit: the first change after opening rebuilds only from the changed feature on, as any later change does. The store is kept for each version of the geometry kernel, so it survives RazorCAD updates that leave the kernel alone.

Each open part keeps a geometry kernel of its own, as many as the PC's processor threads and free memory allow (see Speed on your PC in the Troubleshooting chapter). Switching back to a part is then immediate, and a part still building goes on while you work in another tab.

3.7 Point clouds

File ▸ Import Point Cloud (PLY, XYZ)... brings a scan into the part as a **Point Cloud** feature, shown as dots: a PLY file (ASCII or binary; its faces are not read) or a text file of x y z lines (XYZ, TXT, CSV, PTS). Up to 200 000 points are kept (every so many of a bigger scan). **Surfaces ▸ Fit Surface to Points** makes a surface body fitted to it by least squares: a **Plane**, a **Cylinder**, a **Sphere**, or a **Freeform** surface through a grid of heights over the cloud's best plane (more grid points follow it more closely). The message gives how far the points are from the surface (RMS and the largest). Build the part on the fitted surfaces with the usual tools (Thicken, Knit, Replace Face...).

3.8 Importing STEP, IGES, STL, OBJ and 3MF

There are two ways to bring in a STEP, IGES, STL, OBJ or 3MF file:

- **File ▸ Open...** opens it as a part of its own in a new tab, named after the file, without asking. A STEP file that holds an assembly asks first:

  - **Open as an Assembly** makes a new assembly with the file's structure: a sub-assembly for each of its sub-assemblies, a part for each part, and a component, fixed, where the file places each one. Each part keeps its own piece of the STEP file and is built on its own, and a part used several times is built once.
  - The first build of each part is kept on disk (in `%LOCALAPPDATA%\RazorCAD\modelcache`, up to 1 GB, shared with parts). The next time the assembly opens, unchanged parts are shown straight from there instead of being rebuilt, so a big assembly reopens in seconds.
  - **Open as One Part** brings every solid into one part, as before.
- **File ▸ Import STEP / STL...** (`Ctrl+I`, also on the Features tab, and **Import STEP / STL / OBJ / 3MF** on the Evaluate tab) offers three choices: a **new part**, **insert in place** into the current part, or **insert and locate with mates**. Locating with mates adds a separate "Locate ..." Move Body feature under the import.

How imports behave:

- The file's contents are stored inside the part file, so the part does not depend on the original file.
- A STEP file's appearances come with it: the colour of each solid and of single faces, and transparency. Your own Appearance and Make Transparent / Make Opaque settings on a body win over them.
- STEP imports as an exact solid. Every solid becomes its own body; a solid made of several separate shells is split into one body per shell; loose surfaces become surface bodies.
- The bodies of a multi-body STEP file are grouped in their own folder under Solid Bodies.
- Tick **Merge touching bodies into one body** in the import feature to fuse them instead.
- STL, OBJ and 3MF meshes are converted to a solid made of their triangles, with coplanar triangles merged. A 3MF file's units and build placement are read. A mesh that is not closed comes in as a surface body, with a note saying how many open edges it has; **Repair Mesh** or **Convert to Solid** (Surfaces chapter) can close it.
- Big meshes are simplified on import, by default to 80,000 triangles, keeping their edges and outlines. Set the limit in the import's properties (**Simplify meshes above ... triangles**; 0 never simplifies). The import's note says how far a mesh was simplified.
- Imported bodies can be cut, extended, patterned and (for STEP) filleted like any other body. A cut or fillet on a body of more than 20,000 mesh faces warns that it can take several minutes.

IGES files (.igs, .iges) open and import like STEP files. Trimmed and bounded surfaces, B-rep solids, B-spline, plane, revolved, extruded, cylinder, cone, sphere and torus surfaces, lines, arcs, B-spline and composite curves and polylines are read, with the file's units and transformations. Faces that close up are sewn into a solid; the rest come in as surface bodies. RazorCAD translates IGES into STEP for the geometry kernel, which has no IGES reader of its own.

Volumes of bodies with B-spline faces (imported NURBS, IGES) are measured exactly.

3.9 Exporting

Command	Result
File ▸ Export ▸ STEP...	An exact B-rep. Each body is written as its own named, coloured solid (in its own colour, else its material's). AP242 unless **Settings ▸ Export** says otherwise (AP214, AP203, the file's units, colours, author)
File ▸ Export ▸ STL...	A mesh: binary, or ASCII, and as fine as **Settings ▸ Export ▸ STL** sets

Command	Result
File ▸ Export ▸ glTF (.glb)...	The bodies as the view shows them, for web viewers and AR : each a mesh with its colour and its render look's metalness and roughness, in metres, Y up
Evaluate ▸ **Export OBJ** / **Export 3MF**	A mesh; 3MF is written in millimetres
Right-click a body ▸ **Export Body as STEP...**	One body only
File ▸ Export ▸ Face as DXF...	The outline of a flat face (see below)
File ▸ Export ▸ Flat Pattern as DXF..., Sheet Metal ▸ **Export DXF**	Flat patterns (see the Sheet Metal chapter)
Drawing ▸ **Export PDF** / **Export DXF**	Drawing sheets (see the Drawings chapter)

3.9.1 Export Face as DXF

Click a flat face, then right-click ▸ **Export Face as DXF...** (or **File ▸ Export ▸ Face as DXF...**). The face's outline (layer `OUTLINE`) and holes (layer `HOLES`) are written as a 2D drawing in millimetres, seen from outside the part. Use it for laser or waterjet cutting, or to take a profile into a 2D CAD program.

Lines, arcs and circles are exact; other curves become fine polylines. The file is DXF R12, which practically every program can read.

4 Navigating the 3D View

4.1 Mouse and keyboard

Action	Mouse / keys
Rotate (free, about the point under the cursor where the drag starts: a red dot shows it)	Middle-drag, or left-drag on the model outside a sketch
Pan	**Shift** + right-drag, or **Ctrl** + middle-drag
Zoom	Mouse wheel (toward the cursor), or **Shift** + middle-drag
Roll	**Alt** + middle-drag, or **Alt** + ← / →
Rotate in steps	Arrow keys 15°, **Shift** + arrows 90°
Pan in steps	**Ctrl** + arrow keys
Zoom to fit	**F**
Standard views	**Ctrl+1** … **Ctrl+7**
Normal To	**Ctrl+8** (press again to flip to the back side)

4.2 Standard views

View menu: Front (**Ctrl+1**), Back (**Ctrl+2**), Left (**Ctrl+3**), Right (**Ctrl+4**), Top (**Ctrl+5**), Bottom (**Ctrl+6**), Isometric (**Ctrl+7**) and Trimetric. **Normal To** (**Ctrl+8**) looks straight at the selected face or plane.

4.3 Up axis: Y or Z

A part can be shown **Y up** (the default) or **Z up**: **View ▸ Up Axis**, right-click **Origin** in the feature tree, or click **Y up** / **Z up** at the right of the status bar. With Z up the **Top plane is XY**, the **Front plane XZ** and the **Right plane YZ**, and Z points up; the tree shows each standard plane's axes beside its name. The choice is kept with the part and used for new parts too.

Only the names of the axes change: the model stays where it is and no feature moves. The triad, X / Y / Z directions, Move / Copy Bodies, reference point coordinates, Measure, Mass Properties and 3D sketch coordinates all use the part's axes, and so do exported STEP, STL, OBJ and 3MF files: a part shown Z up is written Z up, and a file imported into it is read Z up. The origin and corner triad always show X in red, Y in green and Z in blue: the vertical axis is green with Y up and blue with Z up.

An **assembly** keeps its own up axis in the same way (View ▸ Up Axis, the status bar, or right-click its **Origin** in the tree; Undo takes it back). Its tree shows the standard planes' axes, and a STEP file exported from an assembly shown Z up is written Z up. A new assembly starts with the up axis used for new parts.

4.4 Display styles

View menu or the heads-up toolbar: **Shaded With Edges**, **Shaded**, **Hidden Lines Removed** and **Wireframe**.

Perspective (the heads-up toolbar's box button, or View ▸ **Perspective** / **Orthographic**, also in the Display Style list) draws the view in perspective: nearer things look bigger, as a camera sees them. Rotating, panning, zooming (in perspective it moves the view in and out), picking and dragging work as before, and

renders are taken in perspective too. **Orthographic** (the default) draws without perspective, so parallel edges stay parallel. The choice is kept in your settings for every part and assembly.

4.5 Visibility

- **View ▸ Show / Hide Planes** toggles the standard planes.
- The **eye column** at the right of the Feature tree shows or hides each sketch, plane, axis, point, curve, body and body-making feature on its own. Hiding a feature hides the bodies it built. Planes, axes, points and curves can also be hidden one at a time with right-click ▸ **Hide**.
- The **eye button** on the heads-up toolbar: **click** it to hide everything ticked in its list at once, and click again to show it all again (the ticks are kept); **right-click** it (or **View ▸ Visibility...**) for the list: standard planes, reference planes, sketches, **construction lines**, axes, points and surfaces, with select-all and deselect-all.
- **Construction lines** : a sketch's construction lines show with the sketch. Once a feature uses a sketch it is hidden, and its construction lines with it; tick **Construction lines** in the eye's list to show every sketch's construction lines even so (to pick one as a revolve's axis, say). Right-click a sketch ▸ **Show Construction Lines** / **Hide Construction Lines** sets it for that sketch alone, whatever the list says.
- Right-click a body (tree or view) ▸ **Hide**. **`Tab`** hides the body under the cursor; **`Shift+Tab`** shows the hidden body under the cursor.
- **View ▸ Show Hidden Bodies** shows hidden bodies so you can click them to show them again. Choose **Done Showing Hidden Bodies** to finish.
- Right-click a body ▸ **Isolate** hides everything else. The isolated body can still be edited. **View ▸ Exit Isolate** returns.
- Right-click a body, or any feature that made or changed it, ▸ **Make Transparent / Make Opaque**. Transparency is saved with the part and can be undone.

4.6 Section view

Tools ▸ Section View... (or the Evaluate tab) cuts the model open on a plane so you can see inside. Choose the plane, an offset (equations are allowed) and flip the side kept. With a flat face, a plane or a sketch selected when you start, the cut starts on it; while the panel is open, click a flat face or a plane to cut from it instead. Drag the blue arrow on the cut to move it along its normal (the offset follows, in round steps; looking straight along it, drag up to bring it toward you). Tick **Section 2** (and **Section 3**) to cut with two or three planes at once: only what all of them cut away is removed, e.g. a quarter cut. The arrow and a click in the view move the section you changed last. It works the same in assemblies (Evaluate ▸ Section View): every component is cut and capped, and a click on a component's flat face cuts from it. ✓ keeps the cut on while you work; the Section View button then opens the panel again, and ✗ removes the cut. Saved with a cut on, a part or assembly opens with it again, and each tab keeps its own cut. Each cut body or component gets its own cap in its colour, outlined and hatched with section lines; the hatch angle and spacing change from body to body so touching parts read apart. Section View can't be used inside a sketch.

5 Selecting

5.1 Selecting in the part

A click selects what is under the cursor, in this order: a **vertex**, then an **edge**, then a **face**, then a **plane**. **Ctrl**+click adds to the selection.

5.2 Selection filters

The filter buttons on the heads-up toolbar limit what can be picked. Several filters can be on at once, and they also apply while picking references for features.

Filter	Key	Picks
Faces	**X**	Model faces
Edges	**E**	Model edges
Vertices	**V**	Corners (edge end points)
Solid Bodies	**B**	Whole bodies (a click on any face picks its body)
Planes		Standard and reference planes
Clear filters	**F5**	Back to picking anything

Several bodies: with the Solid Bodies filter on, click a body to select it and **Ctrl**+click more bodies to select them too (**Ctrl**+click a selected one again to take it away); in the tree, **Ctrl**+click works with or without the filter. Right-click one of them for what to do with them all: Hide, Make Transparent, Isolate, Body Appearance..., **Edit Material...**, Move/Copy, Combine and Delete. Features that take bodies (Delete Body, Move/Copy Body, Split, Combine, Convert to Solid) start with the selected bodies filled in.

5.3 Quick readout of sizes

Clicking an edge shows a small tag in the view: its length, R and length for an arc, or Ø and circumference for a circle. **Ctrl**+click more edges for their total length (the perimeter when they go all the way round). Sketch entities shown in the view can be clicked the same way.

One face shows its area (and Ø when round). **Ctrl**+click a second face for how they lie: "Parallel · 12 mm apart", "Perpendicular · 90°" or the angle; two round faces give their centre distance; three or more faces their total area.

5.4 Highlighting from the tree

Hovering a feature in the Feature tree highlights what it made in orange: the faces of a boss, cut, fillet, chamfer, shell, pattern or moved body; a sketch's curves; a plane, axis or point; a surface feature's surfaces. Clicking it keeps the highlight in blue while it is selected.

5.5 Selecting hidden bodies

Hold **Shift** and **right-click** to step through the bodies and surfaces under the cursor, front to back; **Shift** + left-click steps back. The current one is shown in orange, through whatever is in front of it. Release **Shift** and **left-click** on the same spot to select it. **Esc** or a click elsewhere cancels.

- With only the face filter on, or while a Property panel asks for faces (Shell, Offset Surface), it steps through faces instead.
- While a Property panel asks for bodies (Combine, Split, Move/Copy, Knit, Thicken), the click adds the body.

5.6 Right-click menus

With bodies or faces selected, a right-click anywhere in the view (empty space included) brings up their menu, as if on the selection itself; one face or several picked with **`Ctrl`** (Face Appearance for them all, and the commands of their bodies). To get another item's menu, click it first, or **`Esc`** to clear the selection.

Right-click on	Offers
A face	Sketch, Normal To, Export Face as DXF, Fillet, Chamfer, Face Appearance..., Part Appearance...
An edge	Fillet, Chamfer (started with that edge), Reference Axis
A vertex	Reference Point
A face, edge or body	Body Properties, **Edit Material...** (the body's own material, or back to the part's), Body Appearance..., Export Body as STEP, Hide, Isolate, Make Transparent, folder commands
Several bodies selected	Hide, Make Transparent, Isolate, Body Appearance..., Edit Material... (one material for them all), Move/Copy, Combine, Delete
A sketch in the tree	Edit, Show / Hide, Change Sketch Plane...
A sheet metal body	View Flat Pattern, Export Flat Pattern as DXF
Empty space	Show Hidden Bodies, Measure and view commands

6 Sketching

6.1 Starting and ending a sketch

Click a plane or a flat face, then click **Sketch** (Sketch tab), or right-click a face ▸ **Sketch**. Clicking a sketch tool such as **Circle** with no sketch open also starts a 2D sketch: pick a plane or flat face and the tool is ready. (For sketches in space, see the 3D Sketches chapter.) You can also start a feature such as Extrude without a sketch: RazorCAD asks you to pick a plane or face, you sketch, and when you exit the sketch the feature continues. If the profile is open you can keep editing or abort and keep the sketch.

A sketch on a flat face of the model follows the face: when the model changes so that the face moves (a taller block, a deeper cut), the sketch moves with it along the face's normal and the features made from it follow. Its curves keep their place on the plane: the sketch moves only along the normal, not across. The face is found again by its direction and where it was in the part, as fillet edges are. If it is no longer there, the sketch stays where it was and its tree item says so; right-click it ▸ **Change Sketch Plane...** to put it on a face again. Sketches put on faces before version 0.20.1 keep the plane only and stay where they are, until you pick the face again with Change Sketch Plane.

- **`Esc`** ends the active sketch tool first; pressing it again closes the sketch and keeps it.
- **`Enter`** does **not** leave the sketch.
- The ✓ in the confirm corner, or **Exit Sketch**, closes the sketch.
- **`Ctrl+Z`** / **`Ctrl+Y`** undo and redo inside the sketch.

Typing sizes while you draw. With a shape started (its first point placed), type a number: a small box opens by the cursor with its sizes. The shape follows what you type; **`Tab`** goes to the next size, **`Enter`** makes it at those sizes and adds a dimension for each length you typed (the others follow the mouse, in the direction it went), and **`Esc`** closes the box. A line typed at 0°, 90°, 180° or 270° is made horizontal or vertical; other angles, an arc's sweep and a polygon's sides shape it without a dimension. Sizes can be expressions, as in any dimension. The shape and its dimensions are one undo step.

Tool	Sizes	
Rectangle (corner or centre)	Width, Height	
Circle	Ø	
Polygon	Ø (of the circle through its corners), Sides (3 to 40)	
Line, Centerline	Length, Angle	
Centerpoint Arc	Radius and Sweep (after its centre); Sweep (after its start)	
3 Point Arc	Radius (after both ends; the shorter arc, on the side of the mouse)	
Tangent Arc	Radius, Sweep (turning the way the mouse went)	
Ellipse	Major R and Minor R (after its centre); Minor R (after its major axis)	
Straight, Center Point and Overall Slot	Length and Width (after the first point); Width (after the length)	

The grid. Points snap to the sketch grid, drawn round the middle of the view; it gets finer as you zoom in. The status bar shows its step (**Grid 5 mm**) while a sketch is open.

6.2 The sketch tool wheel

While sketching, press and hold the right mouse button. After a moment, or as soon as you start to drag, a wheel of sketch tools appears around the pointer. Keep the button held, move toward a tool (it highlights) and release to activate it. Releasing in the centre or pressing **Esc** cancels. A quick right click (press and release without moving) still opens the right-click menu.

6.3 Defined and under-defined sketches

Sketches are made of points, lines, arcs and circles held in place by **relations** and **dimensions**. A numerical constraint solver keeps them satisfied.

- Geometry is **blue when under-defined** and **black when fully defined**.
- The Property panel shows the remaining degrees of freedom.
- An under-defined sketch shows **(-)** in the tree. Construction lines only need to be positioned. To rename a sketch, right-click its tree item and choose **Rename**, including when it has the **(-)** marker.

6.4 Automatic relations

With **Automatic relations while sketching** enabled, Line and Centerline infer parallel or collinear directions from other lines in the active sketch, including construction and projected lines. The reference highlights orange and the cursor identifies the proposed relation. Click to create a persistent relation. Starting near a line's extension snaps onto its supporting line; continuing along it creates Collinear. Endpoint, midpoint and on-curve snaps retain priority. Direction inference uses a 4° angular tolerance and a 9-pixel proximity limit; spline control-polygon helpers are excluded. Turning automatic relations off disables these inferences.

6.5 Sketch orientation

In the sketch Property panel, **Rotate 90°** turns the sketch axes about the sketch normal, and **Flip Side** looks at the sketch from the other side of the plane. Both are saved with the sketch.

6.6 Sketch tools

Tool	Key	How it works
Select	**S**	Select entities, drag geometry
Line	**L**	Click a chain of points. Coincident, horizontal, vertical, parallel and collinear relations are inferred automatically; snapping to a midpoint or onto a curve adds those relations too. Double-click, **Enter** or **Esc** ends the chain.
Centerline		Construction line for symmetry, mirroring, gears or as a revolve axis
Midpoint Line		Midpoint, then one endpoint; the opposite endpoint is mirrored and a Midpoint relation preserves the centre
Corner Rectangle	**R**	Built with horizontal and vertical relations
Center Rectangle		Also adds construction diagonals with a midpoint relation

Tool	Key	How it works
Circle	C	Centre, then radius
Centerpoint Arc	A	Centre, start, end; the direction follows the mouse
3 Point Arc		Start, end, then a point it runs through
Tangent Arc		From the end of a line or arc, kept tangent by a Tangent relation
Ellipse		Centre, end of the major axis, then the minor radius. Trim it for an elliptical arc.
Fit Point Spline	N	Through fit points, with editable tangent direction and strength handles; see below
Point		Lone sketch points that stay, for example to place holes
Straight Slot		Built with tangent and equal relations
Center Point Slot		Slot midpoint, one end-arc centre, then width
Overall Slot		Two outermost points along the axis define total length; the third click defines width
Center Point Arc Slot		Centre of the path arc, start centre, end direction, then width
3 Point Arc Slot		Start centre, end centre, third point on the centre arc, then width
Polygon	P	Set the number of sides in the Property panel
Dimension	D	See below
Trim	T	See below
Extend		A line or arc end runs on to the first entity in its way and stays on it
Split		Splits a line or arc in two at the click; the halves stay collinear or equal
Project Edges	E	See below
Offset	O	See below
Mirror		Select the entities plus a centerline, then click Mirror
Import DXF		See below
Sketch Picture		A picture (PNG, JPEG, BMP, GIF or WebP) on the sketch plane, to trace over. See below
Equation Curve		A spline through the points of **y = f(x)** (x from / to) or of **x = f(t), y = g(t)** (t from / to), in a number of **Pieces**. Part variables can be used and the curve follows them; sin, cos and tan take degrees. A parametric curve that ends where it starts is closed. An **Equation** relation holds it (it is fully defined); select it and use Equation Curve again to change it.
Linear Sketch Pattern		Opens a side panel with an orange preview. Select source entities before or after opening it, then choose an angle, line or curve as direction/path. Copies stay linked to their sources, path, quantity and spacing. Along a curve, spacing is measured along it and copies can rotate with it; open paths stop at their end. A straight direction also supports a second direction.
Circular Sketch Pattern		Copies the selected entities with the relations and dimensions among them. A selected point is the pattern's centre.

Tool	Key	How it works
Make Block / Insert Block / Explode Block		Make Block saves the selection (with its relations) under a name in your settings. Insert Block places copies with a click, in any sketch, turned by an angle, and can delete blocks from the library. A **linked** copy (the default) moves and turns as one (a rigid Block relation holds it) and follows the block: Make Block again under the same name updates every copy in the sketch where it is, keeping relations from other geometry to it. Explode Block makes the selected copies plain geometry with the block's own relations and dimensions. Untick **Linked** for plain copies.
Text		See below
Gears		See the Gears chapter
Airfoil		See the Airfoils chapter
Sketch Fillet / Sketch Chamfer		Set the radius or distance, then click corners (a fillet: where two lines, a line and an arc, two arcs, or a fit-point spline and a line or an arc meet: the spline is cut back to the fillet, which stays tangent to it; a chamfer: two lines). The first one made gets the dimension; the others made before you leave the tool are made Equal to it, so one value changes them all.
Intersection Curve		Click earlier sketch curves to create linked intersections with the active sketch plane, or select a model section
Construction Geometry	`Q`	Toggles selected curves to/from construction; with no curves selected, toggles construction mode for newly drawn geometry. The drawing panel also has **For construction**. Preview and resulting geometry are dashed and ignored by features.
Display Relations		Shows or hides the relation glyphs
Delete	`Del`	Deletes the selected entities or relations

Linear patterns and patterns along a path: confirm with the checkmark, or cancel with the cross (Enter / Esc also work outside input fields). The preview does not change the sketch until confirmed. Use **Sketch Patterns ▸ Edit Pattern** in the sketch Property panel to change sources, path, count, spacing or rotation later; count and spacing accept expressions. Edit original geometry to change the copies' shapes. Copies are linked and cannot be independently dragged; deleting a source, path or generated entity removes that pattern's copies. Parameters and links are kept with the part and follow undo/redo and rebuilds.

Patterns made by earlier versions were independent copies and contain no stored pattern definition. Recreate them with the updated command to obtain a parametric pattern.

Pattern copies use the defined-geometry colour: black in the light theme and light in the dark theme. Purple identifies projected geometry; the pattern preview is orange.

Fillet corners can share one endpoint or have endpoints joined by Coincident relations. If exactly two regular lines/arcs meet at a corner, construction geometry there is ignored when choosing the curves to fillet. An invalid radius leaves the sketch unchanged.

6.7 Dimension

Click a line, circle or arc (or two entities), then click to place the dimension.

You can also select a curve or two entities before starting **Dimension**. Move the pointer to position the orange preview, then click to confirm; the dimension is not created until that click. Press **Esc** to cancel the pending dimension. The text is placed at the point you choose.

With **Select** or **Dimension** active, drag a dimension's number to reposition its annotation without changing the measured geometry or value. A plain click still opens the value editor. This also works for driven/reference dimensions; placement is saved with the sketch and follows undo/redo. Press **Esc** during a drag to cancel the move.

Linear dimension text follows the dimension line and stays upright. Extension lines leave a small gap at the geometry and run slightly beyond the dimension line; short dimensions use outside arrows. For a point-to-line distance, dragging the text also moves the perpendicular dimension line along the reference edge. This lets a short horizontal offset such as 4 mm and an inclined perpendicular offset such as 13 mm use conventional drafting layouts without changing what they measure.

- Moving the mouse switches between horizontal, vertical and aligned.
- Two lines give an angle; a point and a line give a perpendicular distance.
- Values accept equations, such as `width/2`.
- A dimension that would over-define the sketch is added as a driven (reference) dimension.
- **Circles and arcs:** dimension from a circle or arc to another circle, arc, point or line by clicking both. The Modify box offers **Center** (centre to centre), **Min** (nearest) and **Max** (farthest). Min also measures the gap between concentric circles. Switching keeps the geometry in place and shows the new value.
- **Outside references:** you can also click model edges, model vertices or entities of earlier sketches. RazorCAD creates linked construction geometry and dimensions to it, so the dimension follows the model. Example: "keep this hole 25 mm from the right edge" still holds after the width changes.

6.8 Trim

Power trim: hover over a segment and it turns red; click to remove it up to the nearest intersections. Lines are shortened or split, circles become arcs, and new end points get on-curve relations.

6.9 Offset

Click a line, arc, circle or spline, choose the options in the Property panel, then click to create its offset. **Select chain** switches between the clicked entity and its connected contour, including mixed line/arc/spline chains. Corners are re-trimmed or extended; an unjoinable or collapsed offset reports an error.

Direction offers **Pick side**, **Inside / left**, **Outside / right** and **Both sides**. Inside and outside follow the winding of a closed contour, including reversed contours. For an open chain they mean left and right along its traversal. Both sides places a copy at the specified distance on each side: one dimension drives both copies, so the total separation is twice the distance.

Enter **Distance** before creating the offset; expressions and parameters are supported. **Use mouse distance** lets the pointer set the initial distance instead. The resulting offset dimension can be edited later. Linked offsets retain their source references through save/reopen and follow source edits. Constrain the source geometry to hold it in place while changing dimensions.

Line and arc offsets remain analytic. Spline parallels are represented by cubic spline pieces, checked against source-curve normals at sampled parameters within **0.001 mm**; this is a sampled numerical check, not a certified global bound. Cusps, detected crossings and folds are rejected. A large source edit that exceeds the stored fit tolerance asks you to recreate that offset. Ellipses are not supported. Copies and blocks retain plain geometry rather than the offset link.

- You can also click a **model edge** (with **Select chain**, the connected edges lying in a plane parallel to the sketch, such as a face outline) or a curve of another sketch.

- **Linked offset** (default): the offset follows its source, including model edges when the model changes, and one dimension drives the chain (relation "Offset ⇉").
- Untick it for a **single offset**: plain geometry with no link.
- While the tool is active, the faces the sketch plane cuts are highlighted and the cut outline is drawn in teal. Clicking it offsets the whole outline.

6.10 Project Edges

While editing a source sketch, sketches that depend on it are temporarily hidden to avoid displaying outdated projections over the current geometry. They return after leaving the sketch and rebuilding; independent reference sketches remain visible and the saved visibility settings are unchanged.

If a source edit such as a fillet separates formerly joined endpoints, their projected endpoints separate on rebuild while retaining their source links.

Splines projected from another sketch remain linked splines, preserving their control points, degree and weights. Earlier projections saved as line segments retain that representation.

Click a model edge, or an entity of an earlier sketch, to project it into this sketch. Or select edges first and then click **Project Edges** (or press **E**): they are projected at once. Projected geometry is drawn in **purple**, is linked and updates when the source changes (relation "On Edge ⇲"). The ends of projected lines are joined to each other and to sketch geometry already there, so the edges of a face give a closed outline ready to extrude. While the tool is active, the outline where the sketch plane cuts the bodies is drawn in teal; clicking it converts the whole outline. This works for solids and surfaces.

6.10.1 Selecting source sketches for Project and Intersection

While editing a sketch, activate **Project Edges** or **Intersection Curve**. Earlier independent sketches are temporarily shown, including hidden and absorbed sketches; their saved visibility does not change. Click the source curves in the canvas (orbit the view if they overlap). Project creates linked projected geometry. Intersection creates linked points where source lines, arcs, circles or splines cross the active plane (a spline's crossings are solved on its exact curve, every one it makes); coplanar curves retain their geometry. Use the points as guide endpoints. For solid/surface sections, click the displayed section outline. Escape returns to Select.

Projection Source → **Source** can restrict picking to one earlier sketch or to model edges/sections. Choose the intended sketch when several sources overlap in the view. Other source sketches are temporarily hidden; leaving the tool restores their saved visibility. Source geometry is gray, while projected entities are purple. A visible source edge joining two projected sides is not automatically added to the active sketch.

6.11 Spline

New **Fit Point Splines** include a tangent handle at each fit point. Drag a handle to change tangent direction and strength; adjacent pieces keep the same tangent through their shared fit point. Dragging a fit point carries its handle with it. Handle lines are construction geometry and accept ordinary line constraints and dimensions (for example horizontal, parallel, length or angle). Their remaining freedoms count towards sketch definition. Copies, mirrors, blocks, patterns and linked projections preserve the handles. Existing saved splines retain their previous representation.

Tangent handles are shown with solid teal stems, small direction arrows and round grips. Fit points use square markers. Selection names identify **Tangent Handle** and **Tangent Handle Line**, so these controls can be distinguished from ordinary construction centre lines.

Click the points the curve runs through (they snap like line points), then double-click, **Enter** or **Esc** to finish, or click the first point again to close it. Drag a point to reshape the curve, or drag the curve to move it; deleting a point takes it out of the spline. **Convert to Control-Point Spline** (in its panel) makes it a control-point spline that runs through the same points. Splines make profiles on their own or with lines and arcs and are exact curves for extrudes, revolves, cuts, lofts and sweep paths. Lines can be trimmed where they cross a spline.

6.12 Control-Point Spline

Control-Point Spline (beside Spline) draws a spline from control points: click them, then double-click, **Enter** or **Esc** to finish (click the first point again to close it: it then runs round smoothly, with no join). An open one starts and ends on the first and last control points and passes near the others; a closed one passes near them all. The **Closed** box in its panel opens or closes it. Choose the **degree** (1 to 7) in the panel before drawing, or select the spline afterwards to change it (at most one less than the control points): a higher degree is smoother and keeps farther from the polygon. The control polygon is drawn dashed and is made of construction lines, so you can dimension it and give it relations like any lines: horizontal, parallel, equal, an angle. The curve leaves each end along the polygon's end leg, so a relation on that leg (tangent to an arc, collinear with a line) sets its direction there. Drag the control points to reshape it. Deleting the spline deletes its polygon.

Select a control point for its own settings. **Weight** (1 is plain) pulls the curve toward that point when larger and lets it go farther when smaller: with three control points at the corners of a square, degree 2 and a weight of 0.7071 on the middle one, the curve is a quarter circle (to within 0.00001 mm). **Insert After** adds a control point half way along the next leg (the curve changes shape a little); **Remove**, or deleting the point, takes it out and joins its two legs.

6.13 Import DXF

Import DXF brings a DXF file into the open sketch: lines, arcs, circles, polylines with arcs, ellipses and elliptical arcs, splines, points and blocks. The file's units are converted to millimetres. You can scale it and place it as in the file, centred, or by its corner.

Sketch Picture puts a picture on the sketch plane behind the sketch, to trace over (a photo, a scan of a drawing). Set its width (the height follows), centre, angle, opacity, or hide it. To scale it by something of known size in it, draw a line along that, select the line, open Sketch Picture again, pick the picture and give the line's **true length**: the picture is scaled about the line's start. Pictures are shown while you edit the sketch, are saved in the part (shrunk to 2000 pixels across at most) and are not geometry: they make no faces and are not exported.

6.14 Text

Click **Text** on the Sketch tab, then click to place. The panel sets the text, font (the PC's installed fonts, and the DejaVu Sans, Sans Mono and Serif fonts that come with RazorCAD), height (capital-letter height), angle, bold and italic. A font a PC doesn't have is drawn in the DejaVu font most like it. Drag the text to move it. Text extrudes as a boss or cut; text inside an outline is cut out of it. To put text on a curved face, use **Wrap** (Features chapter).

6.15 Relations

Select entities (**Ctrl**+click for several, or drag a box from empty space: left to right selects what lies wholly inside it, right to left whatever it touches; **Ctrl** or **Shift** adds). The Property panel lists the relations that

apply:

Selection	Relations offered
1 line	Horizontal, Vertical, Fixed
2 or more points	Coincident, Horizontal, Vertical
2 or more lines	Parallel, Perpendicular, Collinear, Equal, Horizontal, Vertical
Point + line	Coincident (on line), Midpoint
Point + arc	Coincident (on curve), Midpoint (half way round the arc)
Point + circle	Coincident (on curve)
Line + circle/arc	Tangent
Line + open spline	Tangent at a spline endpoint (control-point or fit-point spline)
Circles/arcs	Equal Radius, Equal Curve Length, Concentric, Tangent
2 or more curves (including splines, ellipses or mixed line/curve selections)	Equal Curve Length; for lines only, use Equal
2 points + line	Symmetric (about the line)
3 lines	Symmetric (two lines about a centerline)

Relations are shown as small green glyphs next to the geometry. Click a glyph to select it, then press **`Del`** to remove it. Hovering a glyph highlights the entities it ties together. For several selected entities, relations already implied by existing constraints are skipped while useful relations are added. A relation refused says why: one that is **already satisfied** names the relations that already define the geometry that way (an information message); one that **would over-define** the sketch names the relations or dimensions in its way. Either way those relations are selected, so you can delete one and add the new relation instead. A true conflict rejects the entire request without changing the sketch.

Tangent with a spline: Ctrl-select a line and an open spline, then choose **Tangent**. The shared endpoint, or otherwise the spline endpoint closest to the infinite line, is used. To choose explicitly (required if both ends are equally close), also select the desired spline endpoint. The selected endpoint is stored and does not switch after edits. For control-point splines the first or last control-polygon leg is aligned with the line; fit-point splines use their endpoint tangent. Coincidence is separate: add Coincident between endpoints or Coincident (on line) to make the curves meet. Closed splines and zero-length endpoint directions are unsupported.

Equal Curve Length (≋) constrains the length measured along each selected curve, not its chord or control polygon. Two arcs may have different radii and sweeps while keeping equal lengths; a spline can also match a line or another curve. Select the curves with Ctrl and choose **Equal Curve Length** in **Add Relations**. With more than two curves, each is matched to the first. Circular curves' **Equal Radius (=)** keeps their radii equal and remains a separate relation. Length links survive source edits, undo/redo, save/reopen and rebuild. Ellipse and spline lengths are integrated numerically from their definitions.

6.16 How profiles are found

Profiles are found automatically from all non-construction geometry and sent to the kernel as exact lines, arcs and circles. Intersections are split, dangling lines are ignored and nested loops become holes. A full circle (or arcs that make one) becomes a single closed curve, so an extruded circle has one continuous

cylindrical face.

For canvas region selection, connected lines, arcs and splines can divide a closed outline into separate selectable cells within the same sketch. Click inside the desired cell; shared boundary edges may belong to several cells. Holes remain attached to their enclosing cell and can also be selected as disks. New cell references preserve boundary direction so regions sharing the same source curve IDs remain distinct. Existing whole-sketch and saved outline selections retain their previous extent. Construction geometry does not subdivide regions.

6.17 Showing and hiding sketches

Sketches absorbed by a feature are hidden by default. Right-click a sketch in the tree ▸ **Show / Hide**, or use its eye in the tree.

6.18 Change Sketch Plane

Right-click a sketch ▸ **Change Sketch Plane…** and pick another plane or flat face for it, for example to repair a sketch whose plane was deleted ("The reference plane is missing"). Its curves keep their sketch coordinates. Only planes above the sketch in the tree are offered.

7 3D Sketches

A **3D sketch** holds lines, arcs, circles and splines in space rather than on one plane. Use it for frames (see the Fast Structure chapter), pipe, duct and wiring routes, and sweep paths.

7.1 Starting and ending a 3D sketch

Click **3D Sketch** on the Sketch tab or the Fast Structure tab (or **Insert ▸ 3D Sketch**). The sketch opens empty with the Line tool ready; there is no plane to pick. While it is open the **3D Sketch** button stays highlighted and the first button becomes **Exit 3D Sketch**. Tools that only work in 2D sketches are greyed out. **Ctrl+Z** / **Ctrl+Y** undo and redo inside the sketch.

7.2 Drawing in space

- Click points to draw lines. A line snaps along **X, Y or Z** from the last point whenever the mouse moves along that direction on screen, in any view. A dashed line shows it and an Along X/Y/Z relation is added. Hold **Ctrl** or **Shift** to draw freely.
- Away from the axes, lines follow the **drawing plane** through the last point. **Tab** cycles XY → YZ → ZX; the plane's name shows at the cursor.
- **Ctrl**+click a plane or flat face, or click a plane in the tree, to draw on it. Points placed there get an **On Plane** relation. **Tab** or **Esc** goes back to XY / YZ / ZX.
- Points snap, in this order, to the sketch's own points, to line and arc midpoints (Midpoint relation), to a point along a line (On Line relation), to the origin and model vertices, along X/Y/Z, then to the drawing plane.
- The **Next Point** box in the Property panel takes exact offsets, for example `0, 0, 250`.
- A selected point shows its X, Y and Z; type new values to move it. A selected line shows its length, ΔX/ΔY/ΔZ and end points; an arc its radius, angle and length.

7.3 3D sketch tools

Tool	Key	How it works
Line	**L**	As above
Centerline		Construction lines that shape the sketch (for example a frame envelope) but make no curve
Point		A lone point
Circle, **Centerpoint Arc**, **3 Point Arc**, **Polygon**	**C**, **A**, **P**	Drawn on the active drawing plane through the first click. **Tab** changes the plane before or between clicks. A polygon stays regular.
Tangent Arc		Click the free end of a line, arc or spline, then where the arc ends. Keep clicking for more arcs; **Esc** ends.

Tool	Key	How it works
Spline	`N`	Click the points it runs through; double-click or `Enter` ends it, clicking the first point closes it. A Tangent relation with a line at an end makes it leave the line smoothly. Select a spline to set its **Shape**: tangent-smooth (C1) or **curvature-continuous (C2)**, whose curvature runs on smoothly through every point; and its **end weights** (how strongly the end tangents pull; more leaves the ends straighter). **Show curvature** draws curvature combs on the splines.
Sketch Fillet / Sketch Chamfer		Click a corner where two lines meet. The sharp corner stays as a point on both lines, so lengths measured to it keep their meaning.
Trim	`T`	Click the piece to remove, up to where other entities cross or touch it
Project Edges	`E`	Click model edges to bring them in, linked to the model (see below)
Dimension	`D`	Length, distance between points, angle between lines, arc radius, circle diameter. As in a 2D sketch, the dimension shows in orange as soon as you click the line (or the second point), with its extension and dimension lines, and follows the pointer; click where it goes (it stays there, drawn with its lines). While placing a length, `Tab` cycles true length → ΔX → ΔY → ΔZ.
Construction Geometry	`Q`	Toggles the selected lines to or from construction
Delete	`Del`	Deletes the selection

Move / Copy / Rotate: drag a selected line, arc or circle to move the selection (it snaps along X, Y and Z), or type ΔX/ΔY/ΔZ and click **Move** or **Copy**, or rotate about X, Y, Z or the drawing plane's normal with **Rotate** or **Copy Rotated**. Copies keep the relations among the copied geometry. Fixed and projected geometry stays put.

7.4 Relations in 3D

Along X/Y/Z, Parallel, Perpendicular, Collinear, Equal Length, Equal Radius, Tangent, Midpoint, On Line, Parallel to XY / YZ / ZX, On XY / YZ / ZX plane, On Plane, Regular Polygon, Fix and Merge (coincident). A solver keeps everything held and reports **Fully defined** or the degrees of freedom left. A relation or dimension that conflicts is refused with a message.

7.5 Projected edges in 3D

Project Edges brings model edges into the 3D sketch: lines stay lines, circular edges become arcs or circles and other curves become splines, with shared corner points. The geometry (purple) stays linked to the model: when the part changes, it follows the edges, and lines and dimensions attached to it follow too. **Unlink from the Model** in the Property panel makes it ordinary geometry.

7.6 Curves from a 3D sketch

Each connected run of lines, arcs and splines is one curve; a T-junction splits it into separate runs. A run can be used as straight segments with an optional bend radius at the corners (pipe and duct routes) or as a

smooth spline, and can be closed into a loop. **Sweep**, **Swept Surface**, **Lofted Surface**, **Curve Driven Pattern** and **Structural Member** use 3D sketches, and so do the features below.

7.7 Features that take a 3D sketch

These features take a 3D sketch wherever they take a sketch: choose it in the panel's sketch list, or click it in the feature tree.

Feature	What it takes from the 3D sketch
Hole, Boss, Snap Hook, Nut Pocket, Insert Hole, Screw Hole for Plastic	Its points that stand on their own, and circle centres. Each point must lie on a face of the part: the hole or boss goes square to that face there, so one 3D sketch can place them on several faces, flat or curved. **Reverse** turns them all the other way.
Sketch Pattern	Its points (as above, anywhere in space): a copy at each, moved from the point nearest the feature
Helix	Its circle (or arc): the centre, diameter and axis. The start angle is measured from the level direction square to the axis (the model's X axis when the axis is upright).
Reference Axis, Reference Point, Reference Plane	Its lines, circles, arcs and points, one by one, from the references list (an axis along a line, or a circle's axis). In the view, a click on one of its points picks the point; for an axis a click on a line, circle or arc picks it (a plane or point takes the whole 3D sketch as a curve, for along a path). Picked ones are drawn orange.
Loft	One closed chain or circle as a profile, which may be curved out of a plane (a saddle): the loft is closed there by a surface through it. A thin loft has walls square to the loft's sides; manual connectors attach to its points, or a seam angle on its circle (measured from the level direction square to its axis).
Lofted Bend	One open chain as either edge, which may run out of a plane
Transition / Reducer	One line as the axis (see Transition / Reducer)
Sprue and Runners	Its lines as runners, for runners that follow a parting surface of your own

8 Gears

RazorCAD's **Gears** tool on the Sketch tab draws involute gear trains as closed profiles, ready to extrude.

8.1 Creating a gear train

1. Draw lines between the gear centres (centerlines work well).
2. Select the lines (or a single point for one gear) and click **Gears**. If you are not in a sketch, RazorCAD opens the selected sketch (or starts one) and asks you to click the centre lines, then press **`Enter`** or **Make Gears**.
3. Each line end point becomes a gear. Neighbouring gears mesh along the line, and teeth are turned automatically so they mesh.
4. Extrude the sketch: each gear becomes its own body.

Gear numbers are shown on the gears in the sketch (view labels only, not geometry).

8.2 The Gear Train panel

Click a gear to open it.

Setting	Meaning
Module and pressure angle	The tooth size and shape for the whole train
Teeth per gear, or ratio per mesh	Setting a ratio sets the next gear's teeth. Changing teeth or a ratio changes the centre-line lengths to suit, including your own length dimensions on those lines.
Backlash, tip/root clearance	Tolerances for prototyping and 3D printing
Bore diameter	The centre hole
Turn	Rotates the first gear
Set the line lengths to the mesh distance	Adds driving dimensions so the gears mesh exactly; each mesh shows ✓ or how far off it is

8.3 Rules for a valid train

Centres must be exactly at the mesh distance, gears must not overlap and each gear needs at least 6 teeth. Fixed line lengths get a module and teeth that fit. Edits that would break a train are refused. A broken train is drawn red with a **Fit gears to the lines** button and can't be extruded.

9 Airfoils

The **Airfoil** tool on the Sketch tab draws a wing or blade section on a line.

9.1 Drawing an airfoil

1. Select a line in a sketch and click **Airfoil** (or click **Airfoil**, then the line).
2. The line becomes the **chord**: a construction line with the leading edge at its start. The airfoil is drawn on it.
3. The airfoil follows the line. Drag its ends or dimension it (or set **Chord length** in the panel) and the airfoil scales and turns with it.
4. Extrude the sketch. The profile is one smooth closed curve that follows the exact section to within 0.002 % of the chord.

9.2 Sections

Choice	Sections
Library	28 NACA 4-digit sections (0006 to 6412), 10 NACA 5-digit sections (21012 to 25012, and the reflexed 23112), and Clark Y (11.7 %, 8, 10 and 14 %)
NACA code	Type any code, for example `4415` or `23012`
Custom	4-digit: maximum camber, its position and thickness. 5-digit: design lift coefficient, camber position, reflexed camber line and thickness. Clark Y: thickness. Decimals are allowed.

Options: sharp or blunt trailing edge, camber to the other side, and the leading edge at the other end of the line.

10 Tutorial: a Guided Loft and a Sheet Metal Tray

This tutorial builds a small duct and a tray with tools added in the 0.17 versions: the closed control-point spline, a plane placed with **Placement**, a loft that follows a guide curve, a variable fillet with a point along its edge, and edge flanges that meet corner to corner. The finished part is in **File ▸ Open Example Part (Tutorial)** (also on the start window): open it to look round, or follow the steps in a new part.

10.1 The bottom profile

1. Select the **Top Plane** and click **Sketch**.
2. Click **Control-Point Spline** on the Sketch tab and click eight control points round the origin, roughly at (50, 0), (40, 25), (0, 30), (−40, 25), (−50, 0), (−40, −25), (0, −30) and (40, −25). Click the first point again to close it: the curve runs round smoothly, near the points rather than through them.
3. Select the spline: its panel shows the **Degree** (3) and the **Closed** box. Select a control point to see its **Weight**; leave it at 1 for now. Drag a control point and the curve follows.
4. Exit the sketch.

10.2 A plane, moved and turned

1. Click **Reference Plane**, pick the **Top Plane** and set the offset to 100.
2. Under **Placement**, set **Turn its axes in the plane** to 30°: the plane's axes turn in the plane, so whatever is sketched on it turns with them.
3. Sketch a circle of radius 20 at the origin on this plane and exit the sketch.

10.3 The guide curve

1. Sketch on the **Front Plane**. With **Control-Point Spline**, click three points: the first where the bottom profile crosses the X axis (about 46.67, 0), one out to the side (about 62, 45) and the last on the circle (20, 100). Finish with a double-click or **Enter**.
2. A guide curve has to touch every profile. The first point lies on the bottom profile's curve, not on a control point: with these control points the curve crosses the axis at 140/3 = 46.6667 mm (select the point and type 46.6667 for its X in the panel).

10.4 The guided loft

1. Click **Loft**, pick the bottom sketch and then the circle's sketch as the profiles.
2. Under **Guide Curves**, pick the guide sketch. The loft's side follows the guide between the two profiles, and the profiles stay exactly as drawn. The note reads "2 profiles, 1 guide curve".

10.5 A fillet with a point along its edge

1. Click **Fillet**, set the type to **Variable size** and **Radius** to **At vertices and points along edges**, and click the loft's bottom edge where the guide starts.
2. Set the radius to 3: that is the radius at the edge's vertices.
3. Under **Points along the edges**, click **Add Point**: half way along the edge (50 %) with a radius of 8. The point is marked on the edge; the fillet swells smoothly from 3 at the vertices to 8 there.

10.6 The tray

1. Sketch an 80 × 60 rectangle on the **Top Plane**, beside the duct, and click **Base Flange** on the Sheet Metal tab with a thickness of 1.5 mm.
2. Click **Edge Flange** and pick the four top edges, with a length of 20. Under **Corners**, **Corner to corner** is the default: where two flanges meet, each runs on up to the other's inside face and ends square, the gap apart, and the bends end in a small V cutout. **Mitred** would close them on the diagonal, and **Overlap** would run one flange past the other's end.
3. Click **Flatten** to see the tray laid out, with its bend lines.

> **Tip:** change any value in the tree (the spline's degree, a weight, the plane's turn, the fillet's point) and the part rebuilds from there.

11 Solid Features

11.1 Live previews

While a feature's Property panel is open, the model is rolled back to just before that feature and the feature is shown in colour: **yellow** for material it adds and **red** (seen through the body) for material it removes. Fillets and chamfers show the material they add or remove; reference geometry being edited is drawn in orange. Click ✓ (or **Enter**) to finish.

Every value field accepts numbers or equations (see Equations in the Evaluate chapter).

11.2 Extrude and Extrude Cut

Profiles uses the same canvas selector as Loft. Click inside a closed region or on its contour to add it; click it again or use its remove button to deselect it. Select several coplanar profiles to extrude them together, including profiles from different sketches. Clicking a sketch in the tree selects its whole profile, including text and generated outlines. Picking an individual region afterwards replaces that whole-sketch selection. Nested regions can be selected separately, so you can extrude a ring or just the disk inside its hole.

Model faces as profiles: click a flat face of the model (where no sketch profile is under the cursor) to use the face itself as a profile, holes and all, without drawing a sketch on it; click it again to remove it. A face profile lies in the face's plane and extrudes out of its body (tick **Reverse direction** to go into it, for a cut). It can be combined with sketch profiles in the same plane and takes every end condition, the draft and every operation; it cannot be a Thin Feature. If the face changes when earlier features change, the profile follows it.

Chain selection selects connected curves. Turn it off to pick individual segments, then choose **Add selected contour** or press Enter; Escape clears the unfinished selection. Open contours enable **Thin Feature**. Turn off **Select profiles in canvas** to pick end references or target bodies; choosing an up-to end condition also switches to reference picking. Existing whole-sketch extrusions continue to build as before.

Extrudes a sketch profile into a solid, or cuts it out.

Start and **End** are independent. **Start from** → **Profile plane** preserves the usual extrusion. **Offset from profile** translates the starting plane by a signed distance along the first selected profile's normal; expressions are supported. **Plane / planar face** projects the profiles along the extrusion direction onto the underlying plane of the selected reference, including inclined planes. Use **Pick start surface in canvas** or the Start surface list. End references have their own picker and do not replace the start reference.

Blind depth, symmetric and two-sided distances are measured from the new start. Reversing the extrusion changes its extent direction while preserving the start offset. With an inclined start and a distance end, both caps follow the starting plane, separated along the extrusion direction. **Up To Surface**, **Offset From Surface** and **Up To Vertex** retain independent end references. Crossing boundaries, zero thickness and a starting plane parallel to the extrusion direction are rejected. The options apply to solid extrusions, cuts, Thin Feature and selected profile regions; saved start-plane references follow source edits. Surface Extrude retains its existing start behavior.

End conditions:

End condition	Meaning
Blind	A set depth
Mid Plane	The total depth split equally on both sides of the sketch

End condition	Meaning
Two sides	Independent Depth and Second depth on opposite sides of the profile plane
Through All / Through All – Both	Through every body, in one or both directions
Up To Vertex	Click a model vertex or choose a reference point
Up To Surface	Click a flat model face or choose a plane; angled surfaces are supported and the extrusion is trimmed to the surface
Offset From Surface	As Up To Surface, at a set distance from it

Second Side: with a one-sided end condition (Blind, Through All, Up To Vertex, Up To Surface or Offset From Surface), tick **Second side** to extrude the other way from the start as well, to an end of its own: Blind (its own Depth), Through All, Up To Vertex, Up To Surface or Offset From Surface, with its own vertex or surface picker (**Pick second side surface in canvas**). The second side always goes the opposite way to the first: an end of the second side that lies on the first side's side of the profile is refused. Both sides share the start, the operation and the Thin Feature; the draft is shared too (each side drafted away from the profile) unless **Second side drafted on its own** is ticked (see Draft). Mid Plane, Two sides and Through All – Both already reach both ways and do not offer it.

Options:

- **Operation**: **Join** merges with existing material where possible; **New body** keeps the result separate; **Cut** removes material; **Intersect** keeps only the material inside the extrusion. A Join that does not touch existing material makes a new body. An Intersect changes each body the extrusion overlaps and leaves the others as they are; with **Feature Scope** limited to chosen bodies, a chosen body it does not overlap is removed. An Intersect that overlaps no body is refused, and an Intersect feature cannot be patterned or mirrored.
- **Thin Feature**: makes a wall of the given thickness from the profile; direction one side, other side or mid-plane. Open sketches switch it on automatically.
- **Draft**: tilts the walls by the **Draft angle**, about the plane the extrusion starts from: the solid narrows away from the profile, or widens with **Draft outward**. The walls of holes in the profile tilt the other way with them, so the material keeps one taper (a cut narrows as it goes in). Mid Plane, Two sides, Through All – Both and a second side draft each side away from the profile. Up To and Offset From an angled surface take the draft too: the walls tilt about the profile and the angled end is trimmed with them. From an angled start surface the walls tilt about the start surface, so the profile's outline is kept where the extrusion starts. If the walls would meet before the end, the draft is refused. With a second side, or with Two sides, **Second side drafted on its own** gives the second side (the Second depth) a **Second side draft angle** of its own, inward or (**Second side drafted outward**) outward; 0 leaves it straight.
- **Feature Scope** (cuts and intersects): **All bodies** by default; untick it and click the bodies to cut or intersect.

11.3 Revolve and Revolve Cut

Profiles uses the common canvas selector: choose regions, connected contours or individual entities rather than automatically consuming a whole multi-profile sketch. A flat model face can be a profile too, as for Extrude; its axis must then be a reference axis, a model edge or face, or a line of a sketch (a face has no sketch axes). Coplanar selections can be combined. Whole-sketch selection remains available explicitly and older files remain compatible.

Choose **Select axis in canvas** separately from the profiles. The axis may be a sketch axis, a line in an earlier sketch (including construction geometry), a reference axis, a straight/circular model edge or a cylindrical face. **Project axis into profile plane** projects its origin and direction onto the selected profile plane; a

perpendicular axis cannot be projected. Return to **Select profiles in canvas** to edit the profile selection.

Extent is **Full (360°)**, **Angle**, **Up To Vertex** or **Up To Surface**. Up To Vertex turns the profile round the axis until it reaches the vertex (click a model vertex, or choose a reference point); Up To Surface until it reaches a plane or flat face. A plane through the axis stops the revolution at whichever half of it comes first; another plane or face trims the revolution where it crosses it, keeping what lies on the profile's side. **Reverse direction** turns the other way round. A vertex on the axis, a surface through the profile and a surface the revolution never reaches are refused (a profile touching the axis needs an end plane through the axis). Surface Revolve goes up to them too. Angle supports **One side**, **Two sides** with independent angles, or **Symmetric** (the total angle split equally). Reverse exchanges the signed direction for one/two sides. The total angle must not exceed 360°. Profiles must not cross a coplanar rotation axis. Full ignores stored angle values. Older files whose angles were clamped to 360° keep that geometry and migrate to Full when edited.

Operation offers Join, New body, Cut or Intersect (as for Extrude; Sweep and Loft offer the same). **Thin Feature** revolves a wall along open or closed contours; open contours enable it automatically. Cuts have the same **Feature Scope** as Extrude; **Select target bodies in canvas** switches explicitly from profile/axis selection to target-body picking.

11.4 Sweep and Sweep Cut

Sweeps a profile along a path.

- **Profile**: a selected sketch region or contour, or **Circular** with a diameter (useful for springs).
- **Path**: a selected sketch chain, 3D sketch, curve or helix; or, under **Path from Model Edges → Select model edges**, edges of the model joined end to end into one chain (open or closed, without branches). A chain of model edges is followed from its end in the profile's plane.
- **Orientation**: **Perpendicular to path** keeps the section square to the path; **Parallel to initial profile** moves it along the path without turning it (tight bends can make the sweep invalid); **Square to faces it runs on** keeps the profile's plane through the normal of the faces chosen with **Select faces in canvas**, for a path that lies on them (a groove round a curved part).
- **Twist angle** (with Perpendicular): the profile turns by this total angle along the path, by distance: **Twist → Evenly along the path**, or **Eased in and out** (slowly at both ends, fastest half way). Positive angles turn by the right-hand rule along the path. Equations are allowed; 0 is the usual sweep. A twist needs an open path with no sharp corners (round them first), and the profile in the plane square to the path at its start.
- **End scale** (without a guide): the profile grows or shrinks about the path from its own size at the start to End scale times it at the end, by the length along the path, evenly or (**Scale → Eased in and out**) slowly at both ends; 1 keeps its size. It works with every orientation and with a twist. Checked against the volumes integrated along a line and an arc to 1e-7.
- **Guide (optional)**: a second curve (sketch chain, 3D sketch, curve or helix) the profile follows. It replaces Orientation and Twist; removing it brings them back.
  - **Guide controls → Orientation**: the profile turns to keep facing the guide and keeps its size. The guide doesn't have to touch the profile.
  - **Guide controls → Section size**: the profile is scaled evenly so its edge stays on the guide. The guide starts on the profile's edge and crosses each plane square to the path once.
  - A guided sweep takes one closed profile without holes or Thin, an open path and an open guide. Both curves run between the planes square to the path at its ends, and don't touch. A guide drawn the other way round is turned automatically. Contact with the guide is checked to 0.0001 mm; geometry that doesn't fit gives an error rather than an ordinary sweep.

- **Thin** option (without a guide).

Select Profile / Path / Guide in canvas activates one role at a time. Click inside a closed profile or on its contour; click path and guide curves directly. Each role has its own **Chain selection** checkbox. Turn it off to pick individual connected entities from one sketch, then choose **Use selected contour** or press Enter. Add or clear an unfinished contour before switching roles. Paths and guides may include construction curves. Different contours from the same sketch can be used in different roles, but the same entities cannot be reused across roles. Unrelated sketch geometry is ignored by selected contour references; missing selected curves report an error. **Other sources (whole sketch / curve)** retains explicit whole-sketch and supported derived-curve selection, including existing files.

Solid Sweep uses the same **Join / New body / Cut** operation control as Extrude and Revolve. Cut supports explicit target-body selection. Without a guide, selected regions retain internal holes and open profiles can use Thin Feature.

11.5 Loft and Loft Cut

Blends between two or more section profiles, in order. A third or later profile shapes the middle.

Activate **Select profiles in canvas**, then click a sketch contour or inside a closed region. Multiple regions in one sketch can be selected independently; holes belong to their enclosing region. **Select guides in canvas** switches to guide selection, including construction curves. Source sketches are temporarily shown while selecting, without changing their saved visibility.

Chain selection picks connected curves automatically. Turn it off to select individual segments from one sketch, then choose **Add selected contour** or press Enter. For a sweep's path or guide, tick **From one segment to another** as well: click the first segment and then the last, and the segments between are added (the rest of the sketch is left out). Escape clears an unfinished selection. Profiles and guides have separate ordered lists with move and remove controls, matching numbered highlights in the canvas. **Other sources** retains whole-sketch and supported 3D curve selection. Stored contour references follow source curve IDs through edits and renaming; missing or ambiguous boundaries report an error. Surface Loft uses the same contour selection for open or closed sections.

Guide contacts are checked against the actual CAD profiles. The fit uses the original guide curves and refines intermediate sections until sampled distances to the CAD faces are within **0.01 mm**. If this check fails, the feature reports an error: add a section near a tight bend. These are sampled numerical checks, not a guarantee over every point of the continuous curve. Thin lofts retain the wall's offset from the sketched guide.

Guide construction → CAD constraints (two profiles) builds patches constrained directly by the original 3D guide edges. It checks the kernel's contact error and independent distances against **0.0001 mm**. This method currently requires two profiles without Thin Feature, with guides starting and ending on corresponding patches. Holes in the profiles are lofted between them without the guides (each hole's sections joined directly) and cut out of the guided outline; they must stay inside it. **Across guide seams** selects C0 (contact, the default), G1 (tangent joins) or G2 (curvature-continuous joins along shared guides). Original profile corners are preserved. Unsupported or incompatible constraints report an error without falling back to approximation. **Adaptive approximation** remains available for the broader set of loft options and is the default for existing features.

11.5.1 Profile correspondence

Enable **Edit profile connectors** to display numbered connections between matching profile vertices. Use **Pick start in canvas**, choose a start vertex in the panel, or drag a connector endpoint to another existing vertex on that profile. Moving an attachment shifts the correspondence around the contour while preserving boundary order. For circles, set the seam angle (expressions are supported) or drag the seam in

the canvas. Open surface profiles use their endpoints; reversing their direction exchanges the endpoints.

Settings follow the selected profile through renaming, reordering, save/reopen and source edits. A deleted attachment reports an error and must be reselected. Use a profile's **Automatic for this profile** button to clear its settings, or reset all connectors to return to automatic matching. Closed profiles must share the same traversal direction: reversing only some would fold the loft. Manual matching is passed to the CAD builder without automatic remapping and also applies when guides or endpoint conditions are used.

Manual connectors support sketch profiles, and in a Loft 3D sketch profiles too (their points, or a circle's seam angle; not a start inside a curve). **Start vertex → Inside a curve...** (or dragging a connector onto a curve between vertices) starts the profile at a point inside one of its curves, **Along it** a share of the curve's length in its own direction (expressions allowed): the profile gains a point there, without splitting the sketch, so a triangle can be lofted to a square by starting the triangle inside one of its edges. Every profile must end up with the same number of points.

Profiles with holes list each hole: **Hole 1**, **Hole 2**... choose which of this profile's holes goes with the other profiles' hole of that number (pair them by hand when the automatic pairing joins the wrong ones), and **Hole ... start** sets where each hole starts (a vertex, inside a curve, or a seam angle for a circle), as for the outline.

Thin lofts take the connectors too: each wall starts at its vertex nearest the profile's start. A thin loft with a 3D sketch profile is the solid loft's walls (made square to its sides, sharp at its corners), so its connectors work as a solid loft's. Other 3D curve profiles use automatic matching. **More connectors**: after a profile's start vertex, **Add connector** adds another vertex to pair (Connector 2, 3...), so profiles with different numbers of corners can be matched corner to corner as you choose (a square's four corners to four of a hexagon's six). Every profile needs the same number of connectors, in the same order round it; between two connectors, the profile with fewer sides has its longest ones halved until the sides pair one to one. Connectors in a different order round one profile would make the loft's sides cross, and are refused. Surface Loft uses the same controls.

11.5.2 Start and end conditions

Start Condition and **End Condition** control the first and last section independently:

- **Free** preserves the usual loft when both ends are free.
- **Normal to profile** makes the longitudinal endpoint derivative normal to its planar sketch.
- **Direction** uses a world axis or a reference axis. Create a reference axis from a sketch line for an inclined direction.
- **Tangent to adjacent faces (G1)** matches the tangent planes of existing solid or surface faces along the endpoint profile. Click **Select start support faces** or **Select end support faces**, then click the adjacent side faces in the canvas. Select all faces needed to cover the outer and inner profile edges, including Thin walls: one cylindrical face can cover a circle; a rectangular extrusion needs its four side faces. For a hollow section also select the adjacent inner walls. The profile must coincide with their boundary, not merely lie inside a face. Click **Select profiles in canvas** to return to contour selection.

Curvature to adjacent faces (G2) also matches the curvature of the selected adjacent faces. Set this independently at either end, including G2 at both ends or a G1/G2 combination. It uses the same support-face selection rules as G1. Curvature is measured on the CAD surfaces, including the along-edge, across-edge and mixed components; incompatible geometry reports an error. A sampled CAD normal check also rejects detected local collapses between profiles; reduce Influence or add a section if this happens. This does not certify the absence of every possible self-intersection.

Influence is a positive multiplier of the endpoint derivative length; 1 uses the distance to the neighboring profile as the scale. Larger values extend the endpoint's influence farther into the loft. Expressions are

supported. **Reverse tangent direction** reverses the signed derivative and may create an invalid or overlapping loft.

Endpoint conditions require matching profile curve counts and a smooth loft. Solid lofts support **internal holes and Thin Feature**, including Outside, Inside and Mid-plane sketch offsets and open Thin profiles. Every section must have the same number of corresponding inner loops. Outer and inner walls are constructed together and sewn to end faces with openings, rather than filling the cavities. G1 and G2 support selection must cover all walls. They support two or more sections, and closed profiles for solids or open/closed profiles for surfaces. Intermediate sections use C2 cubic interpolation. Normal to profile requires a planar sketch; Direction also supports curve sources in Surface Loft. Closed surface profiles stay uncapped.

Guide curves with endpoint conditions use **Adaptive approximation**. Guides must meet every section in order, and their endpoint tangents must agree with the requested tangent planes and direction. Conflicting constraints report an error. Refinement uses 8/16/32/64/128 intermediate stations per profile interval and checks independent native guide points against the final CAD side faces to **0.0001 mm**. With Thin, guide checks retain the sketched outline's offset to the outer wall; inner loops follow the guide deformation. Multiple guide deformations blend smoothly between contact locations (C2 blend weights), preserving physical profile corners. Influence remains relative to the original neighboring section, independently of the refinement count. **CAD constraints** with C0/G1/G2 guide seams still requires free endpoint conditions; selecting it with endpoint constraints gives an explicit error instead of switching methods.

The kernel constructs clamped tensor B-splines (cubic, or degree five along the loft when G2 is requested), refining profile interpolation from 8 through 128 intervals. It checks off-node profile contact and endpoint derivatives, including influence and direction. Final sewn CAD faces are checked again at 33 positions per endpoint for contact within **0.0001 mm**, tangent-plane mismatch within **0.5°**, and relative endpoint derivative error within **0.0001**. These are sampled numerical checks, not a certified global bound. Unsupported or incompatible constraints report an error without dropping the endpoint conditions. G2 additionally compares all three components of the second fundamental form on the final CAD faces, with curvature mismatch at most **0.001 1/mm**. Feature notes report measured contact, angular and G2 curvature errors.

- A circle is split into as many equal arcs as the other profile has corners.
- Profiles with different segment counts give an error naming the sketches.
- Options: straight (ruled) sides, Thin. Without straight sides: **Through the profiles** curvature-continuous (C2) or tangent-continuous (C1), and **Smooth the surface**, which evens out its bends.
- **Guide Curves** (optional): curves the loft's outline follows between the profiles, such as a body's outline seen from the side. Each runs from the first profile to the last, through every profile, and is a selected chain of sketch curves or a supported curve source. Separate guides can come from the same sketch. One guide pulls the loft along it; two or more shape it round the outline, each where it touches the profiles. The profiles themselves are kept as drawn. Holes in the profiles, and a thin loft's inside, move with the outline, so a thin wall keeps its thickness; for a thin loft, draw the guide on the sketched line.

11.6 Hole

Hole (Features tab, **Insert** ▸ **Features** ▸ **Hole...**) makes standard holes.

Type	Sizes
Hole (clearance)	ISO 273 close, normal or loose fit, for M2 to M24
Counterbore	For socket head cap screws (ISO 4762)

Type	Sizes
Countersink	ISO 10642, 90°
Tapped hole	ISO metric or Unified (inch), drilled at the tap drill size; the thread is shown as a cosmetic thread and given in drawing callouts
Counterdrill	A wider drill (standard sizes: the counterbore's diameter and depth) with a drill point of its own angle down to the hole
Slot	Two half circles of the hole's diameter joined: **Slot length** between their centres, **Slot angle** from the face's X direction
Custom sizes	Your own diameters and depths

- **End**: **Blind** (with a **Drill point** of **Drill point angle**, 118° unless set), **Through All**, **Up To Next** (each hole until its axis first comes out of the material: through the wall it starts in), **Up To Vertex**, **Up To Surface** and **Offset From Surface** (a plane or flat face, picked in the canvas or chosen in the list; each hole's own depth along its axis); reversible.
- **Far-side countersink** (Through All, Up To Next): a countersink of its own diameter and angle where the hole comes out (the far side of the first wall for Up To Next, of the last for Through All).
- Place holes at clicked points on a face, or exactly at the points and circle centres of a sketch on the face. On a flat face each clicked hole shows its place as **From left** and **From bottom**: its distances from the face's left and bottom edges, looking at the face (its up the part's up). Type them to put the hole exactly there. Each click adds a hole, more than one on the same face too; a click on a hole already placed takes it away. A hole on a flat face keeps to the face as the part changes: when the face moves (a thicker plate) the hole starts on it where it is now, and when the face changes size (a dimension, a configuration) the hole stays as far from the face's nearer edges, across and along, as it was placed. Bosses, snap hooks, the 3D-printing holes and pockets keep to their faces the same way. On a curved face (a shaft, a pipe) a hole is drilled square to the face where it was clicked: a radial hole.
- Holes work on sheet metal too, and a drawing's **Hole Callout** reads their size from the feature.

11.7 Helix

Pick a sketch (or a 3D sketch) with a circle; it sets the axis and diameter. Define the helix by **pitch and revolutions**, **pitch and height**, or **height and revolutions**. Options: start angle, clockwise, reverse. Use a helix as a sweep path for springs and coils.

11.8 Thread Cut and Thread Extrude

Click a bore (Thread Cut) or a shaft (Thread Extrude).

- The size is chosen automatically from the diameter, or choose a **metric** (ISO) or **inch** (Unified) standard size, or custom. Coarse / UNC sizes are preferred.
- Optional pitch; **left-hand** option.
- Length: the whole face, a depth, or up to a plane or face. Threads never reach past the ends of their face.
- The bore or shaft is resized to the thread size if needed (a bore to the minor diameter, a shaft to the major). A bore that would break out of the body is refused.
- Threads run all the way to the ends: grooves run out into open air, otherwise the thread is cut off flush with the end.

11.9 Cosmetic Thread

Cosmetic Thread (Features tab, **Insert ▸ Features ▸ Cosmetic Thread...**) marks a bore or shaft as threaded without modelling the thread, so it costs nothing to rebuild. Use it where the thread only needs to be shown and called out.

- Click the bore (internal thread) or the shaft (external thread). The thread starts at the end nearest the click; **Start from the other end** reverses it.
- Size, pitch, left-hand and length (whole face, depth, or up to a plane or face) as for Thread Cut. The feature's note says when the face is not the size the thread expects (a bore about the minor diameter, a shaft the major).
- In the model it shows as a thin helix on the face. Tapped holes made with the Hole feature show their threads the same way.
- On drawings it is drawn by convention: from the side, thin lines at the thread's other diameter and a line where it ends, dashed for a hidden bore; end on, a three-quarter circle. A view's **Cosmetic threads** box turns them off.
- **Hole Callout** on a bore with a cosmetic thread gives the thread, for example "M10×1.5 THRU ALL / TAP DRILL Ø8.5 THRU ALL".

11.10 Wrap (Emboss / Deboss)

Pick a sketch (outlines and/or text) and a face, flat or curved. The sketch is projected along its normal onto the face and raised (emboss) or sunk (deboss) by the depth.

11.11 Patterns and Mirror

Patterns and mirrors copy features or bodies. Any feature that changes the bodies at a place can be copied: extrudes, cuts, holes, fillets, chamfers, drafts, domes and sheet metal forms. A feature without a shape of its own (such as a fillet) is run again on the matching edges or faces of each copy.

Feature	Copies
Linear Pattern	Along one or two directions (X, Y, Z or any reference axis)
Circular Pattern	Round an axis (X, Y, Z or any reference axis)
Mirror	About a plane or flat face
Sketch Driven Pattern	At every point (or circle centre) of a sketch
Curve Driven Pattern	Along a sketch, 3D sketch or helix, spaced evenly to the end or at a set distance; moved, or turned to follow the curve
Table Driven Pattern	A copy for each row of a table: **x, y** from the feature along a plane's directions (the feature's own sketch plane when none is chosen) and optionally an **angle** turning the copy about the plane's normal through the feature. Rows are separated by `;` or new lines, cells by commas, tabs or spaces, so a table pasted from a spreadsheet works (a header row is skipped); cells can be sums or use variables
Fill Pattern	On a square or staggered grid over a flat face, kept a margin from its edges (for example a perforated panel)

Linear Pattern options : **Distance** → **From the first to the last** spreads the instances over a total length instead of setting the spacing between each two. **Spacing grows by** makes each spacing that much more than the one before (a negative value: less), for copies that spread out or close up. **Symmetric** adds as many again the other way from the feature. With a second direction, **Pattern the feature only** copies just the feature along it, not the whole first row (an L of copies).

Circular Pattern options : **Angle grows by** (without equal spacing) makes each angle between copies that much more than the one before. **Symmetric** adds as many again the other way round (less than a whole turn). **Rings** makes rings of copies further out from the axis, **Ring spacing** apart, out through the feature's middle.

Instances to Skip (Linear and Circular) : instance numbers to leave out, separated by commas: 1 is the feature, 2 the first copy, -2 the first copy the other way (symmetric). With a second direction or rings, row.column: 3.2 is the third along the first direction (or round) in the second row (or ring).

What to copy. Choose the **Feature**, and add any number of **More features**: each is copied the same way. In a Linear, Circular, Sketch Driven or Curve Driven Pattern, **Clicking the model picks** either **Faces**, where each face brings the feature that made it (click a hole's wall to copy the hole), or **Bodies**, which are copied whole to every place as new bodies. Mirror takes bodies too (tick them, or click them on the model). Fill Pattern copies features only.

Feature folders. Right-click a feature ▸ **Add to New Folder...** (or **Move to** a folder already there, or **Remove from Folder**) to group features in the tree; a folder sits where its first feature is, with its features under it. Right-click a folder to rename it, delete it (its features stay), or make a **Linear** or **Circular Pattern of the Folder**. A pattern's **More features** list takes folders too: it copies whatever is in the folder each time it rebuilds, so a feature added to the folder later is copied as well.

Library features. Right-click a feature or a feature folder ▸ **Save as Library Feature...** to keep it under a name in your settings, for any part: the feature with the sketches it uses (and, for a folder, everything in it), and the variables its sizes use. **Features** ▸ **Library Feature** puts it in the part you are in: choose it, set its variables (each copy gets variables of its own: a second copy's `w` becomes `w_2`), the angle to turn it about the face's normal, then click the plane or flat face for its first sketch. The copy goes into a folder of its name, and its features stay ordinary features you can edit. A feature that uses a plane, a body or projected geometry outside the group cannot be saved this way (put that in the group too).

Patterns of cuts are cut out in one operation, and patterns of added features (bosses, pins, ribs) are fused into the body in one operation, so large patterns rebuild quickly. A plate with 200 bosses builds in about 2 seconds.

A pattern or mirror of a sheet metal feature that makes a body (Base-Flange, SM-Extrude, Lofted-Bend, Transition) copies that body as it is at the pattern, with the flanges and cuts added to it so far. Each copy is sheet metal with its own flat pattern. A mirrored copy's flat pattern is the mirror image, with its bends going the other way. A mirrored lofted bend becomes a plain solid.

11.12 Fillet

Open the feature and click edges on the model. The model is rolled back to just before the feature so every edge can be picked; tick **Preview** to see the result. The radius accepts equations.

- Edges are remembered by type, direction and position, and found again after parameter changes.
- Where a fillet can't be added on top of earlier fillets (a corner where several blends meet), RazorCAD recomputes it together with the preceding fillets in one operation.
- Where two blends exactly meet (for example R2 on both edges of a 4 mm wall) the result is a full round.
- **Variable size**: a radius at the start and another at the end of each edge, or **Radius** ▸ **At vertices and points along edges**: the panel lists the vertices of the selected edges (the ends of each run of edges) with a radius for each; empty ones use the fillet's radius. Along each edge the radius changes from the one at its start to the one at its end.
- **Points along the edges** (with radii at the vertices): **Add Point**, then choose the edge, where along it (the share of its length from its start in %, or a distance from its start in mm) and the radius there; or

Add by Clicking and click on the edge where the point goes. The radius runs smoothly from the edge's start through its points to its end. While the fillet is edited each point is marked on its edge, so you can see which end the % counts from; **drag a mark** along its edge to move the point (its % or mm follows, and the fillet rebuilds when you let go).

- **Setback at corners**: where three or more fillets meet, the corner is blended smoothly from this distance out, instead of the usual small corner patch. A little more than the radius works well; 0 keeps the usual corner. The fillet's note says how many corners were blended.
- **Face fillet**: pick faces instead of edges; the edges where they meet are rounded.
- **Full round** : click a side face, the face to round off, then the other side face (flat and parallel: a wall of even thickness). The middle face is replaced by one round meeting both sides, its radius half the wall's thickness.
- **Size by → Chord width** (constant size) : the value is the width across each fillet, from edge to edge; each edge's radius follows from how far its two faces turn (measured at its middle: exact between flat faces).
- **Shape → Curvature continuous (G2)** (constant size) : instead of a round (which meets the faces beside it in tangent but jumps from flat to curved), each blend leaves its faces with the same tangent plane *and* no curvature, then curves gradually: no visible line where it starts, as on styled housings and covers. It sets back from the edge as far as the round of that radius would, but stays closer to the sharp corner, so it takes less material away. The blend is exact (a curve of degree five swept along the edge, not a fitted patch). It is made on straight edges between flat faces whose ends meet flat faces, and on circular edges all the way round between a flat face and a round face on the same axis (the foot or top rim of a boss, a counterbore's edge: the section revolved round the axis); round corners and other curved edges in a fillet of their own. The note reports the tangent and curvature mismatch measured on the CAD faces. For a G2 patch over an opening use **Filled Surface** with Boundary continuity **Curvature (G2)**; for G2 joins between lofted profiles use **Loft → Across guide seams → G2**.
- **An edge's own size** : each edge in the list has a box for a radius of its own (expressions allowed); blank uses the fillet's. One feature can so round edges with different radii in one go, corners and all.
- **Add a face's edges**: then click a face, and all its edges are added (clicked again, removed).

11.13 Chamfer

Click edges as for a fillet. Chamfer types: **equal distance**, **distance + angle** or **two distances**. **Flip direction** swaps the sides. With equal distances each edge can have a distance of its own, and **Add a face's edges** works as for Fillet.

Vertex (cut off a corner) : click corners of the model where three straight edges meet; each is cut off by the plane through the points the distance along each of its edges. **Different distances along X, Y and Z** gives the edges running most along X, Y and Z their own distances. A distance as long as its edge, or a corner in a recess (where the cut would add material), is refused.

11.14 Shell

Click the faces to remove and set the wall thickness, inward or outward. With no faces removed, the body is **hollowed**: a closed void inside, the thickness in from the outside everywhere (a body too thin for it is refused). Hollowed **outward**, the body itself becomes the void and the walls are built round it, the thickness out from every face (edges and corners rounded).

Walls of Their Own Thickness : **Select walls in canvas**, then click flat faces of the body: each wall gets a thickness of its own, more than the shell's (the box beside it). Inward, the material behind the face is kept down to that depth; outward, the face is built out to that thickness over its own outline. A removed face

cannot have one (it is the opening). Shell validates the resulting solid and rejects a failed offset that merely returns the original full body. For inward shells, small fillets or chamfers can be retained outside with sharp corners inside; the feature note reports this construction. If no valid cavity can be made, reduce the thickness, select different removal faces, or place Shell before the fillets and chamfers.

11.15 Draft

Pick a neutral face, then the faces to tilt by an angle. **Reverse** tilts them the other way.

11.16 Rib

An open sketch line is thickened and filled out until it meets the model, either in the sketch plane (for example a gusset across a bracket corner) or square to it.

11.17 Dome

A spherical cap or half ellipsoid on round faces, or a smooth dome filled over any other flat outline. **Reverse** presses it in.

11.18 Plastic part features

Four features for moulded parts (Features tab).

- **Mounting Boss** : posts at clicked points on a flat face (or at a sketch's points and circle centres), rising out of the face: **Outer diameter**, **Height**, a **Draft on the outside**, a **Hole** (diameter, and a depth: empty for the full height), and **Ribs**: how many, their width, how far out from the boss and how high.
- **Snap Hook** : a cantilever beam (**length**, **thickness**, **width**) with a hook at its tip: its **overhang**, its **lead-in angle** from the beam (the slope it slides in on) and its **catch angle** (90°: square, holding; less: it lets go when pulled). **Hook faces** turns it about the face's normal.
- **Lip / Groove** : click the flat rim face along the open edge of a shelled part: a **lip** (raised) or **groove** (sunk), along its **inner** or **outer** edge, of a width and height (depth). For two halves that fit: a lip on one, on the other a groove a little wider and deeper.
- **Vent** : slots across a sketch's closed outline on a face, of a **slot width** with **bars** between them, at an angle, kept a **margin** in from the outline, cut through or to a depth.

11.19 Core and Cavity

Core and Cavity (Features tab) makes the two halves of a simple mould for the part : a block round it (**Block margin** beyond it every way), the part taken out of it grown by the **Shrinkage** (the plastic shrinks as it cools), split at a parting plane square to the **Pull direction**. The plane goes at the part's largest section along the pull (or **At a height up the part**). The half the pull direction points into is the **Cavity**, the other the **Core**; holes along the pull become shut-offs where the plane crosses them. **Parting** can also be **a parting surface** of your own (a surface body reaching past the block all round, made with the surface tools along the part's parting line): the block is split by it, the side the pull points into the cavity. Check the part with Draft Analysis and Undercut Check first.

Side Core (after Core and Cavity) makes a side action for an undercut: click its faces on the part (a side hole, a catch) and the direction it is pulled (the first face's normal, or an axis). A block round the faces (grown by the margin), from the part out past the mould block, is taken out of the core and the cavity and made a body of its own, **Side core**.

Sprue and Runners cuts round runners along the lines of a sketch on the parting plane (half in the core, half in the cavity; they gate into the part where they meet it) and a tapered sprue from the back of the cavity down to the start of the first line.

11.20 Move Face

Push or pull flat faces along their normal, or turn a face about one of its edges. The neighbouring faces follow.

11.21 Delete or Resize Fillet, Defeature

These edit a solid directly, with no history needed: good for imported parts.

Delete / Resize Fillet (Surfaces tab): click fillet or chamfer faces. **Delete** carries the faces either side on until they meet at a sharp edge again; **Resize** then rounds (or chamfers) that edge at the **New radius**. The faces round them are grown or shrunk to close the gap exactly, so it works on most rounds and chamfers, including rounds that run into other rounds at a corner; the message says what the faces were (R2, chamfered, blended). When the faces cannot close the gap, it falls back to the simpler way: rounds along straight edges between flat faces (convex or concave), full rounds on circular edges (between a flat face and a round face, like the foot or top rim of a boss) and chamfers between flat faces, whose ends meet flat faces.

Defeature takes every fillet up to a radius and every hole up to a diameter out of a body (or of every solid), for a simpler part to send to a supplier or to analyse. A hole is filled with its counterbore, countersink and drill point; one whose counterbore is wider than the limit stays whole. They are all taken out at once, exactly, the faces round them grown or shrunk to close the gaps (the message then says "taken out exactly"). If the kernel cannot close a gap, the body is done the simpler way, one at a time, and fillets whose ends do not meet flat faces, and holes that break out of the side of the part, stay; the message says how many were taken out.

11.22 Scale

Scales every body by one factor, about the origin or about the centre of the bodies.

11.23 Long operations

If a feature takes more than a few seconds, a banner above the model says what is being worked on ("Still working on Extrude Cut1 (12 s)...") with a **Stop** button (also next to the progress bar). Stop restarts the geometry kernel: a feature you are still editing is cancelled and a finished feature is suppressed, so you can unsuppress it to try again.

12 Bodies and Multi-Body Parts

12.1 Solid bodies

A part can contain several solid bodies. A boss that doesn't touch the existing material makes a new body; untick **Merge result** to keep a touching boss separate. A cut that slices a body in two leaves two bodies.

The **Solid Bodies (n)** folder at the top of the tree lists the bodies:

- Click a body to select and highlight it.
- Double-click (or right-click ▸ **Rename**) to name it. Unnamed bodies are named after the last feature that changed them.
- Bodies are identified by the feature that created them, so names and materials stay attached when you change dimensions.
- **Folders**: right-click ▸ **Add to New Folder** or **Move to folder** to organise bodies.
- **Identical bodies** are listed once with their count (×N). Open the item to see each body. Right-click for Isolate, Hide or Show all of them. Identical means the same volume, surface area, principal moments of inertia, handedness and material (and for sheet metal, the same thickness and flat pattern).

12.2 Body Properties

Right-click a body (tree or view) ▸ **Body Properties...**. It holds the body's name, **material**, description and any number of **custom properties** (name/value pairs such as part number or finish). It also shows the body's mass, volume, area and centre of mass. For sheet metal bodies it adds the sheet data (Sheet Metal chapter).

12.3 Materials

The **Material** item in the tree sets the part material (**Tools** ▸ **Edit Material...**). A body set to **Use part material** follows it. A body given its own material keeps it when the part material changes and is drawn in that material's colour. Mass Properties adds up every body with its own density. The material list is in Appendix B.

12.4 Appearances

The **Appearances** tab (above the feature tree, next to Configurations) holds the colours and finishes put on the part: on single **faces**, on whole **bodies**, or on the **whole part**. Use them to tell parts and bodies apart at a glance, or to set a part up for a photorealistic render.

Adding one. Select faces or a body (or nothing, for the whole part) and click **+ Add Appearance** in the tab (or **Add Appearance** on the Render tab, or right-click a face ▸ **Face Appearance...** / **Part Appearance...**, or a body ▸ **Body Appearance...**). The panel shows:

- **Apply to**: Faces, Bodies or Whole part. With Faces or Bodies, click them in the view to add them; click one again to take it off.
- **Appearance**: the library, by category. **Metals** (machined, mill-scale, brushed and polished stainless, chrome, galvanised, cast iron, brushed, bead-blasted and cast aluminium, checker plate, brass, copper, bronze, titanium, gold, weathering steel); **Anodised aluminium** (clear, black, red, blue, gold, green, purple, orange, titanium grey); **Paints and coatings** (gloss, satin and matte paint, metallic paint, smooth and textured powder coat, hammertone, safety yellow, red oxide primer, a plain colour); **Plastics** (gloss,

matte, textured, nylon, clear acrylic, frosted); **Glass** (clear, tinted, frosted); **Timber** (oak, pine, walnut, maple, cherry, birch plywood, MDF); **Stone and concrete** (concrete, polished granite, marble, slate, sandstone, brick); **Rubber** (black, grey, red, ribbed matting, translucent silicone); **Meshes and gratings** (bar grating, expanded metal, perforated sheet with round holes or slots, woven wire mesh, welded wire fence mesh, chain-link fence); **Fabrics and composites** (carbon fibre twill, canvas, leather, felt).

- **Colour**: any colour over the look (a red anodised, a green powder coat); **Reset** goes back to the library's colour.
- **Texture** (looks with a pattern): its **Scale** and **Angle**. Patterns are drawn at their real size (a bar grating's bars 30 mm apart, a brick 215 mm long) and laid on each face from the side it faces most, so nothing needs unwrapping.
- **Name**: shown in the tab (empty: the look's name).

✓ keeps it; ✗ cancels (an edit goes back to what it was).

Meshes and gratings are see-through : a plate given bar grating, expanded metal, perforated sheet or a fence mesh shows its holes (and casts the matching shadow in renders) without them being modelled. The part stays light and quick to rebuild, and the face can still be picked, measured and sketched on.

Order matters . The tab lists the appearances top to bottom in the order they apply; each goes over those above it. A face coloured red and, lower down, its body given oak shows oak; drag the oak above the red (or right-click ▸ **Move Up**) and the face is red again. The first row, **Materials and body colours**, is what is under them all: the bodies' materials and colours.

In the tab click an appearance to see what it is on, double-click it to edit it, drag it to change the order, and right-click it to **Rename**, **Suppress** (it stops applying but is kept), **Copy**, **Move Up / Down** or **Delete** it. Undo and Redo take them back like any change. Faces are found again by where they are when the model changes; if a face can no longer be found the appearance shows **!** (hover for how many), and editing it lets you pick again.

Appearances are saved with the part and go with it into assemblies (as colours: textures show in the part itself). Render looks from **Render** (below) still apply to faces without an appearance. The older per-face colours of earlier versions still show, under the appearances.

12.5 The Render tab

The **Render** tab has the rendering tools: **Render**, **Quick Render**, **Realistic View**, **Lighting**, **Add Appearance**, **Appearances** (the tab), **Edit Material**, **Decal**, **Decals** and **Export glTF** (for other viewers and renderers).

Realistic View shows the part in the view as it renders: every face physically based (metals reflect, glass is clear, textures and gratings as they are), lit by the render scene. Click it again to go back to the normal shaded view. **Lighting** sets the **scene** (below), the exposure, the floor's shadow and how far a photographed place is turned, for renders and the realistic view; it is kept with the part. **Quick Render** renders the view at its size in one pass.

Render makes a photorealistic picture of the view as it is now : choose the size (as the view, or up to 3840 × 2160), the **Quality** (Draft: one pass; Good: 16 passes, Best: 64, each with the view and the light moved a little and the passes averaged, for smooth edges and soft shadows), the **Scene**, a soft **shadow on the floor** and the exposure, and a **look** for each body: polished, brushed or cast metal, gloss paint, powder coat, glossy or matte plastic, rubber or glass, or **From the material** (aluminium and stainless brushed, steel cast, brass and copper polished, ABS glossy, PLA and nylon matte). Bodies keep their colours, and their looks are kept with them; faces and bodies with an appearance render with it (its texture, finish and holes). **Save PNG...** writes the picture. The settings are kept with the part.

The **scenes** light the part, show in its reflections and sit behind it:

Scene	What it looks like
Studio	A soft grey room with a large light overhead and panels at the sides, on light grey
Studio on white	The studio, on white (for catalogues and web pages)
Soft gradient (infinity cove)	A brighter, softer studio on a white-to-grey gradient
Dark studio	A black room with a light overhead and tall strip lights at the sides (rim highlights), on a dark gradient
Showroom	A warm room with rows of ceiling spotlights, on a warm cream gradient
Night (blue)	A dark blue room with a cool light on one side and a warm one on the other, on a dark blue gradient
Outdoors: field, clear sky	A mown field with trees under a clear sky and a low sun
Outdoors: garden	A formal garden of clipped hedges and flower beds under a cloudy blue sky
Forest (misty)	A pine forest in the mist: soft, even light
Overcast (hillside)	A rocky hillside above a valley under a grey sky: soft light, faint shadows
Sunset (waterfront)	A stone promenade by the water in Venice at sunset
Workshop (machine shop)	A machine shop: concrete floor, castings, machines and an overhead crane
Warehouse	An empty warehouse: concrete floor, metal-clad walls and strip lights
Office	A sunny office with desks, monitors and big windows
Store (home décor)	A home décor shop with chandeliers, sofas and sample racks

The last nine are **photographed places** : real 360° HDR photographs. The place lights the part (its sun, sky, windows and lamps as bright as they really are), shows in its reflections, and is the backdrop. The part **stands on the place's ground**: the photograph's ground carries on under the part, seen from as high as the view looks from, with the part's shadow and a soft contact shadow where it meets the floor. So the view's angle matters: looking down on the part shows mostly the ground (a concrete floor, grass, paving); a lower view shows more of the place behind it. A perspective view (View ▸ Perspective) suits them best.

The sun (or the brightest light) is put at the view's right, where the key light is, so the part's light, its shadow and the place agree. **Lighting ▸ Turn the place** (degrees) turns the place about the vertical to choose what is behind the part; the sun, the key light and the shadow turn with it. Where the photograph's ground is not flat (hedges, the water's edge), turn the place until the part stands on open ground. The photographs are from Poly Haven (CC0); `app/scenes/CREDITS.md` lists them.

12.5.1 Spotlights

Render ▸ Spotlights (or **Add Spotlight**, or **+ Spotlight** in the Appearances tab) places lights of your own around the part, as in a photo studio: a key light, a fill, a rim light, a coloured accent. They light **renders** and the **realistic view** only; the ordinary view is not changed. While the Spotlights panel is open the realistic view is on, and each light shines on the part as you change or drag it.

- **In the view:** each light shows as a yellow **lamp**, a ring where it is **aimed**, and its beam's cone. Drag a lamp to move the light across the screen; **Shift+drag** moves it nearer to or farther from what it points at, along its beam. Drag the ring onto the part to aim the light at that point of its surface (off the part, the ring moves across the screen).
- **In the panel:** the list of lights (click one to edit it; **+ Add Spotlight**, **Delete**), and for the light picked: its **Name**, **On**, **Colour** (with warm, neutral, daylight and coloured presets), **Brightness**, **Cone angle** (how

wide the beam is), **Soft edge** (a hard-edged or soft-edged pool of light), **Casts shadows**, and where it is (**Light at**) and where it points (**Aimed at**) as X, Y, Z. **Aim at the Part** points it at the part's middle; **Light from the View** puts it where the view looks from, pointing the same way.

- **Scene lighting:** the scene (Lighting…) still lights the part. Turn its **Scene light** down and its **Scene key light** off to light the part with the spotlights alone, as in a dark studio; the **Dark studio** scene suits this.

✓ keeps the lights (with the part; Undo takes them back), ✗ puts them back as they were. The Appearances tab lists them under the appearances: double-click one to edit, right-click to turn it off or delete it. In multi-pass renders each spotlight's shadows are softened, more for a softer edge.

Render ▸ Decal puts a picture (a logo, a label) on a flat face: open the picture, give its width and angle, then click the face where its middle goes. Decals show in the view and in renders, are saved with the part, and are not geometry. **Tools ▸ Decals…** lists them, to delete them.

12.6 Exploded views of a part

A part with two or more bodies can have exploded views: **Insert ▸ Exploded View…**, or right-click the Solid Bodies folder. Only the display moves; the model stays as it is. The steps are made as for an assembly (see Exploded views in the Assemblies chapter).

12.7 Combine and Subtract Bodies

Combine joins bodies with **Add**, **Subtract** or **Common**; the first body clicked is the main body. **Subtract Bodies** picks a cutting body and target body; **Consume cutting body** (on by default) removes the cutting body afterwards.

12.8 Split Body

Split Body separates solid or surface bodies into independent bodies. First select **Bodies to Split**, then activate **Select splitting tool** and choose a sketch, construction plane, model face or surface body. A surface body selected as a target stays a target; it is not automatically treated as the tool. The tool is kept.

- **All intersected bodies** is optional; otherwise only the selected bodies are divided.
- **Extend Splitting Tool** uses the underlying surface of a trimmed face and extends open sketch ends along their endpoint tangents. Construction planes already extend through the model. A bounded spline surface retains its natural parameter limits.
- With a planar tool, **Keep** retains all resulting bodies or only those on its positive or negative side.
- If the tool does not divide any target, the operation reports an error instead of creating an unchanged result.

Both the Features and Surfaces toolbars provide **Split Body** and **Split Face**. **Split Face** keeps the body count unchanged; use it when you only need an extra boundary on a face.

12.9 Move/Copy Bodies

Moves or copies solid and surface bodies.

- **By coordinates**: ΔX/ΔY/ΔZ plus a rotation about X/Y/Z or a reference axis, through the axis or the bodies' centre. Copies repeat the step.
- **With mates**: Coincident, Distance, Parallel, Perpendicular, Angle and Concentric between an entity of the moving bodies (face, edge, vertex) and fixed geometry (another body, a plane, axis or point). Two flat faces made Coincident or Distance start facing the way they sit (a pin's top face on a plate's top face

keeps the pin the right way up); **Flip alignment** turns a mate around. The mates follow what they are mated to: move the hole a pin is mated into and the pin moves with it (a mate on the hole's edge or on its wall).

- Right-click a body (or a face of it) ▸ **Move/Copy Body…** starts it with that body picked.

12.10 Delete Body

Removes a body from the model from that point in the history. Deleting the Delete Body feature brings the body back.

13 Surfaces

The **Surfaces** tab builds surface bodies (zero-thickness faces), listed under **Surface Bodies** in the tree. Cuts and Combine only act on solid bodies; Knit, Thicken or Convert to Solid turn surfaces into solids.

13.1 Making surfaces

Feature	What it does
Extruded Surface	Extrudes selected regions, open/closed contours or individual entities (the sketch just drawn for it, or selected, open curves and all, unless it has one closed region); one side, symmetric total depth or independent depths on both sides; reverse supported
Revolved Surface	Revolves selected regions or open/closed contours with separate axis selection, optional axis projection and the same Full/Angle, one-side/two-side/symmetric controls as solid Revolve
Planar Surface	Creates flat surfaces from selected sketch regions, closed entity contours or coplanar model boundary edges; sketch regions preserve their holes
Lofted Surface	A surface through two or more profiles (sketches or 3D sketches), open or closed, optionally ruled (straight between profiles). Open profiles drawn in opposite directions are lined up automatically. **Guide Curves** (rails, optional): sketches or curves the surface follows between the profiles, e.g. along its edges; each runs from the first profile to the last, through every profile. One guide pulls the surface along it; two or more shape it across, each where it touches the profiles.
Swept Surface	An open or closed sketch profile, or a circle, swept along a sketch, 3D sketch, helix or chain of model edges. **Orientation** (including Square to faces), **Twist angle**, **End scale** and an optional **Guide** work as for Sweep (see Sweep and Sweep Cut); with a guide the profile can be open or closed. Closed profiles give a surface without end caps.
Filled Surface	A smooth patch over a hole bounded by one closed loop of edges of a surface or a solid, touching (G0), tangent (G1) or matching curvature (G2) with the faces round it; the condition can be set for each edge. It can be knit into the surface and close a solid. A flat loop gives a flat patch.
Boundary Surface	A surface through curves in two directions. The first and last curve of each direction bound it and must meet at their ends; the curves between shape it. **Meeting other faces** makes it **tangent (G1)** or **curvature continuous (G2)** to faces you pick (**Select support faces**): each bounding curve that runs along an edge of one of them meets that face so; the others just touch. It is checked on the CAD faces as for Filled Surface (contact 0.0001 mm, 0.5°, 0.001 1/mm) and the note gives what was measured. The curves must agree with the faces where they meet (a side curve leaving a G1 edge must leave along the face; for G2 its first three control points in line with it).
Ruled Surface	Straight-sided surfaces standing on edges, either normal to the surface or along a direction tapered by an angle, on one or both sides (for example mould parting surfaces)
Offset Surface	Offsets faces of any body
Copy Surface	Copies faces of any body as surface bodies, leaving the body unchanged. **Join copies that touch into one surface body** is on by default.

13.1.1 Guide control for Swept Surface

Swept Surface uses the same separate canvas selectors and per-role Chain selection as solid Sweep. Choose one open or closed profile contour without holes; a region with inner loops must be reduced to the desired contour. Whole-sketch sources remain available explicitly for older workflows.

Without a guide, **Orientation** and **Twist angle** work as in solid Sweep: perpendicular sections with an optional signed total twist, or sections parallel to the initial profile. Open and closed profiles remain uncapped. The same open, tangent-continuous path restriction and numerical checks apply to nonzero twist.

Guide (optional) uses the same **Orientation** and **Section size** controls as the solid Sweep. Orientation keeps the profile dimensions; Section size uniformly scales the profile to follow the guide. Use one connected profile chain, open or closed, with an open path and guide. The guide must span the endpoint normal planes of the path; Section size also requires it to start on the profile curve and intersect every path normal plane once. Guide contact on the CAD surface is checked numerically to **0.0001 mm**, using the same bounded scale-law refinement as solid Sweep. These are sampled numerical checks, not a certified global bound.

Open profiles form a sheet. Closed profiles and the Circular option form an uncapped surface shell; no end caps or solid are added. Removing the guide restores the ordinary swept surface.

13.1.2 Loft only the portions created by Split Face

In **Lofted Surface**, choose **Profile Source** → **Model edges**. Select only the actual boundary edges of the split portion, then click **Add selected edges as profile**. Repeat for the second section. A profile can be one edge or a connected, unbranched chain; disconnected edges belong in separate profiles or separate lofts. Each selected edge is kept as its exact trimmed CAD curve, not replaced with its source sketch curve.

For a 3 mm strip, select the boundary segment between the original surface edge and the Split Face boundary. Do not select the whole sketch contour. When bridging the two sides of an open surface model, make a separate loft for each side. Existing sketch profiles can be removed with the × buttons before selecting the split model edges.

Model-edge profiles currently use **Free** start/end conditions, **Automatic** correspondence and no guides. Sketch profiles retain the guide and endpoint-condition tools below. The original bodies and the feature that made the selected edges are kept as dependencies, so changing the split rebuilds the loft.

13.1.3 Start and end conditions for Lofted Surface

Lofted Surface uses the same **Start Condition**, **End Condition**, **Influence** and **Reverse tangent direction** controls described under Loft. Adjacent support faces can belong to solids or surfaces. Open profiles form a sheet; closed profiles form an uncapped shell. Guide curves can be combined with endpoint conditions using **Adaptive approximation**, with the same endpoint-tangent compatibility and sampled CAD contact checks as solid Loft.

13.1.4 Guide construction for Lofted Surface

Select **Guide Curves** → **Guide construction** → **CAD constraints (two profiles)** to use the original guide edges as patch boundaries. This supports open profiles forming a sheet, or closed profiles forming an uncapped surface shell. Guides can lie in different planes, making a nonplanar surface. A closed profile does not add end caps or make a solid.

The kernel checks the patch contact error to **0.0001 mm**, validates the resulting CAD shape and checks points on the guide curves independently. **Across guide seams** offers **C0** (contact), **G1** (tangency) and **G2** (curvature continuity) along shared guide edges. G1 checks native CAD normals at nine interior positions, allowing at most **0.5°** mismatch. G2 constructs a C2 B-spline surface for internal guide networks, trims the resulting patches and checks contact, normals and all three components of the second fundamental form at 33 independent positions on each seam. Its curvature mismatch limit is **0.001 1/mm**. Measured angular and G2 curvature mismatches appear in the feature note. These are numerical tolerance checks, not a symbolic proof that every guide point is exact. The method requires two compatible profiles without holes

or Thin Feature. Original profile corners are preserved. A transverse guide through a sharp corner cannot provide G2 continuity: choose C0 or move the contact onto a smooth profile segment. Circular profile seams are positioned automatically; straight guides already on a cylinder or cone preserve its geometry. Incompatible constraints report an error without lowering continuity or switching methods. Choose **Adaptive approximation** explicitly for other supported loft configurations.

13.1.5 Surface Extrude and Revolve selection

These commands use the same canvas profile/contour selector as Planar Surface and solid features. Turn off **Chain selection** for individual sketch entities and confirm the contour with **Add selected contour**. Unlike a solid profile, an open chain is valid. Selecting a region uses its boundary curves, including holes, without automatically using unrelated curves from its sketch. Closed contours remain surface shells; the command does not turn them into solids. Earlier whole-sketch features still rebuild and can be edited through the explicit whole-source fallback.

Surface Revolve has separate profile and axis slots; the axis can come from another sketch. Surface Extrude's **Symmetric** depth is the total depth, while **Two sides** uses Depth and Second depth independently. Surface Revolve shares Full/Angle and one-side/two-side/symmetric behavior with solid Revolve, and goes **Up To Vertex** or **Up To Surface** too (a plane beside the axis trims it, as for solid Revolve). These surface commands create new surface bodies.

Model edges and faces : besides sketch contours, both commands take edges of the model. **Model Edges → Select model edges**, then click edges on solids or surfaces (click again to remove one; **Chain selection** follows a surface body's open boundary). The edges join end to end into chains, open or closed; edges that branch are refused. Clicking a flat face of the model while choosing contours uses that face's outer and inner edges. Edges in one plane extrude square to it, out of the face they border when they lie on one; edges that do not lie in one plane, or a single straight edge, need **Along** (below). Surface Revolve revolves the chains about its axis, which must then be a reference axis, a model edge or face, or a line of a sketch.

Surface Extrude start, end, direction and draft : **Start from** is the contour plane, an offset from it, or a plane / planar face, as for solid Extrude. **Direction** adds **Up To Vertex**, **Up To Surface** and **Offset From Surface** to One side, Two sides and Symmetric; an angled start or end surface trims the extrusion along it. **Along** chooses an axis to extrude along instead of square to the contours (None); Depth is then measured along it, and an axis that lies in the contours' plane is refused. **Draft angle** tilts the walls of closed contours about where the extrusion starts, each contour narrowing away from it (or widening with **Draft outward**), as for solid Extrude; an open contour of sketch lines and arcs leans to its left as drawn (**Draft outward**: to its right), each line and arc offset in the contour's plane as the extrusion rises, so its faces stay flat and conical.

13.1.6 Planar Surface selection

Use **Sketch Profiles → Select profiles in canvas** to click inside a closed region or select its contour. Each selection adds only that region, including its inner holes, instead of every profile in its sketch. Click a selected profile again to remove it. **Chain selection** follows connected sketch entities; turn it off to pick individual curves, then use **Add selected contour** once they close. Selecting just an outer contour fills that outline; click the actual region to include its holes.

Use **Model Boundary Edges → Select model edges** for existing solid or surface edges. Choose one connected, closed, coplanar loop, with optional chain selection. The kernel rejects open, duplicate, missing or nonplanar boundary edges. Each selected sketch region creates a planar face; multiple independent regions are supported. Profiles remain linked to their source entities when dimensions change, and deleted selected entities require reselection.

Existing whole-sketch features remain compatible. **Other sources (whole sketch / curve)** keeps whole-sketch selection available explicitly. A sketch with multiple regions no longer gets selected in full automatically when starting Planar Surface: choose the desired profiles in the canvas. Planar Surface creates flat sheets; use Filled Surface for a nonplanar patch or continuity to adjacent faces.

13.1.7 Filled Surface / Patch continuity

Select one connected, closed boundary loop. **Chain selection** is enabled for new patches: click an edge to add its connected boundary chain on that body; click a fully selected chain again to remove it. Turn the checkbox off to add or remove individual edges. Partial chains are completed without duplicate edges or changing existing per-edge continuity. Selection stops at branches; for solid edges, an explicitly selected support face can identify its boundary loop. The checkbox is saved with the feature; older files keep individual selection until enabled.

Boundary continuity sets **Contact (G0)**, **Tangent (G1)** or **Curvature (G2)** for the loop. Each edge can override that setting, so a patch can combine G0, G1 and G2 boundaries. Open, disconnected, duplicate or multiple loops report an error.

Tangent reach : a tangent (G1) edge's row has a reach in mm (0 by default). With a reach, the patch also passes through points that far in from the edge along the support's tangent plane, so it keeps to that plane further before it turns towards the middle: a fuller, flatter rim. It applies to G1 edges only.

Pass Through : **Add a curve it passes through** adds sketches, 3D sketches and curves lying inside the opening; **Add points it passes through** adds reference points, or a sketch's own points (those not on its curves). The patch passes through all of them as well as meeting its boundary; each is checked on the result to **0.0001 mm** (the constraints are sampled more finely until it passes, or the patch reports how far off it is).

G1/G2 edges require an adjacent support face. An open surface edge infers its unique adjacent face. Use **Select support faces** when a solid edge has two adjacent faces; exactly one selected adjacent face must cover each constrained edge. **Select boundary edges** returns to picking edges. G0 does not require support faces. Older files with **Tangent** enabled keep G1 behavior, with strict verification instead of a contact-only fallback.

The resulting CAD face is checked for validity and sampled at 33 positions on each boundary: contact within **0.0001 mm**, G1/G2 normal mismatch within **0.5°**, and G2 mismatch of all three curvature components within **0.001 1/mm**. Boundary trimming curves are checked again after knitting. Incompatible supports or failed constraints remain errors; the requested continuity is not downgraded. These checks are sampled numerical bounds, not a global mathematical proof. A latitude opening in a spherical face preserves its exact spherical continuation; other constrained patches use native plate construction with bounded refinement and final CAD checks.

Knit into the surface joins the patch when all selected edges belong to one surface body. **Make a solid when that closes it** closes a watertight result; if other openings remain, the result stays a surface and its note says so. Example: `tests/data/surface-patch-g2.rcad`.

13.2 Changing surfaces

Feature	What it does
Trim Surface	Cuts surfaces with a plane, a surface body or a sketch's curves (square to the sketch). Click the pieces to keep, or to remove. **Mutual** trims the tool surface too.
Extend Surface	Carries open edges straight on by a distance, following the surface's slope at the edge, and knits the extension on

Feature	What it does
Untrim Surface	Fills a surface's holes, or gives it back its natural extent (holes and outline). Pick faces to untrim only those.
Knit Surface	Sews surfaces together within a gap tolerance. When they enclose a volume, **Try to form a solid** turns them into a solid.
Thicken	Gives a surface a wall thickness (one side or both) and makes it a solid
Delete Face	Takes faces out of a solid or surface. Either leave the hole (the body becomes a surface body), patch each hole smoothly, optionally tangent, so the solid stays solid, or **Heal**: the faces round the hole grow or shrink until they meet and close it exactly, with no patch (a boss or a rib deleted as if it was never there). Good for removing fillets and bad faces from imported parts.
Replace Face	Moves flat faces of a solid onto a surface body, so the body grows or shrinks to meet it (for example a curved top). The surface must reach past the faces all round.
Freeform	Reshapes one face of a solid or surface by handles you push in or out (see below)
Flatten Surface	Unrolls faces of a surface or solid flat beside the model as a surface body (see below)
Split Body	Splits solid or surface bodies into independent bodies
Move/Copy Surfaces	As Move/Copy Bodies, for surfaces
Delete Surface	Removes a surface body

13.2.1 Freeform

Surfaces ▸ Freeform reshapes one face by a grid of handles. Click the face: a flat face of any outline, holes and all (the top of a block or a disc, a plate with holes), or a freeform face (a lofted or fitted face, or one Freeform has already reshaped). On a flat face with four straight edges or a freeform face that covers its whole surface, **Keep at the edges** sets what is held there. On any other face (another outline, holes, a trimmed freeform face) only the inside moves: the face stays as it was in a band along every edge and hole, so it always meets them in tangent plane and curvature (G2), and handles that fall outside its outline are left out (the note says how many are on the face). Rounds and other curved faces can't be reshaped yet. Set how many handles there are **Across** and **Along** the face (1 to 6 each): they sit evenly across it, and each **Handle** is how far the face is pushed out there (below 0, in), along its normal. The face passes through every handle exactly. **Drag a handle** on the model, in or out of the face, to move it; the face rebuilds when you let go.

Keep at the edges says what the face keeps where it meets the faces round it: **Contact (G0)** (only the edges stay where they are), **Tangent (G1)** (the tangent plane along them too, so a face that ran smoothly into its neighbours still does; the default) or **Curvature (G2)** (the curvature as well). The more is kept, the closer to the middle the shape changes and the less it bulges. The face is made an exact B-spline surface and its control points are moved (the outer one, two or three rows of them held); the note gives how closely the handles are met and, measured on the CAD faces, how closely the tangent plane and curvature along the edges are kept. A face that would run through another face cannot be closed into the solid: use smaller heights.

13.3 Curves and Split Face

Feature	What it does
Split Face	Select the faces first, then a splitting tool. Projection uses a sketch, along its normal or an explicit vector; Intersection uses a plane, model face or surface body. Automatic chooses from the tool. Extend Splitting Tool controls the tool extent. The body count stays unchanged; only selected faces gain boundaries.
Projected Curve	Projects a sketch along its normal onto faces, as a 3D curve
Composite Curve	Joins model edges end to end into one curve

Projected and composite curves can be used as sweep paths, lofted surface profiles, curve-driven pattern paths and boundary surface curves.

13.4 Flatten Surface

Click the faces to unroll. Faces that unroll (cylinders, cones, bent sheets) come out exact. A doubly curved face (for example a dome) can't lie flat without stretching: it is spread out keeping its edge lengths as well as possible, and the note gives the size, the area change and how much the edge lengths changed. **Relaxing passes** sets how hard it tries. A closed tube needs a seam (Split Face) to open out.

13.5 Meshes and closing surfaces

- **Convert to Solid** (also right-click a surface body): surface bodies that together enclose a volume, or a mesh, are joined (edges closer than the gap tolerance meet) and made into one solid. When they leave gaps it says how many open edges and holes there are.
- **Repair Mesh**: closes the holes of a surface body or a mesh with missing pieces (flat holes with a flat face, curved holes with a filling surface, holes in a mesh with triangles), turns triangles that face the wrong way round, and makes a solid. Option: the largest hole to close.

Surface quality can be checked with Zebra Stripes, Curvature and Draft Analysis (Evaluate and Tools chapter).

14 Reference Geometry

14.1 Reference Plane

Reference Plane, Reference Axis and Reference Point share a **References and Relations** panel with three numbered slots. Pick sketch geometry, model geometry or an earlier reference feature in the canvas or tree. The **Choose reference geometry** list also offers individual sketch lines, arcs, circles, supported splines and points (including the sketch origin), grouped by source sketch. Entries show the entity name and stable ID; suppressed, failed and later sketches are omitted. Hidden source sketches are temporarily displayed without changing their saved visibility. Click a slot to replace its reference; a fourth pick never discards an existing reference. Click a selected reference again, or its remove button, to remove it. References remain linked to their sources.

Each slot offers compatible **Relation** choices. **Automatic** chooses the usual relation for the selected combination; selecting an explicit relation keeps that choice when other references change. Distances and angles accept expressions. References must precede the feature in the tree.

The displayed rectangle of a reference plane is centered near its defining line, point or path location, projected onto the plane. This display position does not change the plane's sketch coordinate system. For rotation about a line, use **Coincident / through** on the line and **At angle** on a reference plane.

Fully Defined means the supported constraints have a finite solution with no remaining continuous freedom. **Under Defined** identifies missing constraints; **Conflicting** identifies incompatible geometry, relations or values. An axis direction alone does not fix its position, and two points alone do not fix a plane. Incomplete and conflicting definitions cannot be accepted. When two distinct geometric solutions exist, **Solution 1 / Solution 2** selects the branch; opposite normals of the same plane are handled by **Reverse its normal** instead.

Reference geometry	Available plane relations and constructions
Planar face or plane	Coincident, Offset, Parallel, Perpendicular, At angle
Point or vertex	Coincident / through
Straight edge, sketch line or reference axis	Coincident / through, Parallel, Perpendicular, At angle; Normal at position
Sketch circle/arc or circular model edge	Its containing plane supplies Coincident, Parallel, Perpendicular and At angle relations
Two planes/faces with Midplane relations	Midplane of parallel planes, or either angular bisector
Cylindrical face	Tangent, with an angle round the cylinder; optionally parallel to one selected plane, with either side available

Examples: three non-collinear points define a plane; a line with **Coincident** plus an external point defines the plane through both; a line with **Perpendicular** plus a point defines a normal plane through that point. A line with **Coincident** and a plane with **At angle** can define two angular solutions. Skew lines, inconsistent angles and invalid tangent combinations report conflicts. **Normal at position** uses the selected segment's start, midpoint, end or a distance within a straight segment. A reference axis supplies a direction and its stored anchor.

The solver uses affine equations and the unit-normal constraint, rather than a tessellated approximation. Numerical rank and consistency checks distinguish free, unique and incompatible definitions. Tangency currently supports a cylinder alone or a cylinder with one parallel plane; midplane supports two plane references. Arbitrary curved-surface relations and additional constraints on these special constructions are

not supported yet.

Placement transforms the constructed plane after its reference relations have been solved: move along its normal, turn its sketch axes in the plane, or reverse its normal. An Offset relation already supplies its signed distance and has **Flip offset direction**.

Existing plane and axis constructions keep their original geometry and anchor when opened for editing. The panel identifies the preserved construction; changing a reference or relation upgrades it to the constraint definition. Cancel restores the original saved definition.

14.1.1 Plane Along Path and point along a curve

In Reference Plane or Reference Point, select a sketch line, arc, circle or spline, a model edge, or an earlier helix/3D/composite/projected curve feature. Choose the reference relation **Along path (arc length)**. The source picker and tree also offer curve features; a feature containing several disconnected routes must be reduced to one connected path first.

Position offers Distance along curve, Fraction (0–1), Start, Half arc length and End. Distances and fractions accept expressions. **Reverse path direction** measures from the opposite end and reverses the tangent used by the plane. A fraction follows changes in the total curve length; a distance retains its physical length when the source changes. Closed paths retain the curve's existing seam; distances outside the path are rejected rather than wrapped.

The plane passes through the computed position, has its origin there and is normal to the local tangent. The point uses that same position. A point can lie at a corner or cusp; a plane cannot be made there without a unique tangent. The preview identifies a conflict at detected corners, singularities or nonsmooth closed seams instead of choosing one tangent arbitrarily.

Locations use adaptive integration of curve speed and bounded inversion of the resulting arc length, independently of display tessellation. Sketch splines retain their original Bézier pieces, degree and rational weights. Model edges and curve features use derivatives of the actual CAD curve and an independent CAD length check. Position inversion targets **0.0000001 mm** or **1e-10 of the path length**, whichever is larger; refinement exhaustion reports an error. These are numerical convergence controls, not a certified global error bound. Existing Midpoint and straight-segment Distance relations keep their previous meanings; choose Along path explicitly for arc-length placement.

14.2 Reference Axis

Uses the same three slots, per-reference relations and definition states. Two points define a through-axis; two intersecting planes define an axis in both. A straight edge or sketch line supplies its line, a circular edge supplies its centre axis, and a cylindrical or conical face supplies its centre axis. Plane **Perpendicular** plus a point defines a normal axis through that point. Parallel and perpendicular relations can constrain the direction while other references constrain the position.

Axes remain linked, are drawn dashed and can drive a revolve or a pattern. Right-click an edge and choose **Reference Axis** to preselect it. Parallel distinct planes, coincident points and zero-length directions cannot define the requested axis.

14.3 Reference Point

Choose **Coordinates** (expressions allowed) or **Reference geometry**. The latter uses the shared slots and states. A point/vertex can be copied; a selected sketch line or model edge can supply its start, end or midpoint, and a circular curve can supply its centre. **Distance from start** currently supports straight segments and rejects distances outside the segment.

Coincident relations intersect straight lines and planes: two crossing lines, a line crossing a plane, or three independent planes can define one point. A point on one plane retains two degrees of freedom; one on a line retains one. Skew lines, incompatible planes and missing references report conflicts. Circular and other curved-edge midpoint positions use the kernel's normalized curve parameter, preserving the existing Edge midpoint behavior; they are not a general arc-length locator. Use **Along path (arc length)** above to locate a point by physical curve distance or fraction. Two references where one is curved also give a point where they meet: a curved model edge (circle, arc, spline) and a plane, a flat face, a curved face or another edge, or a curved face and an axis. When they meet more than once, the panel says how many crossings there are and **Next solution** steps through them, in a fixed order. A curved face and a plane meet along a curve, so that pair is refused: pick an edge of the face instead.

Right-click a vertex and choose **Reference Point** to preselect it. Coordinate mode retains its stored reference choices but does not depend on or evaluate them.

15 The Feature Tree and History

15.1 Editing features

- Double-click a feature, or right-click ▸ **Edit Feature**, to open its Property panel.
- **Double-click a face of the model** to change sizes there and then: a small box by the cursor lists the sizes of the feature that made the face (its depth, radius and so on, and the dimensions of the sketch it was made from), each with its ID. Type a new size and press **`Enter`**; the model rebuilds and the box stays for more. **`Ctrl+Z`** in the box undoes the change; **Edit Feature** opens the feature's panel; **`Esc`** closes the box.
- Select a feature and press **`Del`** (or **Edit ▸ Delete**) to delete it. Only that feature goes: the sketches, 3D sketches and features it was made from stay, shown on their own in the tree again.
- **Edit ▸ Undo / Redo** (**`Ctrl+Z`** / **`Ctrl+Y`**) works on the whole part history.
- **Undo history**: right-click the **Undo** button above the view (or click its small arrow, or **Edit ▸ Undo History...**) for the part's last steps, newest at the top, each said in words ("Added Extrude2", "Edited Hole1", "Renamed Sketch3 to Base"). Click one to go straight back to just after it; steps undone are shown greyed and can be clicked to redo them, until you do something new. In a sketch it lists the sketch's own steps since you opened it ("Added 4 Lines with 2 dimensions", "Dimension 40 → 50", "Deleted Arc", "Moved").
- **Rebuild** (**`Ctrl+B`**, or the quick button) rebuilds the part.
- A feature that fails shows an error in the tree so you can fix or re-pick its references (see Fix References).

15.2 References that follow edits

Features that use picked edges and faces (fillets, chamfers, shells, drafts, axes...) find them again after the model changes. An edge made by an extrusion (a boss, cut or pocket from a sketch, without draft or thin walls) is also named by what made it: the sketch curve it runs along and which end of the extrusion it is at (its top or bottom rim, a pocket's floor), or the sketch point it runs from. After a change it is looked for by that name first, so when you move a circle in a sketch of several, a fillet on its boss goes with it, even if another boss now stands where it was. Other edges, and named ones the name does not settle (an edge split in two by a later cut, say), are found by where they were. An edge remembers the faces on its two sides and whether it is an outside or inside corner. When a dimension moves it so far that other edges now lie nearer its old place, the right edge is still found, and an edge of another kind is not taken for it. A face that moved is found as the face facing the same way that is clearly nearest. A curved face (a hole's wall, a countersink's cone, a fillet's round, a sphere or torus, a freeform face) remembers what it is: its kind, radii and axis direction (a freeform face its area). When it moves (a hole moved in its sketch), it is found again as the face of the same kind and size whose middle is clearly nearest where it was. A reference found again this way is updated, so it keeps following the model.

Found references to check. When a reference was found again as the nearest of faces or edges that look the same (two holes of the same size, say), it may be the wrong one. The feature then shows an orange **?** in the tree; hover it for why. Check the feature, then right-click it ▸ **Accept Found References**, or edit it (its panel says so too) and click ✓; pick again if it took the wrong one. When the look-alikes are too close to call, the reference is not taken at all: the feature fails, and Fix References lists it.

15.3 Fix References

Tools ▸ Fix References... lists every feature that failed to rebuild, what went wrong and which of its edges or faces are no longer in the model. Open it also by clicking the red error count in the status bar, or by right-clicking a failed feature. For each feature:

Button	What it does
Edit	Opens the feature (or sketch) to pick its references again
Remove missing	Keeps the references that are still found and drops the others
Suppress	Leaves the feature out for now

Fix them from the top: a feature can fail only because one above it did. The list follows every rebuild while it is open.

15.4 Rollback bar

The blue line at the bottom of the tree. Drag it up, or right-click a feature ▸ **Roll Back** / **Roll to End**. Features below the bar are suppressed, and new features are inserted above it.

15.5 Reordering features

Drag a feature up or down the tree. Its absorbed sketch moves with it. A move that would break a reference is refused with the reason.

15.6 Reference arrows

Select a feature in the tree. Blue arrows come from what it uses; purple arrows go to what uses it.

15.7 Tree symbols

Sheet metal features and bodies have their own icons. Under-defined sketches show **(-)**. Sketches absorbed by features are hidden by default.

16 Sheet Metal

The **Sheet Metal** tab holds the sheet metal features. Sheet metal bodies are listed in the **Sheet Metal Bodies** folder below Solid Bodies, with sheet metal icons.

Group	Features
Making a sheet	Base Flange, Sheet Metal Extrude, Lofted Bend, Transition / Reducer, Convert to Sheet Metal
Flanges and bends	Edge Flange, Miter Flange, Hem, Sketched Bend, Jog
Forms and details	Cross Break, Tab and Slot, Forming Tools, Gusset
Corners	Corner Relief, Closed Corner
Flat	Unbend, Bend, Flatten
Checking and output	Sheet Metal Check, Export DXF, Export All DXF, Nest Flat Patterns, DXF Settings

16.1 Base Flange

A closed sketch (holes allowed) becomes a sheet of the given **thickness**. The **bend radius** and **K-factor** set here become the part's defaults.

Bend calculation (Base Flange, Sheet Metal Extrude, Lofted Bend, Transition): **K-factor**, or, from your bend table for this thickness and radius, the **Bend allowance** (the length a 90° bend takes flat) or the **Bend deduction** (how much shorter the flat is than the two outside lengths to the sharp, for a 90° bend). RazorCAD works out the K-factor it gives (refused when outside 0 to 1), so every flat length agrees with the table at 90° and other angles follow in proportion. **Bend table (by angle)** takes the allowance for several angles ("30: 1.5; 45: 2.0; 90: 4.2", for this thickness and the default radius): each bend's allowance is read from it, straight between its rows and in proportion past its ends; bends of another radius use the K-factor its 90° allowance gives.

16.2 Gauge tables

Base Flange, Sheet Metal Extrude and Transition / Reducer have a **Gauge** list: mild steel, steel gauges, stainless 304, and aluminium 5052 and 6061-T6. Picking a gauge sets the thickness, inside bend radius and K-factor. **Save...** keeps your own values as a gauge.

16.3 Sheet Metal Extrude

Extrudes an open sketch of lines and arcs into a sheet metal part. Arcs in the sketch are bends of their own radius; sharp corners get a bend of the inside radius set in the panel. Material goes on one side, the other side or both sides of the sketch; depth one way, reversed or mid-plane. A circle on its own is rolled into a cylinder, leaving a seam gap. Flanges, hems, Flatten and DXF export work on it.

16.4 Lofted Bend

A sheet metal part between two open profile sketches (each one chain of lines and arcs), for example a round-to-square transition, hopper or duct piece. Points are matched along both profiles by length, keeping every corner of either profile.

- **Method**: **Bent** makes flat facets folded along straight lines, each fold a bend line in the flat pattern and DXF; set the number of divisions. **Formed** makes a smooth sheet (rolled or pressed), with the set

number of forming lines along the profiles.

- The flat pattern is developed at the neutral layer (K-factor) and works with Flatten, DXF export and flat pattern drawings. **Material on the other side** flips the thickness.
- A cut through a lofted bend is carried into its flat pattern, and the cut edges are square to the sheet, as a laser-cut blank formed to shape would be. This makes trimmed conical parts possible: loft the cone between two arcs, then cut it to the outline with a revolved cut.
- Edge flanges, corner reliefs and closed corners can't be added to a lofted bend; its shape comes from the profiles.
- **Flanges** : tick **Flange at the first profile** and/or **at the second profile** for a flat bolting ring there, as on a Transition: a flat sheet metal plate of its own (with its flat pattern, holes and all), out past the profile's plane, away from the other profile, by its **Thickness**, and **Width** wide all round, sharp at the corners. It goes round the profile with its gap closed straight across, so it suits a part made in one piece, and the profile must lie flat (a sketch, or a 3D sketch that lies in a plane). Its opening is the inside of the part there: the profile when the material is outside it, the profile less the thickness when inside. Bolt holes (**Bolt hole diameter**, 0: none) go at each corner of a line round the middle of the flange and evenly between, at most the set distance apart; round a smooth profile, evenly round it.

16.5 Transition / Reducer

Cones, concentric and eccentric reducers, square-, rectangle-, triangle-, pentagon- and hexagon-to-round transitions (either way up) and polygon frustums.

- Sizes, offset and rotation are set in the panel; sizes can be inside, outside or mean. A line from bottom centre to top centre, in a sketch or a 3D sketch, places the part and sets its height; change the line and the part follows. In a sketch, X lies in the sketch plane, square to the line; from a 3D sketch (which has no plane), X is level, square to the line (the model's X axis when the line is upright), and Y points toward the top. Started with no sketch, it asks for a plane and opens the Line tool.
- **Axis line** : the line can be one of several in a sketch or 3D sketch (a duct route, a layout): while the panel is open, earlier sketches and 3D sketches show, and a click on a line picks it (drawn orange); or choose it under **Axis line**. Starting Transition from inside a sketch or 3D sketch with one line selected takes that line. With several lines and none picked, the panel says so.
- The part is made from a number of equal pieces, each a sheet metal body with its own flat pattern: a cone from one piece or more, a flat-sided transition from two or more, with seams at the middle of the flat sides or at the corners.
- **Formed** (rolled, with a set number of forming lines per corner or round the whole part) or **bent** (bump formed with a set number of bends).
- **Spigots** : a **Straight spigot** at the bottom and/or top end is a straight length of that end's shape, from the flange's inner face if there is a flange, or from the end of the line. Spigots are sheet metal parts of their own, each with its flat pattern: a round end's spigot is rolled true round, a flat-sided one bent at its corners to the **Spigot bend radius** (empty: the sheet thickness). **Bottom spigot parts** / **Top spigot parts** set how many parts each is made from (empty: 1 for a round end, 2 for a flat-sided one, which must divide its sides); the first seam is where the taper's is, and a one-part spigot has a small gap at its seam. The taper is what is left: spigots and flanges are inside the axis line's length, so the whole part is never longer than the line (if they would be, it says so).
- **Flanges** : tick **Flange at the bottom end** and/or **at the top end** for a flat bolting ring there. Each is a flat sheet metal plate of its own (with its flat pattern, holes and all, for cutting from plate), with its outer face at the end of the axis line: its **Thickness** is inside the line's length, so the part with its flanges is as long as the line. **Mounted**: **On the end (butt)**, the flange's opening is the inside of the part at that end (inside sizes as given; outside or mean sizes less the thickness, across the wall's slope where the taper

meets it); **Slipped over the end**, its opening is the outside of the part there and a **Clearance** all round (1 mm unless set), and the part's straight end goes through it to sit flush with its outer face, or **in from the flange's face** by the offset set. Either way it is **Width** wide all round. **Bolt hole diameter** (0: no holes): a round flange has **Bolt holes** evenly on the **Bolt circle diameter** (0: the middle of the flange), straddling X; a flat-sided flange has a hole at each corner of a line round the middle of the flange and more along each side, evenly, at most the set distance apart (a 400 square end with a 40 wide flange and 100 apart: 5 a side, 20 in all). The note under the feature counts the flanges and holes. Holes as wide as the flange, or a bolt circle that puts them off it, are refused.

16.6 End Flange

Sheet Metal ▸ End Flange puts a flat bolting ring on the end of a part, as the Transition's flanges are: a duct, a pipe, a sheet metal part, round or flat-sided, made by any feature. The flange is a flat sheet metal plate of its own, with its flat pattern (holes and all) for cutting.

- **End**: click the flat end face of the part (the end of its wall). For a part made in pieces (a transition's parts, a spigot in two), click each piece's end face. Gaps between the pieces (a rolled part's seam gap, say) are closed up to **Gaps between the pieces closed up to** (10 mm unless set). An end's faces lie in one plane.
- **Several ends at once**: click more ends, and each gets a flange of its own, all with the same settings: both ends of a duct, or the ends of other parts. The ends must all be the same size outside (one flange size); one of another size is refused, so add another End Flange for it. Each flange is a body of its own.
- **Mounted**: **On the end (butt)**: the flange sits on the end, out past it by its thickness, its opening the inside of the end. **Slipped over the end**: its opening is the outside of the end and a **Clearance** all round (1 mm unless set), and the end goes through it to sit flush with the flange's outer face, or **in from the flange's face** by the offset set (less than the flange's thickness).
- Small rounded corners (bends tighter than the flange is wide) are squared off, so a bent square duct gets a square flange.
- **Width** (out from the opening), **Thickness**, **Bolt hole diameter** (0: no holes). A round end's flange has its **Bolt holes** evenly on the **Bolt circle** (0: the middle of the flange); a flat-sided one has a hole at each corner of a line round the middle of the flange and more along each side, evenly, at most the **pitch** apart.

16.7 Edge Flange

Click straight edges on the top or bottom face. Where the top and bottom edges of a thin sheet lie almost on top of each other on the screen, a click takes the one nearer you. With several edges, every flange folds the same way as the first edge's (**Every flange the same way as the first edge's**), even when an edge was picked on the other face of the thin sheet; untick it for each to fold away from its own face.

Setting	Options
Length	Measured to the outer virtual sharp, inner virtual sharp or tangent
Flange width	**Along the edge**: the whole edge, **Offsets from its ends** (from the start, from the end; **Swap the ends**), or **A width in its middle**; the rest of the edge stays as it is, with reliefs where the bend ends
Angle	Flange angle
Position	Material inside, material outside, bend outside, plus an offset; flip
Bend radius	Its own radius, or the default
Auto relief	Rectangular, obround or tear relief where a bend ends next to material

Setting	Options
Corners	Flanges picked together that meet at a corner: **Corner to corner** (the default: each runs on up to the other's inside face and ends square, their inside edges the gap apart; the outside corner is left open and the bends end in a V cutout), **Mitred** (they meet on the diagonal), **Overlap** (one runs on to cover the other's end) or **Open**. Closed corners keep a gap (default 0.5 mm). For Overlap, **Swap** reverses which covers, and the Bend region is a V cutout or Closed.

16.8 Miter Flange

A flange with a profile of up to four segments along a chain of edges. Each segment has a length (to the outside of the bends) and an angle; a negative angle folds back the other way, so 90 / 90 makes a C and 90 / −90 a Z. The corners are mitred (closed, with a gap) at every level. **First bend** sets where the material sits on the edge, as for an edge flange.

16.9 Hem

Folds a straight edge back on itself. Hems meeting at a corner are mitred.

Setting	Options
Type	**Closed** (pressed flat), **Open** (with a gap), **Teardrop** (its own angle, default 225°) or **Rolled** (a curl)
Length	From the outside of the fold
Gap, inside radius, angle	As the type needs
Position	**Inside** (the fold at the edge) or **Outside** (added beyond the edge); flip direction

16.10 Sketched Bend

Draw lines on the top or bottom face of a flat part of the sheet; the sheet is bent along each one. Set the angle, the radius (or the part's), whether the bend is centred on the line or starts or ends at it, the fold direction and which side stays put. The flat pattern stays the plate it was.

16.11 Jog

One sketch line on a face makes two opposite bends that step the sheet over by a height. The height is measured face to face, outside to outside or inside to inside; the jog says when the height is too small for the bends.

16.12 Cross Break

Click flat faces to mark an X of shallow stiffening folds. It shows in Flatten, in the DXF (on the Forming marks layer, with a note) and in flat pattern drawings.

16.13 Tab and Slot

Click the edge of one sheet metal part where it sits on another, then the face of the other part. Tabs are added along the edge and matching slots, with clearance all round, are cut through the other part, so the parts locate each other for welding.

- Settings: number of tabs, tab length, tab height (empty: through the other part), margin at each end (empty: evenly spread) and clearance.
- Both flat patterns and DXFs get the tabs and slots.

16.14 Forming Tools

Forming Tools presses a form into the sheet. Place forms at clicked points, or exactly at the points and circle centres of a sketch on the face (for a louver, its lines). They are pressed toward the face they are placed on, or the other way.

Form	Settings
Dimple	Diameter, depth
Extruded (flanged) hole	Hole diameter inside the collar, collar height, bend radius
Louver	Width back from the slit, opening height, hood side; drawn as a line on the face (the slit)
Bead	A round-topped stiffening rib along each sketch line, its width and height (at most half its width), its ends rounded
Lance	A bridge: slits along the sketch line and its width from it, the strip between pressed out by the height and ramped back at 45° at its ends; both slits are cut lines in the flat pattern
Emboss	Each closed outline of the sketch raised (or pressed, with Flip) by the height, with straight walls the sheet's thickness; its outline is marked in the flat pattern

The flat pattern gets a centre mark and a note for each form, an extruded hole's pre-hole (the collar laid flat) and a louver's slit as a cut. Forms stay with the part through later sheet metal features and are left off, rather than breaking the part, if one can't be made.

16.15 Gusset

Click a bend (its round face) to press a stiffening dart across it at that point. Set the size of each leg from the inside corner and the width across the bend; tick **Pressed down** to press it the other way. The flat pattern marks it. (For gusset plates between structural members, see the Fast Structure chapter.)

16.16 Corner Relief

Adds a rectangular, circular or tear relief at every corner where two bends meet.

16.17 Closed Corner

Closes the corners between 90° flanges: one flange covers the other's end, with a gap.

- **Corner type**: Overlap, Underlap, Butt (square ends: each flange runs on up to the other's inside face, the gap between their inside edges, the bends in a V cutout) or Mitre (the gap across the mitre, bends included).
- **Which corners**: click a face of a flange next to a corner and the matching flange is found automatically; the clicked flange is the one that overlaps or tucks under. With no corners picked, every corner of the body is closed.
- **Swap** reverses which flange covers.

- **Bend region**: **V cutout** (default) runs each bend into the corner as it would form from the flat, leaving a small teardrop-shaped opening with a relief at the corner point. **Closed** keeps the bends meeting on the diagonal.
- The flat pattern matches the folded part at closed corners, including each bend's run-on, the relief and the corner round.

16.18 Unbend and Bend

Unbend keeps the picked face in place and flattens the rest. Features after it (such as cuts across bends) are made in the flat. **Bend** folds it back to where it was. Both appear in the tree.

16.19 Cuts and other features on sheet metal

Cuts on sheet metal (extruded cut and so on) become holes in the flat pattern, so the flat pattern always matches. Holes (the Hole feature) work on sheet metal too. Other features that change a sheet metal body are allowed, but the flat pattern can't follow them; this is shown in Body Properties and the DXF preview.

16.20 Flatten

Click **Flatten** on the tab, or right-click the body ▸ **View Flat Pattern**. The flat pattern is shown on its own; **Esc** or Flatten again restores the view as it was. While it is shown, **Export DXF…** on the flat pattern banner (or right-click the view) exports it straight away.

16.21 Convert to Sheet Metal

Turns an ordinary body into sheet metal. Pick a flat face of a thin-walled body (for example a STEP export of a sheet metal part, or a shelled box), or of a knitted surface of flat faces (enter the thickness).

- RazorCAD follows that side of the sheet across every smooth edge, through flat faces and cylindrical bend faces, so the whole body is found. Each bend keeps its own radius and angle.
- Bends that run straight into each other with no flat between (rolled shapes such as a fan scroll) are joined automatically. Cut-outs in bends are kept and holes stay true circles in the DXF.
- Sharp-cornered bodies (shelled boxes) get bends of the chosen radius at every edge between two flat faces.
- A flange whose ends run right up against the material beside it, with no relief (a flange off a cross bar between the sides of a window, or off the edges of a slot), is ripped there: the flat pattern (and its DXF) has a cut line along each flange end, across the bend and on to the flange's edge, on the cut-outs layer, as the part is cut and bent.
- The panel shows what was found in the body, for example "thickness 3 mm · bend radii R300, R227, R13 (inside, 3 bends)" or "4 sharp edges".
- **Use the body's thickness** and **Use the body's bend radii** are on by default. Untick either to set your own value (pre-filled with the detected one).
- A body curved other than in plain bends (a square-to-round transition or a reducer from another program, a cone) is **unrolled**: laid out flat at the neutral layer, triangle by triangle, so its flat pattern keeps every length; a ring is cut open at a seam. For a cone, click its curved face. A closed tube (bends all the way round) is unrolled the same way, cut open at a seam. A face curved both ways (a dished end, a spherical cap) can't be laid flat exactly: it is spread out as nearly as it can be (from its middle out along its profile, then relaxed so its lengths change as little as they can), and scaled to the dish's own area (pressed, the sheet keeps its volume). The note gives how far it stretches (the most over 98 % of it, and on average): use it as a blank size to press from, and check it against a trial. Cuts in such a sheet are

unrolled with it (the hole is in the flat pattern); flanges, hems and corners can't be added (as on a lofted bend).

- A body that RazorCAD folded itself (for example its STEP export) gets its closed corners back as RazorCAD made them: the corner type (V cutout, closed mitre or butt), which flange covers which and the gap are worked out from the body, and the relief and curved V lines are made again exactly. Other corners are traced from the body as they are.
- After converting, the sheet's volume is compared with the body's. The feature note warns when they differ by more than 0.5 % (for example a nut or motor part that isn't sheet metal).
- A **curved surface** that is curved one way only (a cylinder, or a profile swept straight) is converted into a rolled sheet: its section is followed by smoothly joined arcs, each a bend of its own radius. **Arc fit tolerance** (default 0.05 mm) sets how closely it follows the surface. **Thickness** goes behind the surface, in front of it, or both sides.
- **Trim where the sheet would run into itself** (on by default) trims a rolled sheet back where its two ends would meet, leaving a small clearance.

16.22 Etch Lines

Sheet Metal ▸ Etch Lines: pick a sketch drawn on the top or bottom face of a flat part of the sheet. Its lines, arcs, circles, splines and text are marked in the flat pattern, for etching, scribing or marking part numbers and bend marks. They go on the DXF's **ETCH** layer (DXF Settings can leave them out) and show in flat-pattern drawing views (View Properties ▸ Etch lines), clipped to the face they are on. The folded part is not changed; the etch lines show on the model as thin lines.

16.23 Sheet Metal Check

Sheet Metal Check lists every bend of the sheet metal body: direction, angle, inside radius, length and the straight flange on each side. It warns about flanges shorter than 3 × thickness past their bend, and holes closer than 2 × thickness + radius to a bend.

16.24 Sheet metal Body Properties

Body Properties for a sheet metal body shows thickness, bend radius and K-factor, number of bends, overall flat size, flat area, mass and **quantity**, plus:

- **Cut length**: the total length of the flat outline plus every hole and cut-out, and the number of **pierces** (closed contours), for laser-cutting estimates. Also shown in the DXF part information and under flat pattern drawing views.
- **Flat pattern seen from**: top face or bottom face. The bottom face mirrors the pattern and swaps UP / DOWN bend lines and notes. Flatten, DXF export and drawing views all follow it.

16.25 Exporting flat patterns as DXF

- Right-click a sheet metal body (tree or view) ▸ **Export Flat Pattern as DXF**, or **Export DXF** on the tab.
- **Export All DXF**: choose a target folder; one file per body, named after the body. A body you have not named goes by the part's name when it is the part's only sheet metal body (an unnamed body is otherwise called after the last feature that changed it); the same name is on the flat pattern's banner and its PART line. Identical bodies give one file with the quantity multiplied by the count.
- **Nest Flat Patterns** : lays every sheet metal body's flat pattern out on stock sheets (width × height, a margin round the sheet, a gap between parts, parts turned 90° where that fits better). Identical bodies are counted, times each body's quantity and the number of **Sets**. Parts are placed by **their true**

outlines (the default: bottom left first, the biggest first, into each other's gaps and inside the holes of others) or by **the rectangles round them** (fast). They can be **turned** not at all, by 90° steps, or (by outlines) by 45° or 15° steps. The result shows each sheet and how much of it is used; **Export DXF...** writes one DXF with the sheets side by side, their outlines on layer SHEET and each part's outline and holes, named at its middle.

- Each file is shown in a **preview window** with its layer colours and line types, which can be changed there. Buttons: **Export**, **Skip This One** and **Abort Export**.
- In the preview, zoom with the wheel, pan by dragging and double-click or **Fit** to see it all. Click lines to select them (**`Ctrl`** or **`Shift`** adds, **`Shift`**+drag for a box, **`Ctrl+A`** for all), then **Delete** them, give them their own **Line type**, or move them to another **Layer** (for example make a line an etch line). **Undo** and **Restore All** put them back. These edits apply to that one export.
- Options: bend lines, bend notes, bend region (hidden) lines, etch lines and part information.

16.26 DXF Settings

DXF Settings on the Sheet Metal tab sets the defaults for every export: layer mapping (colours and line types) and options. Defaults: **up bends red**, **down bends blue**, **etch lines yellow dashed**. **Reset to Defaults** restores them.

17 Fast Structure

The **Fast Structure** tab builds welded frames (weldments) from standard steel profiles: Structural Member, Trim/Extend, End Cap, Gusset, Weld Bead and Cut List. It also has a **3D Sketch** button, since frames are often laid out in 3D.

17.1 Structural Member

Pick a profile and a sketch or 3D sketch: every line and arc of it becomes a member, each its own body named after its profile. With nothing drawn yet, **Structural Member** opens a sketch on the selected plane (or asks for one): draw the frame, a line for each member, and exit the sketch; the members follow on it. (The same goes for Extrude, Base Flange and Extruded Surface: a banner says the feature continues when you exit the sketch, and the ribbon comes back to the tab it was started from.) The Fast Structure tab has **Sketch** and **3D Sketch** for the frame. Add more sketches to the **Path** list for a frame in 3D, for example the top of a table and its legs; the joints between them are treated too. Arcs, and the bends of a 3D sketch with a bend radius, become bent members.

Profile	Types
Hollow sections	Square (SHS), rectangular (RHS), circular (CHS / pipe)
Angles	Equal and unequal angle
Channels and beams	Channel (PFC / UPN); I-beam (UB, UC, IPE, HEA)
Bars	Flat bar, round bar

Sizes come from AS/NZS or EN/ISO tables, or enter your own dimensions.

- **Groups** : instead of every line of the Path sketches, pick the lines yourself. **Add Group**, then click lines and arcs of sketches or 3D sketches in the view (they show while you pick; the group's lines are orange, other groups' highlighted); click one again to take it out. Click a group in the list to pick for it again. Each group has its own **Corners**, **Line runs through**, **Rotate profile** and **Mirror profile** (blank: as the feature). Where two groups meet, the later group is trimmed to the earlier one, which runs on over the corner. A line picked that is no longer in its sketch is left out, and the note says so.
- **Corners** where two members meet end to end: **Mitre**, **Butt** (first or second member trimmed) or **None** (members meet at the sketch points). **Trimmed ends**: square cut on the face, or coped to the other member's outside.
- A member ending on the middle of another (a T-joint), or under a corner (a table leg), is trimmed to it, square on its face. Pipes and round bars are always coped (fish-mouthed).
- **Alignment**: which point of the profile the sketch line runs through (centre, corners or edges), a profile rotation, and mirror.
- **Which way a profile faces** : along a chain of members (lines and arcs joined end to end) every member runs the same way, whichever way each line was drawn, so a channel or angle faces the same way round the whole frame (all its webs out, or all in; **Mirror** turns them all). A member's height is kept upright; in a sketch on an upright plane (a frame standing up, such as one on Front) it lies in the plane, square to the member, so every member's profile faces the same side out of the plane, the upright ones as the level ones.

17.2 Trim/Extend

Click the members to trim on the model, then the boundary: a plane or another body. Each member is trimmed (or, with **Extend members that stop short of it**, extended) at its end nearer the boundary, either cut **square** on the boundary's face or **coped** to the boundary body.

17.3 End Cap

Click the open end faces of tubes or pipes. A plate of the set thickness is added **on the end** (outside) or **in the end** (flush).

17.4 Gusset

Click two flat faces that meet at a corner, one on each member. A triangular plate fills the corner. Set the two leg lengths, the thickness, a **corner chamfer** for weld clearance, and where the plate sits along the corner (centred or to one side, plus an offset).

End caps and gussets are bodies of their own, named by size for the cut list.

17.5 Weld Bead

Set the weld size and click the bodies to weld (none: every weldment member). A bead runs along every line where two of them touch; each joint is its own bead body.

17.6 Cut List

Cut List lists identical members together, with their profile, length, end angles, quantity, total length and mass (from the bodies' materials), then plates and welds (with the total weld length). It also gives the total length of each profile. **Export CSV…** saves it for a spreadsheet. A drawing can show it as a **Cut List Table** (Drawings chapter).

17.7 Duct Fitting

Duct Fitting (Weldments tab) makes a sheet metal duct fitting of outside sizes and a sheet thickness: a **segmented elbow** (angle, centre line radius, number of gores: two end gores and the rest full ones), a **cone** (reducer: the other diameter and the length), a **round to rectangle** transition, an **offset** (two parallel round ends side by side) or a **lateral** (a branch joining a main duct at an angle). Each is a body at the origin (move it with Move Body). Its flat patterns are developed on the middle of the sheet, with a **seam allowance** along the long seam: the message lists them and their sizes, and **File ▸ Export ▸ Duct Patterns (DXF)…** writes every fitting's patterns side by side, named with their quantities. Elbow gores are worked out exactly; transitions and offsets are unfolded by triangles, every edge at its true length.

17.8 Pipe Route and Cable

Pipe Route runs a pipe along the straight lines of a sketch or 3D sketch. Where a line branches square off a straight run, an equal **tee** goes in (ASME B16.9 sizes); other branches and crosses are reported. Choose a size (DN15 to DN300, Schedule 40 walls) or a tube of your own outside diameter and wall. **Corners**: **Elbows** puts a long radius elbow at each corner (1.5 × the nominal size, or the **Bend radius** you give), each elbow a body of its own; **Bends** makes one bent tube (3 × its diameter, or your radius). **Reducer** puts a concentric reducer (to a smaller size, its B16.9 length) at the first end, the last end (in the order the lines were drawn) or both. **Flanges at the (other) ends** adds a flange ring at each open end without a reducer. The straight pipes are weldment members, so the **Cut List** gives their lengths; the message sums them up. A line too short for the elbows at its ends is reported.

Part of a sketch : with the Pipe Route or Cable panel open, click a segment (or a point where segments meet) of a sketch or 3D sketch in the view, then the last segment (or point) of the run you want: the segments between are the route, drawn orange. Click another segment and the last of its run to add more; click a picked segment to take it off; **Use the Whole Sketch** goes back to all of it. Select a sketch in the Feature tree before starting the command (or choose it in the panel's list, or click it in the tree) to use the whole sketch.

Spool Drawing (Weldments tab, in a part with a pipe route) makes a drawing with one isometric view of the pipework, the cut list of the straight pipes and a **FITTINGS** table (each elbow, tee, reducer and flange, and how many).

Cable sweeps a round cable or bundle along a sketch or 3D sketch (a 3D sketch spline through the points it must pass follows them smoothly). Give its **Diameter**, or the number of **Wires** and their diameter (the bundle is about 1.15 × d × √n across). The message gives the route length and the cut length with the **Slack** added.

Harness Formboard lays the part's cables flat at their cut lengths, joined where their ends meet: each cable straight on from the one before it, branches fanned out at 45° steps. It shows the board with each cable's name and length and the points where cables meet, and **Export DXF…** writes it to lay out the board full size.

18 Designing for 3D Printing

The **3D Print** tab has what a part made on a 3D printer needs: features that add a nut pocket or a printable hole in a click or two, a check of the part against what the printer can do, and an export laid out on the printer's bed. Every size it uses comes from the part's **print profile**: a printer and a material.

Its buttons run left to right in the order you work: set up (profile, orientation), add features, check, export.

18.1 Print Profile

3D Print ▸ **Print Profile** picks the profile this part uses and edits your own. A profile holds:

Setting	What it sets
Bed X × Y × Z	The printer's build volume: the bed check, Show Bed, Split for Printing
Nozzle diameter, extrusion width, layer height	The smallest walls (2 extrusion widths), the bridging steps
Overhang limit	How steep a face may face down, from vertical, without support (45° on FDM printers)
Longest bridge	The widest ceiling the printer can bridge
Hole clearance	Added each side to nut pockets
Fits: press, snug, sliding, loose	The gaps Fit Clearance uses (0.05, 0.15, 0.3, 0.5 mm to start)
Density, filament diameter	The mass and filament length in the check
Smallest hole, pin and gap	The small-feature checks
The material shrinks	A warning about warping on long bases (ABS, ASA, nylon)

Four profiles come with RazorCAD: PLA, PETG and ABS on a 0.4 mm nozzle, and a resin printer. They can't be changed: **Copy...** one and change the copy. **Use for this part** keeps the choice with the part (a copy of the profile is saved in the part file too); **Make default** is the profile for parts that have none. A change to one of your profiles applies to every part that uses it.

Printer sets the bed from a list of popular printers (Prusa MK4 / MK4S, CORE One, CORE One L, XL and MINI+; Bambu Lab A1 mini, A1, P1S, P2S, X1C, X2D, H2S and H2D; Creality Ender-3 V3, Ender-3 V3 Plus and K1 Max; Elegoo Neptune 4 Pro; Anycubic Kobra 3). Chosen on a starting profile, it makes a copy of it for that printer, used for this part. The sizes are from the makers' own pages; a printer with two nozzles has the size one nozzle prints (a part in one material). The X1C's size is from retailers' and reviewers' listings, and the dialog says so when it is chosen: check it against your printer's specifications. Typing a bed size of your own clears the printer.

Slicer program (Windows app): **Choose slicer program...** picks the program **Open in Slicer** starts.

18.2 Place on Bed, Auto-Orient and Show Bed

The checks, the overhang map and the export all use the part's **build orientation** : which way up it prints. Until you set one, the part's biggest flat outside face lies on the bed.

- **Place on Bed** : select a flat face, then click it: that face goes down on the bed.
- **Auto-Orient** : tries the part with each of its outside flat faces down, and in the 26 directions of a cube's faces, edges and corners, and lists the best three: least support first, then the most on the bed, then the lowest. **Use** sets one.
- **Show Bed** : draws the printer's bed (a 10 mm grid) and its build volume under the part.

Features that depend on which way is up (Printable Hole, Bridging Layer, First-Layer Chamfer, Printable Overhangs) keep the orientation the part had when they were added.

18.3 Check Printability

Check Printability checks the solid bodies in their build orientation against the profile, numbers what it finds on the model, and lists it:

Check	Fails when
Overhangs	A face facing down steeper than the overhang limit, off the bed (amber within 10° of it)
Bridges	A flat ceiling wider, the narrow way across, than the longest bridge
Walls	Thinner than 2 extrusion widths (amber under 3)
Holes and pins	A hole standing up under the smallest hole, a pin under the smallest pin
Closed hollows	A void sealed inside the part (a warning on FDM printers; a fail on resin, which is trapped in it)
Bed contact	Less than 5 % of the part's footprint on the bed
Bed fit	Bigger than the printer
Warping	A base over 150 mm long in a material that shrinks (a warning)

Below the list: the support it would need (an estimate: columns from each overhang down to the part or the bed), the part's volume, its mass in the profile's material, and the filament it takes. Where a print feature fixes what was found, a button starts it: **Printable Overhangs**, **Split for Printing**, **Auto-Orient**.

18.4 Overhang Map and Thickness Map

Overhang Map colours the model up its build direction: green fine (or on the bed), amber within 10° of the profile's overhang limit, red over it (needs support). **Thickness Map** colours walls red under 2 extrusion widths, green to yellow above. Click again to turn them off.

18.5 Nut Pocket

Nut Pocket makes a hex pocket for an ISO 4032 nut, with the screw's clearance hole through (ISO 273 normal fit; untick it to leave it out). Click points on a face, or pick a sketch with points: each gets one. The pocket is the nut's size plus the profile's clearance each side, as deep as the nut plus the clearance (or **Pocket depth**). **Turn the flats by** turns it.

Captive slot: the pocket starts **Slot below the face** inside the part and a slot as wide as the nut runs out sideways past the part's edge (**Slot direction**), to slide the nut in from the side.

18.6 Insert Hole

Insert Hole makes a hole for a threaded insert pressed or melted in (heat-set): the insert's hole diameter, as deep as the insert plus a **Relief** below it, with a lead-in **Chamfer** at the top. The sizes come from a table of typical inserts; makers' sizes differ, so check yours and type its **Hole diameter** and **Insert length** if they do.

18.7 Screw Hole for Plastic

Screw Hole for Plastic makes a pilot hole for a screw cut straight into the print: 0.85 × the screw's size (or a **Pilot hole** of your own), to a **Depth**, with a cap screw's **Counterbore** or a **Countersink** if wanted (ISO

sizes).

18.8 Printable Hole

Printable Hole reshapes holes lying on their side so they print without support: their top comes to a 45° point up the build direction (a **Teardrop**), or the point is cut flat at the top of the hole. Every hole whose axis lies within **Holes within this of level** is changed, or click the holes to change only those.

18.9 Bridging Layer

Bridging Layer helps a counterbore or pocket printed upside down: the printer can't bridge round the smaller hole above it. Click the flat face round the hole (the counterbore's floor). **Steps**: that face is stepped up one layer at a time, first a strip as wide as the hole across the opening, then a square, so each layer bridges onto the one before. **A skin**: one layer closes the hole; drill it out after printing.

18.10 First-Layer Chamfer

First-Layer Chamfer chamfers every edge on the bed (0.4 mm to start), so the first layer, squashed wider, doesn't stick out.

18.11 Printable Overhangs

Printable Overhangs turns rounds facing down, up the build direction, into 45° chamfers of the same size, which print without support.

18.12 Fit Clearance

Fit Clearance moves the faces of a part that fit against another part back by a fit's gap: **Press**, **Snug**, **Sliding** or **Loose** from the profile, or a **Gap** of your own. A flat face moves in; a pin gets thinner and a hole wider, by the gap all round.

18.13 Print-in-Place Hinge

Print-in-Place Hinge makes a hinge that prints assembled. Click the straight edge it turns about: knuckles (**Knuckle diameter**) are spaced along it, the gap apart; the first, third… carry the **Pin** all along, the others are bored the gap wider all round and turn on it. Each set is a body of its own: join each to its part with **Combine**.

18.14 Split for Printing

Split for Printing cuts a part too big for the printer in two at a plane (choose a plane, or click a flat face), with **Dowels** across the cut: holes for dowels (**Dowel diameter**, the profile's snug fit each side, **Hole depth each side**), spaced along the cut's longest side and kept clear of its edges. Glue the halves on the dowels.

18.15 Export for Printing and Open in Slicer

Export for Printing writes the solid bodies as 3MF (it keeps millimetres and each body as its own object), turned into their build orientation, their lowest point on the bed, over the bed's middle. **Export STL for Printing** does the same as STL. **Open in Slicer** (Windows app) opens the last export, or a new one, in the slicer program named in Print Profile; RazorCAD only starts the program with the file.

18.16 Calibrate Fits

Calibrate Fits saves a test piece (3MF): a plate with four Ø6 holes and four pegs, 6 mm less twice the profile's press, snug, sliding and loose gaps, left to right. Print it, try the pegs in the holes, and set each fit to the gap of the peg that fitted that way. **Save the fits** keeps them in the profile (a starting profile: in a copy of it, used by this part).

19 Configurations

A configuration is a variant of a part or assembly: other values for the global variables and other features suppressed in a part, other part configurations and components shown in an assembly.

19.1 The Configurations tab

The **Configurations** tab sits next to the Feature tree and the Property panel at the top of the left panel. It lists **Default** and every configuration, the one shown ticked, each with a summary of what it changes.

- **Double-click** a configuration to show it. The tree shows its name next to the part.
- **+ Add Configuration** makes a new one starting from the one shown.
- Right-click for **Show Configuration**, **Rename**, **Copy Configuration**, **Properties (Design Table)** and **Delete**.

19.2 What a part configuration changes

- **Variable values**: each configuration can give the global variables (Tools ▸ Equations) other values.
- **Suppressed features**: while a configuration is shown, **Suppress / Unsuppress** in the tree changes that configuration only. The tree greys out what it suppresses.

19.3 Design table

Design Table… in the Configurations tab (or right-click ▸ **Properties (Design Table)**, or **Configurations** on the Evaluate tab) edits all the configurations in one table. The table can be copied to and pasted from a spreadsheet as CSV.

19.4 Configurations in assemblies

Assemblies have the same tab. While an assembly configuration is shown, these change that configuration only: a component's part configuration, **Hide / Show**, **Suppress** (a suppressed component is left out, with its mates) and a mate's Suppress or value. See the Assemblies chapter for how components use part configurations.

20 Assemblies

An assembly places parts and other assemblies as **components** and holds them together with **mates**. Assemblies are saved as **.rasm** files and open in a tab of their own.

20.1 Starting an assembly

File ▸ New Assembly, or **New Assembly** in the start window. Every assembly has the standard **Front, Top and Right Planes** and the **Origin** at the top of its tree. They show in the view as a part's do (the eye in the tree, **View ▸ Show Planes**), and mates can use them; they stay where they are.

The **Assembly** tab of the ribbon holds Insert Components, Standard Parts, Equations, Assembly Hole, Assembly Cut, Mate, Fix, Float, Hide Component, Suppress, Open Part, Edit in Context, Exploded View, Positions, Explode / Collapse, Motion Study, Bill of Materials, Interference Check, Collision Detection, Speed Mode and Zoom to Fit.

20.2 Inserting components

Insert Components lists the parts and assemblies open in other tabs, or **Browse for a Part or Assembly...** takes a .rcad or .rasm file.

- The first component is **fixed** at the origin; the others are placed beside what is there.
- If the part has configurations, you choose which one to use. Right-click a component later to switch it to another configuration. The same part can be in the assembly several times in different configurations.
- The assembly keeps a copy of each part, refreshed from its tab or file whenever you come back to the assembly, so edit a part and the assembly follows.
- Several instances of one part share it: it is built once and drawn once.

20.3 Standard parts

Standard Parts (Assembly tab, or **Insert ▸ Standard Parts...**) adds parts made to size from their standards:

- hex bolts (ISO 4017) and socket head cap screws (ISO 4762), M3–M24, in the standard lengths; hex nuts (ISO 4032) and washers (ISO 7089);
- inch hex bolts (ASME B18.2.1), socket head cap screws (ASME B18.3), hex nuts (ASME B18.2.2) and SAE flat washers, 1/4" to 1", with UNC threads and inch lengths;
- socket set screws (ISO 4026), dowel pins (ISO 8734), round head rivets (ISO 1051), circlips for shafts and bores (DIN 471 / 472), parallel keys (DIN 6885 A) and 6000 / 6200 series ball bearings.

Each is an ordinary part with its threads as cosmetic threads, so it builds fast. Choose its material: steel, stainless, brass or aluminium. The same type, size, length and material is one part however many times it is used, so the bill of materials counts them.

Select a hole first (its round face or edge) and the size is picked to fit it (a clearance or tapped hole for bolts, the bore for a bearing, the shaft for a key), the length to suit the hole, and the part goes into the hole at the end facing you, mated to it (Concentric, plus Coincident on the face round the hole). For a bolt or screw:

- **Washer under the head**, and **washer and nut on the far side**: the whole set is mated, and the length is picked for it (a 10 mm plate: M8 × 25). When the hole lines up with holes in other components (plates bolted together), the far side is the far side of them all: the washer and nut go under the last plate, and

the length is picked for the whole stack.

- **All matching holes**: the same part or set goes into every hole of the same size parallel to the selected one, in one go.

A hole that ends on a curved face (a cross hole in a round bar) gets a seat plane of its own, which follows the hole when the part changes.

20.4 Mates

Click **Mate** (`M`), then two faces, edges or vertices on different components (or a standard plane), choose the type and click ✓. The panel stays open for the next mate; `Esc` closes it.

The panel lists the types the two entities can take under **Standard Mates**, **Advanced Mates** and **Mechanical Mates**. Mechanical mates make an assembly move as the real machine does: turn or drag one part and the others follow, and a motion study plays it.

Mate	Holds
Coincident	Faces, edges or points together
Concentric	Axes in line
Parallel / **Perpendicular**	Directions
Tangent	A face touching a round face
Distance / **Angle**	At a set value
Lock	One component fixed to another
Gear	Two gears turning together: pick their cylinders (pitch circles)
Rack Pinion	A pinion turning as a rack slides: pick the pinion's cylinder and a straight edge or the flat face of the rack
Screw	On one axis, the second component advancing the **pitch** along it per turn it makes relative to the first (a nut on a lead screw)
Hinge	Concentric and touching in one mate: pick the two round faces (or circular edges), then a flat face of the first component and the face of the second that sits on it. It can only turn about the axis; **Limit the turn** keeps it between a minimum and a maximum angle (0° where it is when made: a lid that opens −10° to 110°)
Universal Joint	Two shafts joined by a cross at the point where their axes meet: pick the two shafts (round faces or their axes). Mate each shaft so it can only turn (Concentric or Hinge); turn one and the other follows. With the shafts at an angle the second turns unevenly, as a real universal (Cardan) joint does: at 30°, about 4° ahead or behind at a quarter turn, level again every half turn
Linear Coupler	As one component slides, another slides the **Ratio** of it, along its own axis, edge or face normal (two slides linked by a lever or a cable). **Reverse** makes it slide the other way
Width	The second component centred between two faces of the first: pick a face of each, then the first component's other face, then the second's (two parallel faces each: a tongue in a slot)
Slot	A pin (a cylinder's axis) or a point kept the set distance from a straight edge (0: on it; the slot's side edge with the pin's radius), free to slide between the edge's ends and stopped there

Mate	Holds
Path	A point (a vertex, an edge's middle, a sphere's centre) kept on a curved or circular edge of another component, free to move along it
Cam	A follower kept touching a cam's curved edge (a plate cam's outline): a roller (a cylinder's axis) its radius from the edge, seen along the roller's axis, or a point at the set distance. Guide the follower with another mate (a Slot or a Concentric on a pivot) and turn the cam: the follower rides on it
Symmetric	Two faces, edges or vertices kept mirror images of each other about a plane: pick them, then the plane (or a flat face) between them

- Flat faces count as planes; cylinders, cones and circular edges as axes; spheres and vertices as points.
- Each mate is shown applied while you choose. **Flip** swaps Aligned / Anti-Aligned (or the side for Distance and Tangent). When the alignment chosen from how the faces sit can't be reached, the other one is used.
- **Limit** (Distance, Angle and Hinge) : free between a **Minimum** and a **Maximum** instead of one value (a slide with end stops, a door that opens 0 to 110°). Inside the limits the component moves freely; dragged past one, it stays there exactly.
- A mate that conflicts with the others is refused with a message.
- A mate keeps its faces and edges by their geometry, not only by their number in the part. When a part gets a new feature or a dimension moves a face, the mate stays on the same face. A mate whose face or edge is gone gets a red **!** in the tree ("1 mate lost a face"). **Tools ▸ Fix Mates...** lists them with Edit, Suppress and Delete.

20.4.1 Gear and rack-pinion mates

These relate one component's motion to another's, from where they are when the mate is made, so set the teeth in mesh first. Hold each gear on its axle with the usual mates (Concentric and Coincident); the gear mate then takes one more degree of freedom.

- **Gear**: turn one gear and the other turns the other way, in the ratio of their radii. **Ratio** is how many turns entity 2 makes per turn of entity 1; change it to the teeth ratio if the cylinders are not the pitch circles. **Reverse the direction** is for an internal gear, or pulleys on a belt and sprockets on a chain, where both turn the same way. The tree shows it as for example "Gear1 1:2".
- **Rack Pinion**: turning the pinion slides the rack along the edge (or face) picked, and sliding the rack turns the pinion. **Rack travel per pinion turn** starts as the pinion's circumference (π × diameter). **Reverse** changes the direction.
- They follow turns of any size made step by step, as when you drag a gear. One jump of more than half a turn can't be told from a turn the other way.

20.5 Moving components

Left-drag a component to move it, **Alt**+drag to turn it. It moves only as its mates allow. The view turns, pans and zooms as in a part (see Navigating the 3D View).

Selecting works as in a part and follows the **selection filter**: with no filter a click picks the vertex, edge, face or plane under the cursor; with the Bodies filter it picks whole components. **Ctrl**+click adds or takes away. Right-click a face, edge or vertex for **Mate...** and the component's commands; whatever is selected goes into the mate. Dragging a component moves it, whatever is selected.

20.6 Fasten All Holes

Fasten All Holes (Assembly tab) finds every stack of lined-up round holes through two or more components (tick **Also holes through one component only** for single ones too) and puts a fastener in each: the type you choose (hex bolt, socket head cap screw, inch sizes…), its size fitted to the holes, its length to the whole stack, with a washer under the head and a washer and nut on the far side if ticked, each mated in place (concentric, seated on the stack's outer faces). Stacks that have a fastener already are left alone, so it can be run again after adding parts.

20.7 Component patterns and mirrors

Component Pattern, **Circular Component Pattern** and **Mirror Components** (Assembly tab) copy the selected component (or the components of the selected faces and edges) :

- **Component Pattern**: along X, Y, Z or a selected straight edge or axis, a spacing and a number of instances, and optionally a second direction.
- **Circular Component Pattern**: round X, Y, Z or a selected axis (a cylinder, a circular edge), a total angle (equal spacing) or the angle between copies.
- **Mirror Components**: about the YZ, XZ or XY plane or a selected flat face. The copy is placed as the mirror image, the part itself the same way round (for parts that are the same both ways round, such as screws, pins and brackets symmetric about the plane).

The copies are components of their own (in the bill of materials, mateable), fixed, and follow their seed when it moves. The pattern is listed in the tree: double-click it to change it, or delete it with its copies.

20.8 Collision detection

Collision Detection on the Assembly tab (or **Tools ▸ Collision Detection Off / Show Collisions While Dragging / Stop at Collisions While Dragging**) checks for parts running into each other as you drag.

- **Show**: the parts that collide turn red while you drag, and the status bar names them.
- **Stop**: the dragged component (and what its mates carry along) stops where it meets the other part, right against it.
- Faces that only touch (mated faces) are not collisions, nor is a shaft in a hole of its size, nor a screw in a nut or tapped hole (threads are cosmetic, so the screw is its full diameter in a bore of about its minor diameter). Parts that already overlap when the drag starts are left out.
- It works from the parts' display meshes, so it keeps up with the mouse. **Interference Check** gives the exact overlap volumes.

20.9 The assembly tree

- Each component shows **(f)** when fixed or **(-)** while still free to move, its configuration, and an eye to hide or show it.
- The **Mates** folder lists the mates; right-click one for **Edit**, **Flip**, **Suppress** or **Delete**.
- Right-click a component for **Mate…**, **Open Part** (or double-click it), **Edit in Context…**, **Fix / Float**, **Hide / Show**, **Suppress / Unsuppress**, **Insert Another Instance**, **Rename** and **Delete**.
- **Make Flexible** (a sub-assembly component) : its parts move in this assembly as its own mates let them (a hinge opening, a slide moving), solved with this assembly's mates on them; where they end up is kept with this component, so the sub-assembly itself and its other uses do not change. **Make Rigid** puts it back as one piece. A component fixed inside it stays fixed only when the sub-assembly component is fixed.

- **Number Parts** (Assembly tab) gives each part and sub-assembly a **Part Number** property in the order of the bill of materials: a prefix, a first number, a step and a number of digits (P-0001, P-0002…). Those with a part number keep it unless you ask to number them again. Save parts kept in files of their own to keep their numbers there.
- **Edit in Context** opens the part with the other parts you tick brought into it as transparent **context bodies** (the features *Context:* …), where they are in the assembly from it. Design the part to fit them: sketch on their faces, convert their edges, extrude up to them. Context bodies are references only: they are left out of the part where it is used (in the assembly, drawings, the bill of materials and exports). The context stays up to date: each time the part comes to the front with the assembly open, context bodies whose parts changed, changed configuration or moved are brought in again (a part taken out of the assembly has its context feature suppressed). Right-click a context feature ▸ **Update Context** to bring them all in again. Delete or suppress the context features when they are no longer needed. Save the part to update the assembly.
- **Edit Part** : right-click a part in the tree or the view ▸ **Edit Part**. It opens the part with every other part shown in the assembly brought in as context bodies (as Edit in Context, without asking which), and its feature tree is the one new features go into. A banner says which part you are editing in which assembly; **Back to the Assembly** returns to it.
- **Cross references and circular references** : a part can take its shape from others (sketch on their context bodies, use their edges, extrude up to them), and it follows them when they change. It can't take its shape from a part that already takes its shape from it, directly or through others, nor from another instance of itself: that would be a circular reference. Edit Part and Edit in Context leave such parts out and say why in a warning; remove the reference in the other part first (its *Context* feature) to use it.
- **Parts expand in the tree** : click ▶ by a part (or a part in a sub-assembly) to see its features under it; double-click one to edit the part.
- Undo and redo work in the assembly.

20.10 Sub-assemblies

An assembly can hold other assemblies as components. **Insert Components ▸ Browse for a Part or Assembly…** takes a .rasm file, and STEP assemblies keep their sub-assemblies when they open as an assembly. An assembly can't go inside itself.

- A sub-assembly is kept whole, with its own parts, placements and configurations, and moves, hides, suppresses and explodes as one. Clicking any of its parts picks it.
- Mates can use the faces, edges and vertices of the parts inside it.
- In the tree, ▶ opens a sub-assembly to show its components. Double-click one, or right-click the sub-assembly ▸ **Open Sub-assembly**, to change it in its own tab; the changes come back to the assembly.

20.11 Equations shared with the parts

Assembly ▸ Equations… (also on the Assembly tab) holds the assembly's own variables, for example `base = 6` and `thickness = 2*base`. A variable that **drives the parts** sets the variable of the same name in every part that has one, through the sub-assemblies, so related parts are sized together from the top.

- The table shows each variable's value, the parts it drives and a **Drives the parts** check box (off for a variable used only to work out others).
- The values go into the assembly's copies of the parts, so the assembly, its drawings, the BOM and exports use them. They are applied again when a part is read again from its file.

- **Update Open Parts** writes them into the parts open in tabs too, to be saved there.

20.12 Assembly holes and cuts

Assembly Hole and **Assembly Cut** (Assembly tab, or the Insert menu) make holes and cuts in the assembly, through every component they pass.

1. Click points on a flat face of a component. More points on the same face make more holes; clicking another face starts again.
2. Set the hole: simple, counterbore, countersink or tapped, a standard size (ISO 273 close, normal or loose fit) or a custom one. The first component gets the hole as chosen, the ones behind it a plain hole of the drill size. **Tap the last component** makes the last one tapped for the size. A cut is a rectangle: width, length and angle.
3. Choose **Through all** or a depth. **Components cut** lists the components it passes through; untick one to leave it out.
4. **Fasteners**: hex bolts or socket head cap screws put into each hole as it is made, with washers and a nut on the far side if wanted, as long as the stack they hold (to a standard length) and mated in the hole. Into a tapped last component, a screw goes about one diameter into the thread, without a nut.

The parts' own files are not changed: each component cut gets the feature in its copy of the part, so the assembly's drawings (hole callouts too), BOM and exports have the holes. The holes stay with the component they were clicked on: move a component and its holes are worked out again. The tree's **Assembly Features** folder lists them; right-click one to edit, suppress, rename or delete it. Fasteners put in a hole are kept when the hole is deleted.

20.13 Exploded views

Exploded View on the ribbon (or **Insert ▸ Exploded View...**) spreads components apart for display; the model and its mates stay put.

1. Click components to pick them (click again to drop one).
2. Pick a direction: ±X, ±Y, ±Z, or along a face's normal.
3. Drag the picked items along the direction, or type the distance, then click **Add Step**. **Spread Out** spaces the picked items out along the direction, one step each.
4. Click a step to change it, ✕ to delete it. **Explode lines** (dashed, from each item's place to where it is shown) can be switched off.

The tree's **Exploded Views** folder lists them. Double-click one to explode or collapse it (animated); right-click for Explode / Collapse, Animate, Edit, Rename and Delete. The **Explode / Collapse** button on the ribbon does the same for the last one. Dragging a component, adding a mate or editing collapses the view first. Exploded views are saved with the file.

20.14 Motion studies

Motion Study (Assembly tab) shows how an assembly moves. Choose a driver:

- **turning a component** about the axis of one of its mates (360°, for example);
- **sliding a component** along one of its mates' axes or faces;
- **changing a Distance or Angle mate's value** from where it is to a new one.

More Drivers ▸ + Add a driver turns or slides further components at the same time, each by its own amount over the same steps (two motors, a crank and a slide). A component can have only one driver.

The motion is worked out step by step and the mates carry the other components along, the mechanical mates (gear, rack-pinion, screw, hinge, universal joint, linear coupler, slot and cam) included. **Play / Stop** (**Space**), back to the start, a slider to go through the steps, the number of steps, the duration and **Loop**. Where the mates can't follow, the panel says at which step.

Detect collisions (on for new studies) checks every step: parts that collide show in red as the motion plays, and the panel names the first collision and its step. **Stop at the first collision** ends the motion at the step before it. **Clearance** between two components (choose them in the panel) works out, at every step, the least distance between them from their display meshes, and the panel gives the least of all and its step (in red when they touch).

✓ keeps the study in the assembly's **Motion Studies** folder: double-click one to play it; right-click to edit, rename or delete it. Closing the panel puts the assembly back as it was: a motion study never moves the assembly itself.

20.15 Interference Check

Interference Check (Assembly tab, or **Tools ▸ Interference Check...**) finds where components overlap. Each overlap shows in red and is listed with its two components (inside sub-assemblies as "Sub-assembly ▸ Part") and its volume. Faces that only touch, such as mated faces, are not counted, and nor are the bodies of one multi-body part against each other. An exploded assembly is collapsed first, so the check is of where the mates put the parts.

20.16 Bill of Materials

Bill of Materials lists the parts with their quantities and materials: one row per part and configuration used, with suppressed components left out. It is indented: a sub-assembly's parts are listed under it (1.1, 1.2...), with the quantities in one of it. **Copy as CSV** or **Save CSV...** takes it to a spreadsheet.

- **Columns**: click a column heading to change what it shows (an item number, quantity, part, material, file, configuration, any custom property, mass, cost, or values typed in), insert a column left or right, move, rename or remove it. Typed values are edited right in the table. The columns are kept with the assembly and are part of undo; **Default Columns** goes back to the usual ones.
- **Mass and cost**: **Mass** and **Total mass** come from each part's material and volume; **Cost** and **Total cost** from a part's **Cost** property (for example 12.50, $3 or R12,50). A **TOTAL** row adds up the quantity, mass and cost; a sub-assembly adds up its parts.

20.17 STEP assemblies

Opening a STEP file that holds an assembly asks whether to open it **as an assembly** or **as one part** (see Importing in the Working with Files chapter). As an assembly, each part of the file becomes a part, each sub-assembly a sub-assembly, and each component is placed, fixed, where the file places it, with the names and colours from the file.

File ▸ Export ▸ STEP... in an assembly saves the whole assembly as one STEP file with its structure: each part once, in the configuration used, every component placed where it is and named after it, sub-assemblies kept as assemblies, and the parts' colours (a body's own colour, from Appearances, over its material's). Suppressed components are left out.

20.18 Saving parts as separate files

When an assembly holds parts or sub-assemblies that are only inside it (as a STEP assembly does when it opens), Save asks how to save them:

- **Save Parts as Separate Files** saves each part as a .rcad file and each sub-assembly as a .rasm file in the assembly's folder, named after them. The assembly (and each sub-assembly) then uses those files, so each part can be opened and changed on its own and the change shows in the assembly. A part used in several places is one file. If files of those names are already there, you are asked before they are replaced.
- **Save as One File** keeps everything inside the assembly file. You are not asked again, except on Save As.

20.19 Collect Files

File ▸ Collect Files... (in an assembly) copies the assembly with everything in it into one folder: each part and sub-assembly a file of its own (as the assembly has it now), the assembly last, using those files. Names can be given something **before** and **after** them (e.g. `P1-`), to start a variant of the design or send it all to someone. The assembly open here is left as it is.

20.20 Big assemblies

Each part an assembly builds is kept in the model cache (`%LOCALAPPDATA%\RazorCAD\modelcache`). Opening the assembly again shows unchanged parts straight from there, and the kernel only builds parts that changed, so a large layout reopens in seconds. Parts are built side by side in several geometry kernels at once, as many as the PC's processor threads and memory allow, and each shows as soon as it is built. Each part is sent to the graphics card once and placed at every component that uses it, so moving, hiding or exploding components is instant.

Speed Mode (Assembly tab, or **View ▸ Speed Mode**) is for very large assemblies: the parts are shown from simplified display meshes, with far fewer triangles to draw and keep in memory. The faces and edges stay the same, so mates, measuring, picking and collision detection work as before. A component you select is loaded in full; turning speed mode off shows everything in full again. It is saved with the assembly, and a large assembly suggests it in the status bar. Drawings, exports and Interference Check always use the exact geometry.

21 Drawings

RazorCAD makes 2D engineering drawings of parts and assemblies, saved as **.rdrw** files and opened in their own tab.

21.1 Starting a drawing

- **Start window ▸ New Drawing**, or **File ▸ New Drawing**, opens the set-up dialog (below) and then an empty sheet.
- **File ▸ Make Drawing from Part** (in a part), or **Make Drawing from Assembly** (in an assembly): after the set-up dialog, places the standard three views at a scale that fits, clear of the title block.
- A view you place (a model, projected, section or detail view) that lands on the title block, on another view or on another view's dimensions and notes is moved to the nearest clear spot inside the frame, its own dimensions with it: a projected or section view only along its line from the view it comes from, so it stays lined up. When there is no clear spot, you are offered the largest scale that gives it one (of the view, or of the sheet when the view uses the sheet's scale) or the smallest larger sheet that does, or to leave it where it is and drag it yourself.
- A drawing can show assemblies as well as parts: every component is drawn where it sits, through its sub-assemblies, in the configurations shown; suppressed and hidden components are left out. Big assemblies (more than about 200 solids) are drawn fast from the parts' display meshes: lines, circles and arcs as exact as the 3D view's, and where an edge turns hidden to about two thousandths of the view's size. A site layout of thousands of solids then takes seconds a view instead of many minutes. In a view's properties, **Hidden lines** chooses Automatic, Exact or Fast; section and flat-pattern views are always exact.

Drawings are opened, saved and listed in the recent files like parts.

21.2 The sheet and its template

The default template is A3 with an ISO 5457 frame (zones and centring marks) and an ISO 7200-style title block: title, drawing number, material, scale, size, units, projection symbol, drawn / checked / approved, date, mass, company, revision, sheet n of m and general tolerances. Sheet sizes A0 to A4, landscape or portrait.

21.3 Drawing templates

A **drawing template** (.rdrt file) keeps a drawing's set-up: its sheets (size, orientation, scale, projection), the drawing settings (text height, decimals…), the title block's standing entries (company, drawn / checked / approved by, finish, tolerances, revision) and notes placed on the sheet. Views, parts and part-specific entries (title, drawing number, material, mass, date) are not kept.

- **File ▸ Save as Template…** in a drawing saves one, by default to `Documents\RazorCAD\Templates`. Opening a .rdrt file edits the template in its own tab.
- **The New Drawing dialog** lists your templates (from the Templates folder, or **Browse…** for one anywhere) and the standard ones: A4 portrait and landscape, A3, A2, A1 and A0 landscape, and A3 and A4 in third angle projection. Set the drawing name, title, drawing number, drawn by and company, and the first sheet's size, orientation, scale and projection. Tick **Use this template as my default** to have it chosen first next time. Double-click a template to create the drawing at once.
- **Change Template…** (File menu, Sheet tab, or right-click the drawing in the tree) switches an existing drawing to another template. Views, dimensions and annotations stay; company, tolerance and finish

come from the new template, and the old template's notes are replaced. Undo goes back.

- **Company sheet formats**: a template can carry its own border, title block and logo in place of the standard ISO one, used on sheets of its size and orientation. Its text shows the drawing's properties, and can be set in a font installed on the PC; PDF exports embed the font. Drawing Properties lists the title block's own fields.

Changing the projection (Change Template or Sheet Properties) lays the views out again for the new projection, so a top view stays a top view.

21.4 Working on the sheet

Action	How
Zoom	Mouse wheel
Pan	Middle or right drag, or left drag on empty paper
Fit the sheet	`F`
Move views and annotations	Drag them
Edit	Double-click a view for its properties, a dimension or note to edit it
Menus	Right-click
Delete	`Del`
Undo / Redo	`Ctrl+Z` / `Ctrl+Y`

21.5 View Layout tab

Command	What it does
Model View	Pick an open part or assembly, or browse for its file (.rcad or .rasm), pick the orientation (front, back, top, bottom, left, right, isometric, dimetric, trimetric or flat pattern) and click on the sheet. Then move around it to place projected views.
Standard Views	Three views laid out for first- or third-angle projection
Projected View	A view aligned with its parent
Flat Pattern	A sheet metal body unfolded (see below)
Section View	Horizontal, vertical or **At an angle** (degrees from horizontal, counter-clockwise) cutting line, hatched cut faces, A-A labels and arrows, flip; an angled section looks square to its line and can be placed anywhere
Detail View	A circle with a letter, 2:1 by default
Auxiliary View	Looks square at an inclined edge, so it shows its true shape
Break Lines	Leaves out the middle of a long part between two break lines; dimensions across the break keep the true length. Right-click the view ▸ Remove Break Lines puts it back
Crop View	Shows only the area you draw round
Broken-out Section	Click a view, then the corners of an outline round what to show inside; click the first corner again to close it and give the **depth** from the front of the model. The material inside the outline is taken away to that depth, the faces cut there hatched, the outline drawn thin. Right-click the view ▸ Remove Broken-out Section takes it away
Compare With	This drawing's sheets against another drawing open in RazorCAD (an earlier version, say), sheet by sheet: lines only in this one in red, only in the other in green, in both in grey

Command	What it does
Update Views	Re-reads the parts and redraws the views

A broken or cropped view can be put back to whole from its right-click menu. Section views hatch each body or component with its own angle and spacing.

Views are hidden-line projections computed by the geometry kernel.

21.6 View properties

Click a view to open its panel:

- Orientation, changeable after placing (also to and from Flat Pattern; projected and section views made from it are removed when it becomes a flat pattern, and **`Ctrl+Z`** brings them back).
- Display style: hidden lines removed, hidden lines visible, or wireframe; tangent edges and cosmetic threads on or off.
- **Exploded** (a view of an assembly that has exploded views): shows the view exploded as one of them. The components are where that exploded view moves them, the view keeps its place on the sheet, and its projected, section, auxiliary and detail views follow. Balloons on it pick the moved parts.
- **Alternate position** (a view of an assembly): draws the assembly again as a **position** kept in it, in thin phantom lines (long dash, two short), over the view: a lid open, a lever at its other end. Keep positions in the assembly with **Assembly ▸ Positions** (move the components, e.g. by changing a mate, keep the position, then put them back).
- **Scale**: the sheet scale or a scale of the view's own, whatever the sheet's; labels. A view and the views made from it (projected, section and auxiliary views) share one scale: change it on any of them and they all follow. The first view keeps its middle, and each view made from it keeps its gap to the view it was made from. A **detail view** has a scale of its own (2:1 of its parent's when placed), which a change of its parent's scale leaves alone. Changing the sheet's scale (Sheet Properties) keeps the gaps the same way for the views that use it.
- **Break from Parent View** (right-click a projected, section or auxiliary view): the view stays as it is (the same direction, cut, scale and place) but no longer depends on the view it was made from. It can be dragged anywhere, given a scale of its own and moved to another sheet, and it stays when that view is deleted. A projected view becomes a model view of its own; a section or auxiliary view keeps its cutting line or arrow on the view it came from. Its own projected and section views go with it. **`Ctrl+Z`** puts it back. This is not **Break Lines**, which leaves out the middle of a long part.
- **Move to Sheet…** (right-click a model, flat pattern, detail or independent view): moves it, with the views made from it and their dimensions and notes, to another sheet or a new one. It keeps its place on the sheet, or goes to the middle when it would fall outside the frame.
- **Rotate View**: −90°, +90°, 180° and 0° buttons and an angle box (degrees, counter-clockwise). Section lines, detail circles and annotations turn with the view.

Aligned views slide only along their alignment. Deleting a view deletes the views made from it, except those broken from it.

Drawings saved before views shared their scale open looking as they did: a projected, section or auxiliary view that had a scale of its own, different from its parent's, opens broken from its parent.

21.7 Flat pattern views

A flat pattern view shows a sheet metal body unfolded with its outline and holes, bend lines (chain lines) with notes (UP / DOWN, angle, radius), optional bend region lines, and its flat size, thickness, number of

bends and cut length underneath. It can be dimensioned (bend lines too) and exported to PDF. If the part has several sheet metal bodies, pick the body in the view properties. Instead of notes, the view can show a **bend table**: the bend lines numbered B1, B2… and a table of direction, angle and inside radius under the view. Forms, cross breaks and tabs and slots show as in the DXF.

21.8 Annotation tab

Command	How	
Dimension	Click a line for its length, a circle for its diameter (an arc: radius), two points, lines or circles for the distance between them (two lines at an angle: the angle), then click where the text goes. Horizontal, vertical or aligned depends on where you place it. Prefix, suffix, tolerance and override text can be set.	
Auto Dimension	Dimensions the selected views, or every view on the sheet (see below)	
Check Drawing	Lists what the drawing still lacks (see below)	
Note	Click on the sheet; click an edge first to add a leader	
Baseline Dimension	Click the base edge or point, then each point or edge to dimension from it; the dimensions stack one above the other	
Ordinate Dimension	Click the origin, then where the values line up, then the points to dimension	
Chamfer Dimension	Click the chamfer edge and an edge next to it, then place the text ("3 × 45°")	
Hole Callout	Click a hole. The callout gives its size and depth from the Hole feature (counterbore, countersink, tapped thread and tap drill) or its cosmetic thread, and how many of the same holes the view shows, for example "4× Ø6.6 THRU ALL / CBORE Ø11 DEEP 6.5".	
Datum Feature	A datum letter in a box, with a triangle on the feature	
Geometric Tolerance	A feature control frame: one of the 14 characteristic symbols, tolerance with Ø zone and material condition, and up to three datums	
Surface Finish	A surface finish symbol with its roughness	
Weld Symbol	Fillet, square, V, bevel or plug weld on a reference line with an arrow to the joint; arrow, other or both sides; all-around, field weld and tail	
Center Mark	Click circles or arcs	
Centerline	Click two lines (or two points); the centerline runs between them	
Revision Table	Rows edited in one dialog; the latest revision goes into the title block	

Command	How	
Table	A general table : a title, rows (one a line, cells split by `, or pasted from a spreadsheet with tabs), the first row a bold header if ticked; each column as wide as its widest cell. {material}, {title}` and the drawing's other properties fill in. Double-click it to change it or delete it	
Cut List Table	The cut list of the part in the selected view (see the Fast Structure chapter); it updates when the part changes	
Hole Table	The holes of the selected view (circles seen end on) tagged in it, a letter for each size, largest first, numbered from the bottom left (A1, A2… B1), with their X and Y from the view's lower-left corner and their size; it updates when the part changes	
Balloon	Click a part in a view, then where the balloon goes. The number is that part's item in the bill of materials, found from the part under the click (in a sub-assembly, the sub-assembly's item). Keep clicking for more; **Esc** ends. Where the part can't be told, it asks for the number. Right-click a balloon ▸ **Item Number…** to change it.	
BOM Table	The bill of materials of the drawing's assembly: ITEM, QTY, PART and MATERIAL, at the top right of the sheet (or the nearest spot clear of the views, their dimensions and the title block; a hole table and a cut list are placed clear the same way). Drag it anywhere; right-click it to show the sub-assemblies' parts (indented). It follows the assembly. Right-click a column to change what it shows (any field, custom property, mass, cost or typed values), insert, move, rename or remove it; double-click a typed value to edit it. **Columns…** (right-click the table, or double-click it) has everything in one dialog. With mass or cost columns a TOTAL row is added.	

Datum features, geometric tolerances, surface finish and weld symbols can each have a leader. Right-click one ▸ **Edit…** to change it.

21.8.1 Auto Dimension

A drawing made from a part or assembly can be dimensioned as it is made: tick **Dimension the views (Auto Dimension)** in the New Drawing dialog (it remembers your choice); the views are dimensioned once they are drawn, with the options last used.

Annotation ▸ **Auto Dimension** (also **Insert** ▸ **Annotations**) adds the dimensions the selected views need, or every view on the sheet's: the overall sizes, outermost; hole diameters, one per size ("4× Ø9"); arc radii; and hole positions. Each size is given once in the drawing, and nothing is added that a view's own dimensions already give. Only views looking along the part's axes are dimensioned.

It asks how, and keeps your choices for next time:

- **Positions**: baseline (each from the view's edge), chain (edge to hole to hole on one row) or ordinate (values from the view's bottom left corner).
- **Tolerance**, for example ±0.1, added to the new lengths and diameters.
- **Spread labels that would overlap**.
- **Add**: which kinds (overall sizes, hole diameters, hole positions, arc radii).

Everything it adds is one undo step; drag the dimensions to tidy them up.

21.8.2 Check Drawing

Annotation ▸ Check Drawing (or **Tools ▸ Check Drawing...**) lists what the drawing still lacks before it goes out, most important first:

- **Title block**: fields left empty. Title, drawing number, drawn by, date, revision and material are to fix; checked and approved are noted, to be filled in at sign-off.
- **Dimensions per view**: part views with no dimensions (a view whose sizes all show in other views is only noted), crowded views, annotations whose view was deleted and dimensions that lost what they measured.
- **Undimensioned features**: overall sizes, hole diameters, radii and hole positions that no dimension gives. A size given in one view is not asked for again in the part's other views, a hole callout gives a hole's diameter, and hole positions count in baseline, chain or ordinate style.

Click an item to show its view or annotation. **Add Missing Dimensions** puts in what Auto Dimension would; **Title Block...** opens the drawing's properties.

21.8.3 DXF drawings on the sheet

Insert ▸ DXF Drawing... places a DXF file's lines, circles, arcs, polylines, ellipses, splines and texts on the sheet as one item, in the middle of the frame at full size or scaled down to a standard scale that fits. Drag it, right-click ▸ **Scale...**, or delete it; it prints and exports with the sheet.

21.9 Sheet tab

- **Sheet Properties**: size, orientation, scale, first or third angle.
- **Title Block** (Drawing Properties): title block fields, text height and decimals.
- **Units**: click **MMGS / IPS** in the status bar. An inch drawing gives lengths in inches with three decimals and its title block says "in"; angles are not converted.
- **Change Template**: see Drawing templates above.
- **Add Sheet**: drawings can have several sheets, shown as tabs under the sheet.
- **Export PDF**: vector PDF, one page per sheet.
- **Export DXF**: the current sheet as a DXF file, with layers for visible, thin, hidden and centre lines, text and hatching. A sheet format's logo is left out.

21.10 Views follow the part

Views and dimensions follow changes to the part or assembly, whether it is open in another tab or read from its file. A copy of the part is kept in the drawing so it still opens if the part file is missing.

21.11 Not yet available

DWG import and export: DWG is a closed format with no free reader or writer that could run in RazorCAD, so drawings are exported, and DXF files placed, as DXF, which practically every CAD program reads and

writes.

22 Evaluate and Tools

22.1 Measure

Evaluate ▸ Measure, **Tools ▸ Measure...** or the view's right-click menu. Click vertices, edges, faces, planes or sketch lines; with two items picked it shows the distance between them plus dX, dY and dZ. Use the body filter (**B**) to pick whole bodies. Sizes are in the document's units.

- **Snap To**: besides vertices, a click picks **sketch points**, the **midpoints of edges** and the **centres of circles and arcs** (a hole's centre too); each can be turned off. As you move, the point a click would pick is marked on the model with its name.
- The items picked are numbered **1** and **2** on the model.
- **Keep the first item** measures from the first item to each next click (a third click replaces the second, not the first).
- **Measured** lists the distances measured so far, most recent first.

22.2 DFM Check

Evaluate ▸ DFM Check checks the solids for machining : **sharp inside corners** (a concave straight edge between flat faces: a cutter leaves its radius there), inside rounds smaller than the **smallest cutter radius**, holes smaller than the **smallest hole** or deeper than a number of times their diameter, hole sizes that are not standard drills (whole or half millimetres up to 20, whole millimetres above: reamed, bored or clearance holes are reported as notes), and very small faces (often import slivers). Each finding is listed and numbered on the model; **Clear the marks** removes them. For thin walls use Thickness Analysis, for moulded parts Draft Analysis and Undercut Check, for sheet metal Sheet Metal Check.

22.3 3D Annotations

Evaluate ▸ 3D Annotation puts product and manufacturing information on the model itself (model based definition) : a **Note**, a **Dimension** between two points (its value measured, or given) with its tolerances, a **Geometric tolerance** (a feature control frame: the characteristic, a Ø zone if ticked, the tolerance and the datums), a **Datum** letter or a **Surface finish** (Ra). Fill in the dialog, then click the point on the model it is about (two points for a dimension). Each shows as a label with a leader to its point and follows the view. **Tools ▸ 3D Annotations...** lists them, to delete them or hide them all. Exported to STEP (whole part, as AP242 in millimetres: the default of Settings ▸ Export), datums, geometric tolerances and dimensions go into the file as semantic PMI (AP242), each on the face it points at, for programs that read AP242 PMI; notes and surface finishes go in as properties of the part. A STEP file with PMI opened in RazorCAD shows its datums, tolerances and dimensions as 3D annotations.

They are saved with the part, and a **STEP export** of the part carries them: each annotation is written as a property of the part with its kind, its text and the points it refers to (in AP242, AP214 and AP203 files alike). Programs that read STEP properties show them; they are not semantic GD&T that a measuring machine program can use directly.

22.4 Tolerance Stack-Up

Evaluate ▸ Tolerance Stack-Up (also in an assembly's ribbon, and Tools ▸ Tolerance Stack-Up...) works out a gap from a chain of dimensions in one direction . Each row is a dimension: its nominal size, its upper and lower tolerance (both as positive numbers: 50 +0.1 / −0) and whether it **adds to** or **takes from** the

gap. The results update as you type: the gap's nominal and mean, its **worst case** limits (every tolerance at its limit together) and its **statistical** limits (root sum of squares: what most assemblies see when the parts vary independently), checked against the gap required (at least / at most). The share column shows which tolerances matter most. Stack-ups are kept in the part or assembly; New stack-up adds another.

22.5 Mass Properties

Evaluate ▸ Mass Properties shows mass, volume, surface area and centre of mass. It adds up every body with its own density and lists the bodies. A body with no material uses 1.00 g/cm³.

22.6 Interference Check

Evaluate ▸ Interference Check finds where the solid bodies of a part overlap. It lists each pair with its overlap volume and shows the overlaps in red. Assemblies have their own (see the Assemblies chapter).

22.7 Surface analysis

These displays colour the model to show the quality of its faces. Click the button again to turn its display off.

Command	Shows
Zebra Stripes	Reflected stripes on every body. Across a smooth join the stripes run on without a kink.
Curvature	Bodies coloured from blue (flat) through green and yellow to red (tightest). The status bar gives the radius red stands for.
Draft Analysis	Choose a pull direction (an axis or a selected face's normal) and the draft needed. Faces turn green with enough draft, yellow where they need more, and red with draft the other way.
Thickness Analysis	Give the thinnest and thickest walls allowed. Each point is coloured by the wall thickness measured from it straight in through the material: red thinner than allowed, blue thicker (sinks and voids in a moulding), green to yellow between, grey where no wall was found. The status bar gives the thinnest and thickest walls and the share of the surface outside the limits.
Undercut Check	Choose a pull direction. Each point is looked at along it and against it: green can be pulled out along it (one mould half), blue against it (the other), red neither way (an undercut, needing a side action or a change). The status bar gives the share of the surface undercut.

22.8 Configurations

Evaluate ▸ Configurations opens the design table (see the Configurations chapter).

22.9 Equations and global variables

Tools ▸ Equations… holds global variables that can be used in any dimension or feature value. Example: `width = 120`, then set a depth to `width/4`. Each row shows the value it evaluates to; invalid or duplicate names are flagged.

22.9.1 Dimension IDs

Every dimension has a short ID of its own: each sketch and 3D sketch dimension (reference ones too) and each value of a feature (an extrude's depth, a fillet's radius…). IDs are given when the dimension is made:

D1, D2… from a counter that never goes back, so an ID is never given twice in a part. The ID shows as a small tag by the value in a feature's panel, in a 2D sketch dimension's Modify box, and in the tooltip of a dimension in the sketch. Click the tag to **rename** it (e.g. `Width`); every value that uses it is changed to the new name.

Use an ID in any value as you would a global variable: set a depth to `D3 / 2` or `Width / 2` and it follows that dimension. (A reference dimension's ID can't drive others yet: its value is measured, not set.) IDs are also what a design table can list.

Deleting a dimension others use : a warning lists what uses it, with **Undo** to keep it. Deleted anyway, the values that used it fail: they hold its last value (so nothing moves), they show in red (a sketch dimension, a feature's value), and a feature with such a value shows as failing in the tree. Change them to drop the deleted ID and the failure goes. Hover a failing feature in the tree to see why; several reasons are shown as bullet points.

Every value field accepts:

Item	Examples
Operators	`+ - * / ^ %` and parentheses
Variables	Any name from Equations
Trigonometry (degrees)	`sin cos tan asin acos atan`
Other functions	`sqrt abs min max round floor ceil pow`
Constants	`pi`
Units of length	`2in`, `30 mm`, `1.5cm`, `2ft`, `(w/2)*in` (see Units)

A value may start with `=`. Variable names start with a letter or underscore and can't be a function name.

22.10 Macros

Tools ▸ Macros… writes, runs and saves JavaScript macros that drive RazorCAD.

- In parts they can sketch (lines, rectangles, circles, arcs, polygons), extrude, cut, revolve, add any feature, change or delete features, set equations and the material, and rebuild. In assemblies and drawings they can add, change and remove components, mates, views and annotations.
- Anywhere they can run any command by its name in the command search, ask you for a value, and run other macros.
- The **API** button lists what a macro can use; a sample macro shows the idea.
- A macro's changes are one undo step. Errors show their line, which is selected in the editor. **`Ctrl+Enter`** runs; **`Ctrl+S`** saves.

Tools ▸ Record Macro records what you do until **Stop Recording Macro** (or the red "● Recording macro" in the status bar): the features you add, change, delete or reorder, the commands you run, and in assemblies and drawings what you change there (parts inserted, components moved, mates, views, annotations, sheets). The recording opens in the Macros dialog to be saved or edited. Items made during the recording are kept in variables and others are found by name, so a recording plays back into another part, assembly or drawing.

Saved macros are in the Tools menu (**Macro: name**) and the command search, so they can have keyboard shortcuts. **Import…** / **Export…** share them as .js files.

> **Warning:** Macros can do anything RazorCAD can. Run only macros you trust.

22.11 Edit Material

Tools ▸ Edit Material… (or the Material item in the tree) sets the part material. See Appendix B for the list.

23 Modelling with Claude

RazorCAD can build features from a plain-language description using Anthropic's Claude.

23.1 Setting up

Open **Tools** ▸ **Claude Settings...** (or the Claude tab ▸ **Claude Settings**) and choose how RazorCAD reaches Claude:

- **My Claude subscription (Claude Code on this PC)**: each request goes through the Claude Code command-line program installed on this PC, signed in to your own Claude Pro or Max plan. It counts toward that plan's usage limits, with no separate bill. The dialog shows where Claude Code was found, its version and whether it is signed in; **Sign In...** starts the sign-in, which you finish in the browser, and **Check Again** reads the status again. Claude Code runs with no tools, settings or memory, from an empty folder, so it sees only RazorCAD's modelling guide and your request.
- **An Anthropic API key**: enter your own key (create one at console.anthropic.com ▸ API Keys; requests are billed to that account). The key is stored on this PC in `%APPDATA%\RazorCAD\claude.json` and is only sent to api.anthropic.com. **Remove Key** deletes it.

Then choose a model: Opus, Sonnet or Haiku.

23.2 Ask Claude panel

Open it with the **Claude** tab ▸ **Ask Claude**, or **Tools** ▸ **Ask Claude...**.

- Describe a part or a change. Claude adds the features to the tree; each answer is **one undo step**.
- If the features don't build, Claude is told the error and corrects it, up to two more tries.
- Claude sees the feature list and body sizes of the part, not the picture.
- **New Conversation** (or **New chat** in the panel) starts fresh; the part stays as it is.
- The model can be switched in the panel.

23.3 RazorCAD Part Builder on claude.ai

A companion page on claude.ai lets you describe a part, see the feature list with sketch previews, and download it as a .rcad file to open in RazorCAD. It uses the same modelling guide as the in-app panel.

24 Troubleshooting

24.1 Crash reports

RazorCAD writes a crash report whenever something goes wrong, to `%APPDATA%\RazorCAD\crashes\<date_time>_<kind>\`. Open the folder with **Help ▸ Open Crash Reports Folder**. The start window mentions new reports. The newest 30 are kept.

File	Contents
`report.json`	The kind of failure, the error message and stack trace, what RazorCAD was doing (mode, Property panel, sketch tool, selection, feature errors) and system information
`session.log`	The recent actions of the session, newest last
`part.rcad`	The part that was open, so the problem can be reproduced

Kinds of report: `ui-error`, `kernel-error`, `kernel-worker-crash`, `python-exception`, `native-crash` and `unexpected-shutdown` (a session that ended without closing normally, found at the next launch).

24.2 Reporting a problem

Help ▸ Report a Problem..., or **Report on GitHub...** in the start window's crash notice, turns a crash report (or a plain description, without a crash) into a GitHub issue. The report holds RazorCAD's version, Windows, what went wrong and the last lines of the session log; you see the whole text first and can edit it. Your user name and folders are taken out, and the part you were working on is not included. **Open GitHub Issue** opens RazorCAD's new-issue page in your browser with the text filled in; nothing is sent until you submit it there. **Copy Text** copies it instead.

24.3 Speed on your PC

How many geometry kernels work at once comes from the PC RazorCAD runs on: one per processor thread beyond two (left for the window and the rest of the PC), and about 1 GB of free memory each beyond 2 GB, read from Windows when RazorCAD starts; at least 2, at most 16. Assemblies build their parts side by side in them, drawings draw their views in them, and each open part keeps one (up to half of them). Kernels that are idle for a few minutes are stopped to give their memory back. A big imported STEP file's bodies are meshed by several kernels at once when that is quicker (64 bodies with 6784 faces: 1.8 s instead of 5.4 s). Crash reports note the figures, for example "up to 12 at once (24 threads, 14.8 GB free)".

Each open part's kernel is **multi-threaded**: Open CASCADE splits booleans (cuts, joins and the features built on them) and meshing over several threads, and faces and edges are read and boxed in parallel too. The first part you open gets all the processor threads but two (at most 22); each further part, assembly or drawing kernel gets half the threads beyond two, from 2 to 8. On a PC with 4 or 5 threads the first part gets all but one and the others one thread; with 2 or 3 threads the first part gets two (about 30 % quicker than one on two cores) and the others one. Parts full of cuts and large imports gain most: 16 test parts rebuild about three times faster than on one thread (on a PC held to 4 threads, 9.2 s instead of 16.3 s). The browser version at razorcad.com/app does the same in a current browser (the website serves it cross-origin isolated, which shared memory needs). The session log says which kernel started ("Geometry kernel ready: multi-threaded, 22 threads").

Cuts made together: a run of cut features (extrude and revolve cuts with a blind, mid-plane or through end, holes other than tapped ones, and patterns and mirrors of cuts) is made in one boolean instead of one after another, each on a body with more faces than the last. The cuts wait until a feature needs the

body as it really is (a fillet, a hole ending on a face, anything that picks model geometry) or the rebuild ends. The Giant Golfball (178 dimples) rebuilds in 2.5 s instead of 4.7 s, with exactly the same result. Editing the last cut feature makes only its own cut again: the rest is remembered.

Editing a feature meshes again only the faces and edges it changed, and a part reopened from the model cache picks up where it left off without being built again.

24.4 Common problems

Problem	What to do
A fillet or projected edge loses its edge after a big change	Edges made by an extrusion are found by what made them; others by their geometry and the faces beside them (see References that follow edits). If one is still lost, **Tools ▸ Fix References** lists it: Edit to re-pick it, or Remove missing.
A mate moved to the wrong face, or shows a red !	Mates find their faces again by geometry. A face that is gone is listed in **Tools ▸ Fix Mates**: edit the mate to pick it again.
A sketch on a model face doesn't follow the face	Sketches put on faces before version 0.20.1 keep only the plane: right-click the sketch ▸ Change Sketch Plane… and click the face again. If the note says the face is no longer in the model, pick the face it should be on.
Project Edges can't see a sketch	It only sees visible sketches earlier in the history. Show the sketch first.
An STL body can't be filleted	STL bodies with thousands of faces can't be filleted or chamfered and don't offer pickable edges.
A fillet won't blend across two touching bodies	Combine the bodies first.
Open fails with a file filter error	Fixed in 0.9.0-beta. Update RazorCAD.
Unsaved work after a crash	Reopen the part and accept the auto-save recovery.

24.5 Settings and data files

Location	Contents
`%APPDATA%\RazorCAD\recent.json`	Recent files list
`%APPDATA%\RazorCAD\claude.json`	Claude settings, API key and model
`%APPDATA%\RazorCAD\prefs.json`	Toolbar layout, icon size, your keyboard shortcuts, whether the tour was taken, and other preferences
`%APPDATA%\RazorCAD\update.json`	The update check: on or off, when it last looked, a version you chose to skip
`%APPDATA%\RazorCAD\crashes\`	Crash reports
`%LOCALAPPDATA%\RazorCAD\modelcache\`	Built models of parts and assemblies, and the geometry after each feature, for fast opening and editing, one folder per version of the geometry kernel (up to 1 GB; safe to delete)
`Documents\RazorCAD\Templates\`	Your drawing templates
`Name.autosave.rcad`	Auto-save, next to the part file

25 For Developers

25.1 Rebuilding the executable

```
python -m pip install pywebview pyinstaller pillow
build.bat
```

`build.bat` bundles `app/` (including the 23 MB OpenCascade WebAssembly kernel) and `razorcad.py` into `RazorCAD.exe` with PyInstaller. If RazorCAD is open, the running copy is renamed to `RazorCAD.old.exe` and the new build is used from the next launch.

`installer\build_installer.bat` builds `RazorCAD.exe` and then `RazorCAD-Setup.exe`. Set the version in `version.txt` first.

25.2 Code signing

Both builds sign RazorCAD.exe, the thumbnail DLL and the setup once a code-signing certificate is set up; until then they say "Not signed" and carry on. `tools\sign.py --check` shows how signing is set up. `installer\SIGNING.md` explains how to get a certificate and set it up (a certificate on a USB token or in Windows by its thumbprint, a .pfx file, or a cloud signing service).

25.3 Regression tests

```
python tests/run_tests.py
```

This rebuilds the test parts in `tests/parts` with the real geometry kernel in headless Microsoft Edge. It compares every body's volume, area, faces and edges (and flat patterns) with `tests/expected.json`, and the drawings of six test parts with their stored line counts and sizes. It also runs checks done as a user does them: sketching, dimensioning and extruding, drawings, assemblies, mates, references, standard parts, collisions, assembly features, macros, accessibility and more. It needs Python's playwright package and exits with 1 on any difference. `--update` stores new expected results, `--only` runs some of them, and `--make-parts` makes the test parts again from `tests/make_parts.js`. On a PC with 12 or more processor threads the run is split over several pages side by side (`--jobs N`; `--jobs 1` runs everything in one page): about 27 s instead of 65 s on 24 threads. See `tests/README.md`.

- **Timings**: every run's times are kept per machine in `tests/perf-history.json` (not in git). A test markedly slower than its usual time is listed under **Performance**; `--perf` lists every time and `--perf-strict` fails the run on a slowdown.
- **On GitHub** the tests run for every push and pull request (`.github/workflows/tests.yml`), split over 6 machines side by side (`--shard I/N`, shared out by the times in `tests/test-weights.json`): some 15 minutes.

25.4 Releases

Pushing a version tag (for example `v0.14.0-beta`, matching `version.txt`) runs `.github/workflows/release.yml`: the tests, then RazorCAD.exe and the setup are built and a draft release is made with the setup, its read-me, checksums and notes from `CHANGELOG.md` (`tools/release_notes.py`). A version that already has a release is left alone. See `installer/RELEASING.md`.

25.5 The geometry kernel builds

`app/vendor/replicad` holds two builds of the same Open CASCADE bindings: `replicad_single` (one thread) and `replicad_multi` (threads, with shared memory). Both are built from source: Open CASCADE 8.0.1 with two speed patches (the same results), opencascade.js 3.1.0 (Emscripten 6.0.5) with whole-program optimisation, replicad's bindings (commit a0bc495) with RazorCAD's additions, and `tools/native/surface-patch.cpp` (`tools/native/README.md`); `tools/patch_surface_kernel_loader.py` adjusts their loaders. The kernel worker loads the multi-threaded one when the app asks for more than one thread and the page is cross-origin isolated: the desktop app's server, `serve.py` and the tests' server send `Cross-Origin-Opener-Policy: same-origin` and `Cross-Origin-Embedder-Policy: require-corp` for that. The website takes files up to 25 MiB, so `tools/build_web_app.py` publishes a bigger kernel in parts, which the worker joins.

25.6 The website and the browser version

`python tools/build_website_docs.py` builds the online manual (`website/docs`) from this manual's source and the browser version (`tools/build_web_app.py`). The browser version is built in its own folder, `RazorCAD Web` next to the project folder, with both kernels (`website/_headers` serves it cross-origin isolated, so it runs the multi-threaded one where the browser can), and copied from there into `website/app`; it carries the geometry kernel's signature so the browser keeps stored models per kernel.

`python tools/check_site.py` checks the website as it is served (razorcad.com, or another address): its pages, the browser version's isolation headers, and the browser version itself in headless Edge (it starts, and a multi-threaded kernel builds a part). GitHub runs it after each push that changes the website (`.github/workflows/site-check.yml`).

25.7 Working on the UI

Run `python serve.py` and open http://localhost:8765/ in a browser. In that mode files are downloaded instead of saved through dialogs. Set the `RAZORCAD_DATA_DIR` environment variable to keep a test run's recent-files list separate from your own.

25.8 Project layout

Path	Purpose
`RazorCAD.exe`	The application (built by build.bat)
`razorcad.py`	Desktop shell: native window, file dialogs, auto-save (pywebview)
`app/js/main.js`	Application controller: documents, commands, selection, files
`app/js/ui.js`	Menus, Command ribbon, Feature tree, Property panel, dialogs
`app/js/model.js`	Feature definitions, undo history, materials, sample part
`app/js/kernel.worker.js`	Geometry kernel worker: replays the history with OpenCascade
`app/js/sketcher.js`, `sketchmodel.js`, `solver.js`	Sketch mode, sketch data and the constraint solver
`app/js/sketch3d.js`, `sketch3dmodel.js`	3D sketches
`app/js/sheetmetal.js`	Sheet metal features and flat patterns

Path	Purpose
`app/js/weldments.js`, `profiles.js`	Fast Structure members and the profile tables
`app/js/assembly.js`, `asmview.js`, `explode.js`	Assemblies, their display and exploded views
`app/js/drawing*.js`, `drawtemplates.js`, `pdf.js`, `dxf.js`	Drawings, templates, sheet formats, PDF and DXF export
`app/js/claudespec.js`	Modelling guide used by Ask Claude
`app/js/collide.js`, `asmfeatures.js`, `asmcuts.js`, `motion.js`, `stdparts.js`	Collision detection, assembly holes and cuts, motion studies, standard parts
`app/js/macros.js`, `drawcheck.js`, `autodim.js`, `compare.js`	Macros, Check Drawing, Auto Dimension, Compare With
`app/js/a11y.js`, `machine.js`, `kernelclient.js`	Keyboard use and screen-reader labels; how many kernels the PC can run; the kernels
`kernelsig.py`	The geometry kernel's signature (the model cache is kept per kernel)
`Manual/`	This manual: source, build script and change tracking

Appendix A: Keyboard and Mouse Reference

General

Action	Keys
New / Open / Save	**Ctrl+N** / **Ctrl+O** / **Ctrl+S**
Save As	**Ctrl+Shift+S**
Import STEP / STL	**Ctrl+I**
Undo / Redo	**Ctrl+Z** / **Ctrl+Y**
Rebuild	**Ctrl+B**
Delete feature or entity	**Del**
Accept / cancel Property panel	**Enter** / **Esc**
Edit a feature	Double-click it in the tree
Search commands	**Ctrl+K** or **/**
The manual (at the feature being edited)	**F1**
Your own shortcuts	Help ▸ Customize Shortcuts
Focus to the panel, ribbon or view	**F6** / **Shift+F6**
In a tree: move / close or open / open the item / first or last	↑ ↓ / ← → / **Enter** / **Home End**

View

Action	Keys
Zoom to fit	**F**
Front, Back, Left, Right, Top, Bottom, Isometric	**Ctrl+1** … **Ctrl+7**
Normal To	**Ctrl+8**
Rotate in steps	Arrows 15°, **Shift** + arrows 90°
Pan in steps	**Ctrl** + arrows
Roll	**Alt** + ← / →
Hide body under cursor / show hidden body under cursor	**Tab** / **Shift+Tab**

Selection filters

Filter	Key
Faces / Edges / Vertices / Bodies	**X** / **E** / **V** / **B**
Clear filters	**F5**

Sketch

Tool	Key
Select	**S**
Line	**L**
Rectangle	**R**
Circle	**C**
Arc	**A**
Spline	**N**
Polygon	**P**
Trim	**T**
Dimension	**D**
Project Edges	**E**
Offset	**O**
Construction Geometry	**Q**
Delete	**Del**
End tool, then exit sketch	**Esc**
Tool wheel	Hold the right button (or drag with it)

3D sketch

Action	Keys
Next drawing plane (XY → YZ → ZX), or back from a picked plane	**Tab**
Draw off the X/Y/Z directions	Hold **Ctrl** or **Shift**
Draw on a plane or flat face	**Ctrl**+click it
Length dimension: true length → ΔX → ΔY → ΔZ	**Tab** while placing
Tools	Same keys as 2D sketches: **S**, **L**, **C**, **A**, **N**, **P**, **D**, **T**, **E**, **Q**

Assembly

Action	Keys
Mate	**M**
Move a component / turn it	Left-drag / **Alt**+drag
Motion study: play / stop	**Space**
Zoom to fit	**F**

Appendix B: Materials

Material	Density (g/cm³)
Plain Carbon Steel	7.80
AISI 304 Stainless	8.00
6061 Aluminum Alloy	2.70
Brass	8.50
Copper	8.96
Ti-6Al-4V	4.43
ABS Plastic	1.04
PLA Plastic	1.24
Nylon 6/6	1.14

A body or part with no material uses 1.00 g/cm³ for mass calculations.

Appendix C: Manual Revision History

Date	Software version	Changes
2026-11-02	0.21.3-beta	Mechanical mates grouped under Standard, Advanced and Mechanical Mates: Hinge (with a turn limit), Universal Joint and Linear Coupler, with icons for Screw, Slot and Cam. Motion studies with more than one driver. Flexible sub-assemblies: mates stay on their parts and dragging moves the parts. Holes placed on the floor of a cut.
2026-11-01	0.21.2-beta	Holes, bosses and snap hooks keep to their faces as the part changes; references found again as the nearest look-alike marked to check; curved faces followed when they move. Sizes typed while drawing arcs, ellipses, slots and polygons; the sketch's own undo history. Drawings: views and their projected views scaled together, detail views on their own scale, Break from Parent View, Move to Sheet, Break Lines (was Break View), views and tables placed clear of dimensions, a scale or sheet offered when a view doesn't fit. Bolts through stacked plates; flat patterns named after the part; body colours in STEP. Exact spline surfaces, Split Body and Split Face, lofted surfaces from model edges.
2026-10-31	0.21.1-beta	Appearances and Render tabs: more scenes and places, photographed scenes, spotlights, the perspective view. Transition / Reducer spigots (round ones rolled, flat-sided ones bent, in parts of their own) and slip-on flanges. End Flange: bolting flanges on the ends of any parts, several at once. Several bodies selected with Ctrl, Edit Material on the body right-click menu, and a right-click anywhere for the selection's menu. A Settings button with export settings; deleting a feature keeps its sketches; partial route paths; construction lines.
2026-10-30	0.21.0-beta	Dimension IDs: every dimension and feature value has an ID that other values can use, renameable, with a warning before deleting one that is used. Assemblies: Edit Part from the 3D view or the tree, parts and sub-assemblies expand to their features, references between parts, no circular references. Structural Member: segments picked in groups, each with its own profile and settings, and profiles that agree around a frame. Transition / Reducer and Lofted Bend with bolted flanges; a Transition on one line of a sketch or 3D sketch. Sketches on faces follow their face; references to edges follow by name. Features that fail are not added to the tree. 3D sketch dimensions shown while placed. More printers in the 3D Print list.
2026-10-29	0.20.0-beta	A 3D Print tab: print profiles with a list of popular printers, the build orientation (Place on Bed, Auto-Orient, Show Bed), Check Printability, Overhang and Thickness Maps, Export for Printing laid out on the bed and Open in Slicer; Nut Pocket, Insert Hole, Screw Hole for Plastic, Printable Hole, Bridging Layer, First-Layer Chamfer, Printable Overhangs, Fit Clearance, Print-in-Place Hinge, Split for Printing and Calibrate Fits. Intersection Curve through splines, reference points on curved edges and faces, outward shells, G2 fillets on circular edges, several loft connectors, Freeform on faces of any outline, and dished ends laid out flat in sheet metal.
2026-10-29	0.19.0-beta	A new geometry kernel (k14): Delete and Resize Fillet, Defeature and Delete Face Heal take faces out exactly, and STEP files carry datums, geometric tolerances and dimensions as semantic PMI. The manual inside RazorCAD ([F1]), docked or in a window of its own. Freeform faces, G2 fillets, Boundary Surface tangent or curvature continuous to faces. Plastic parts and moulds (bosses, snap hooks, core and cavity, parting surfaces, side cores, runners), direct editing, library features, pipe routes with tees, reducers and spool drawings, duct fittings with flat patterns, cables and harness formboards. Assemblies: Edit in Context, slot, path and cam mates, clearance along a motion study, flexible sub-assemblies, Where Used, Number Parts. Drawings: hole tables, broken-out sections, Compare With. 3D annotations, Render, revisions, true-shape nesting, glTF export, point clouds. Extrude and Revolve: face profiles, second sides, draft and Intersect.

Date	Software version	Changes
2026-10-28	0.18.2-beta	The browser version never uses an older cached geometry kernel after an update. Line and Centerline infer parallel and collinear relations, and Project and Intersection can pick from one chosen sketch. The updater checks that the new version installed before it restarts RazorCAD. STEP products with their own solids as well as child parts keep those solids. A repeated Measure on a curved face answers at once.
2026-10-28	0.18.1-beta	Starts reliably on PCs with few processor threads. On PCs with 4 or 5 threads the open part's geometry kernel is multi-threaded (rebuilds about 40 % faster). Guided lofts with CAD guide constraints build up to 16 times faster. The kernel chapter for developers brought up to date.
2026-10-28	0.18.0-beta	A faster geometry kernel: full rebuilds about three times quicker, edits about twice. Guided sweeps and lofts, loft end conditions up to G2, guide constraints. Filled Surface with G0, G1 or G2 on each edge. Sketch offsets, tangency where lines meet splines. Extrude and Revolve: profiles picked in the view, two sides, start offsets. Reference plane, axis and point constraints; planes and points along paths.
2026-10-26	0.17.4-beta	Settings window. Variable fillet points dragged along their edges. Assemblies keep their own up axis. Sketch Fillet where a spline meets a line or an arc. Feature folders, and patterns of them. Developed sheets: the neutral layer on the inside of each face's curve. An experimental Linux build.
2026-10-26	0.17.3-beta	Lofted Surface guide curves (rails). Patterns of more features, faces and bodies. Measure snaps (sketch points, midpoints, centres), numbered picks, keep the first item, history. Refused relations say why. Guided Loft follows its guides more closely; Bézier profile reversal and the Lofted Surface closed check fixed. Contributions merged: parametric sketch patterns with a side panel, draggable dimension labels, Equal Curve Length and Coincident Midpoints relations, loft profiles and guides picked on the model, grouped sketch tools and slots, parameter autocomplete.
2026-10-25	0.17.2-beta	Edge-Flange: corners corner to corner with square ends by default (each flange to the other's inside face, gap 0.5 mm, bends in straight V cuts); mitred kept as an option; no split or sliver faces at the corners. Up axis Y or Z per part (View, status bar, the Origin's right-click menu). Variable fillets: points along the edges. Rips as one cut line. Sketch box selection, splines to degree 7, Sketch Fillet on arcs, sketch patterns along a curve, plane references from sketch entities. An example part and a tutorial. One file dialog at a time.
2026-10-25	0.17.1-beta	Edge-Flange: flanges meet corner to corner by default. Control-point splines: closed smoothly, weights, control points inserted and removed, splines converted. Loft guide curves move holes and thin walls too. Convert to Sheet Metal: flanges flush against the material beside them ripped; a wrapped blade unrolled.
2026-10-24	0.17.0-beta	Control-Point Spline (degree 1 to 5, its polygon dimensioned and constrained). Midpoint of an arc. Reference planes: through an edge and a point, through two edges, at a distance along an edge, tangent to a cylindrical face; placement for every plane. Loft guide curves. Convert to Sheet Metal: square-to-round transitions, cones and closed tubes unrolled, exact corners. Section views saved with the part. STEP, IGES and mesh files opened from Windows are imported.
2026-10-21	0.16.0-beta	Section view: from a selected face or plane, a drag arrow, in assemblies, two or three planes at once. The view turns about the point under the cursor. Big imports kept and read back quicker; Measure quicker on sheet metal.
2026-10-21	0.15.0-beta	Speed on your PC: the multi-threaded geometry kernel and cuts made together. For Developers: the kernel builds; the browser version built in its own folder.

Date	Software version	Changes
2026-10-21	0.14.0-beta	New sections: RazorCAD in your browser, Update Now, Language and theme, Using the keyboard, Properties, Compare With, IGES import, Etch Lines, Standard parts, Collision detection, Equations shared with the parts, Assembly holes and cuts, Motion studies, Speed Mode, assembly STEP export, Auto Dimension, Check Drawing, DXF drawings on the sheet, Macros, Reporting a problem, Speed on your PC, Releases and the website. Updated: bill of materials columns, mass and cost; fillets set at each vertex and setback corners; curvature-continuous 3D splines and loft continuity; the fonts that come with RazorCAD; the model cache; tests.
2026-10-20	0.13.0-beta	New sections: Updates, Getting Started Tour, Search commands, Your own keyboard shortcuts, Units (inches), Cosmetic Thread, References that follow edits, Fix References, gear and rack-pinion mates, Fix Mates, assembly Interference Check, Code signing and Regression tests. Drawings add Balloon and BOM Table, exploded views and units; Hole callouts give cosmetic threads. Faster opening (the geometry after each feature is kept) and faster patterns of added features.
2026-10-20	0.12.0-beta	Brought up to date with the software. New chapters: 3D Sketches, Airfoils, Fast Structure, Configurations and Assemblies. Sheet Metal adds Sheet Metal Extrude, Lofted Bend, Transition / Reducer, Miter Flange, Hem, Sketched Bend, Jog, Cross Break, Tab and Slot, Forming Tools, Gusset, gauge tables and Sheet Metal Check. Drawings add templates, auxiliary, break and crop views, baseline, ordinate and chamfer dimensions, hole callouts, GD&T, surface finish and weld symbols, revision and cut list tables, bend tables and DXF export. Surfaces add lofted, swept, filled, boundary and ruled surfaces, trim, extend, untrim, delete and replace face, split line, projected and composite curves, flatten surface, convert to solid and repair mesh. Also: new sketch tools, Hole, Draft, Rib, Dome, Move Face, Scale, new patterns, appearances, exploded views, interference check, surface analysis, OBJ and 3MF, toolbar customising and the Claude subscription option. Feature and tool names follow the software's new names.
2026-10-13	0.9.0-beta	First edition. Covers parts, sketching, gears, solid features, bodies, surfaces, reference geometry, the feature tree, sheet metal (including Convert to Sheet Metal with detected thickness and bend radii), drawings, evaluate tools, Ask Claude, troubleshooting and developer notes.

Index

Page numbers are links: click one to go to that page.

.

.rasm . . . 19, 87
.rcad . . . 19
.rdrt . . . 95
.rdrw . . . 95

3

3D annotations . . . 102
3D Print tab . . . 82
3D printing . . . 82
3D sketch . . . 36
features that use it . . . 38
relations . . . 37
tools . . . 37
3MF . . . 20

A

Accept Found References . . . 69
Accessibility . . . 17
Airfoil . . . 40
Alternate position . . . 97
API key . . . 106
Appearance
imported . . . 21
Appearances . . . 55
order . . . 56
Appearances tab . . . 55
Arc . . . 30
Ask Claude . . . 106
Assemblies . . . 87
Assembly
edit part . . . 91
feature tree . . . 91
moving components . . . 89
opening quickly . . . 21, 94
planes . . . 87
saving parts as files . . . 93
selecting . . . 89
Assembly Cut . . . 92
Assembly Hole . . . 92
Auto Dimension . . . 99
Auto-Orient . . . 82
Auto-save . . . 20
Auxiliary View . . . 97

B

Balloon . . . 99
Base Flange . . . 71
Baseline dimension . . . 99
Bead
sheet metal . . . 75
Bend . . . 76
Bend allowance . . . 71
Bend deduction . . . 71
Bend table . . . 97
Bill of Materials . . . 93
Blocks . . . 30
Bodies . . . 55
selecting several . . . 25
Body Properties . . . 55
BOM Table . . . 99
Boundary Surface . . . 60
Break from Parent View . . . 97
Bridging Layer . . . 84
Broken view . . . 97
Broken-out section . . . 97
Browser version . . . 13
Build orientation . . . 82
build.bat . . . 109

C

Cable . . . 81
Calibrate Fits . . . 85
Cam mate . . . 89
Captive nut . . . 83
Center mark . . . 99
Chamfer . . . 52
vertex . . . 52
Change Sketch Plane . . . 35

Check Drawing . . . 100
Check for Updates . . . 14
Check Printability . . . 83
Circle . . . 30
Circular Pattern
  options . . . 51
Circular reference . . . 91
Claude . . . 106
Claude subscription . . . 106
Clearance . . . 93
Closed Corner . . . 75
Code signing . . . 109
Collect Files . . . 94
Collision detection . . . 90
Combine . . . 58
Command ribbon . . . 15
Command search . . . 16
Compare With . . . 19
Component pattern . . . 90
Composite Curve . . . 65
Configurations . . . 86
Configurations tab . . . 86
Construction geometry . . . 30
Construction lines
  showing . . . 24
Context menus . . . 26
Control-point spline . . . 33
Convert to Sheet Metal . . . 76
  rips . . . 76
Convert to Solid . . . 65
Copy Surface . . . 60
Copying an assembly . . . 94
Core and Cavity . . . 53
Corner Relief . . . 75
Cosmetic Thread . . . 50
Cost . . . 93
Counterbore . . . 48
Countersink . . . 48
Crash reports . . . 107
Crop View . . . 97
Cross Break . . . 74
Curvature . . . 103
Curvature continuous fillet . . . 52
Curve Driven Pattern . . . 50
Custom properties . . . 55
Cut length . . . 77
Cut list . . . 80
Cut List Table . . . 99
Cuts made together . . . 107

D

Dark theme . . . 16
Datum . . . 99
Decal . . . 58
Defeature . . . 54
Delete Body . . . 59
Delete Face . . . 64
Delete Fillet . . . 54
Design table . . . 86
Detail View . . . 97
DFM check . . . 102
Dimension . . . 30
Dimension IDs . . . 103
  deleted . . . 104
Dimensions
  drawing . . . 99
Dimple . . . 75
Direct editing . . . 54
Display styles . . . 23
Dome . . . 53
Double-click
  sizes . . . 69
Draft . . . 53
Draft (extrude) . . . 44
Draft Analysis . . . 103
Drawing
  fast hidden lines . . . 95
  of an assembly . . . 95
  units . . . 100
Drawing compare . . . 96
Drawing plane . . . 36
Drawing templates . . . 95
Drawing views

placed clear . . . 95
scale . . . 97
Drawings . . . 95
Duct fittings . . . 80
DXF
drawing export . . . 100
face export . . . 22
flat pattern . . . 77
import . . . 33
into drawings . . . 100
DXF preview . . . 78
DXF Settings . . . 78

E

Edge Flange . . . 73
partial . . . 73
Edit feature . . . 69
Edit in Context . . . 91
Edit Material . . . 105
Edit Part . . . 91
Elbows . . . 80
Elephant foot . . . 84
Ellipse . . . 30
Emboss . . . 50
sheet metal . . . 75
End Cap . . . 80
End Flange . . . 73
Equation curve . . . 29
Equations . . . 103
assembly . . . 91
Etch Lines . . . 77
Example part . . . 41
Exploded view . . . 92
drawing . . . 97
part . . . 58
Export for Printing . . . 84
Export settings . . . 16
Extend . . . 30
Extend Surface . . . 64
Extrude . . . 43
Extrude Cut . . . 43
Extruded hole . . . 75

F

F1 . . . 17
Face profile . . . 43
Fast Structure . . . 79
Fasten All Holes . . . 90
Fasteners . . . 87
Feature
left open . . . 15
refused . . . 15
Feature scope . . . 44
Feature tree . . . 15
File associations . . . 20
File previews . . . 20
Fill Pattern . . . 50
Filled Surface . . . 60
Fillet . . . 51
chord width . . . 52
edge radius . . . 52
full round . . . 52
G2 . . . 52
setback . . . 52
variable radius . . . 51
First-Layer Chamfer . . . 84
Fit Clearance . . . 84
Fit Surface to Points . . . 20
Fix Mates . . . 89
Fix References . . . 70
Flange
bolted . . . 72
on the end of a part . . . 73
Flat pattern . . . 76
Flat pattern view . . . 97
Flat patterns
ducts . . . 80
Flatten . . . 76
Flatten Surface . . . 65
Flexible sub-assembly . . . 90
Folder
features . . . 51
Formboard . . . 81
Forming Tools . . . 75

Freeform 64
Freeform face 64
Fully defined 28

G

Gauge tables 71
GD&T 99, 102
Gear mate 89
Gears 39
Geometric tolerance 99
Geometry kernels 107
Getting Started Tour 16
Global variables 103
GlTF 22
Grating 56
Grid 28
Gusset
  sheet metal 75
  weldment 80

H

Harness 81
  formboard 81
HDRI 57
Heads-up toolbar 15
Helix 49
Help 17
Hem 74
Hidden bodies
  selecting 25
Highlight 25
Hinge 84
Hinge mate 88
Hole 48
  end conditions 49
  far-side countersink 49
  follows the face 49
  on a curved face 49
  position 49
Hole Callout 99
Hole Table 99

I

Identical bodies 55
IGES 21
Inches 18
Insert Components 87
Insert Hole 83
Installer 13
Interference Check 103
  assembly 93
Intersect 44
Involute gear 39
Isolate 24

J

Jog 74

K

K-factor 71
Kernel builds 110
Keyboard shortcuts
  customising 17
Keyboard use 17
Knit 64

L

Lance 75
Language 16
Library features 51
Limit mate 89
Line tool 30
Linear coupler mate 88
Linear Pattern
  options 50
Lip / Groove 53
Loft 46
  connector inside a curve 47
  continuity 48
  guide curves 48
  holes paired by hand 47
Lofted Bend 71
  flanges 72
Lofted Surface 60
  model edges 61

Louver 75

M

Macros 104
Manual
in RazorCAD 17
Manufacturability 102
Mass
bill of materials 93
Mass properties 103
Materials 55
Mates 58
assembly 88
MBD 102
Measure 102
Mechanical mates 88
Mesh appearance 56
Mesh simplification 21
Millimetres 18
Mirror 30, 50
Mirror components 90
Miter Flange 74
Model cache 20
Model edges (surface extrude) 62
Model View 97
Motion study 92
Mould 53
Mounting Boss 53
Move Face 54
Move to Sheet 97
Move/Copy Bodies 58
Multi-body 55
Multi-threaded kernel 107

N

NACA 40
Nest Flat Patterns 77
Nesting 77
Normal To 23
Note 99
Number Parts 91
Nut Pocket 83

O

OBJ 20
Offset 31
OK and start another 15
OpenCascade 11
Ordinate dimension 99
Orthographic view 23
Overhang Map 83

P

Pan 23
Parametric modelling 11
Part Number 19
Part numbers 91
Patch
pass through 63
tangent reach 63
Path mate 89
Pattern 50
of a folder 51
of bodies 51
of faces 51
sheet metal 51
skip instances 51
PDF export 100
Performance 107
Perspective view 23
Photographed places 57
Photorealistic 56
Picture
tracing 33
Pipe Route 80
Place on Bed 82
Plastic features 53
PLY 20
PMI 102
Point cloud 20
Polygon 30
Positions
assembly 97
Preferences 15

Preview . . . 43
Print profile . . . 82
Print-in-Place Hinge . . . 84
Printability . . . 83
Printable Hole . . . 84
Printable Overhangs . . . 84
Printer bed sizes . . . 82
Profiles . . . 34
  structural . . . 79
Project Edges . . . 32
Projected Curve . . . 65
Projected View . . . 97
Properties
  custom . . . 19
Property panel . . . 15
Push and pull a face . . . 64

Q

Quick readout . . . 25

R

Rack and pinion mate . . . 89
RazorCAD.exe . . . 13
Realistic view . . . 56
Recovery . . . 20
Rectangle . . . 30
Reducer . . . 72
Reducers . . . 80
Reference arrows . . . 70
Reference axis . . . 67
Reference plane . . . 66
Reference point . . . 67
References . . . 69
  to check . . . 69
Regression tests . . . 109
Relations . . . 33
Releases . . . 109
Render . . . 56
Render scenes . . . 56
Render tab . . . 56
Reorder features . . . 70
Repair Mesh . . . 65
Replace Face . . . 64
Report a Problem . . . 107
Resize Fillet . . . 54
Reverse engineering . . . 20
Revision table . . . 99
Revisions . . . 19
Revolve . . . 45
Rib . . . 53
Ribbon
  More . . . 15
Right-click
  on the selection . . . 26
Rollback bar . . . 70
Rotate view . . . 23
Routing
  cables . . . 81
  part of a sketch . . . 81
  pipes . . . 80
RSS . . . 102
Ruled Surface . . . 60
Runners . . . 54

S

Scale . . . 54
Scan . . . 20
Screw Hole for Plastic . . . 83
Search commands . . . 16
Second side (extrude) . . . 44
Section view . . . 24
Section View
  drawing . . . 97
Segmented elbow . . . 80
Selection . . . 25
Selection box . . . 33
Selection filters . . . 25
serve.py . . . 110
Settings . . . 15
Sheet metal . . . 71
Sheet Metal Check . . . 77
Sheet Metal Extrude . . . 71
Sheet properties . . . 100
Shell . . . 52

hollow . . . 52
wall thickness . . . 52
Shortcuts
your own . . . 17
Show Bed . . . 82
Side core . . . 53
Sizes
changing in the view . . . 69
typed while drawing . . . 27
Sketch
on a face . . . 27
show/hide . . . 35
start . . . 27
typing sizes . . . 27
Sketch Driven Pattern . . . 50
Sketch fillet . . . 30
Sketch orientation . . . 28
Sketch pattern . . . 30
Sketch picture . . . 33
Sketch tools . . . 30
Sketched Bend . . . 74
Slide
mould . . . 53
Slot . . . 30
Slot mate . . . 88
Smart fasteners . . . 90
Snap Hook . . . 53
Speed Mode . . . 94
Spigot . . . 72
Spline . . . 33
degree . . . 33
weights . . . 33
Split . . . 58
sketch . . . 30
Split Body . . . 58
Split Face . . . 65
lofting divided portions . . . 61
Split for Printing . . . 84
Split Line . . . 65
Spool drawing . . . 81
Spotlights . . . 57
Sprue . . . 54
Standard parts . . . 87
Standard Views . . . 97
Standard views . . . 23
Start window . . . 14
STEP
AP203 . . . 16
AP214 . . . 16
AP242 . . . 16, 102
assembly . . . 20, 93
assembly export . . . 93
export . . . 22
import . . . 20
STL
export . . . 22
import . . . 20
Stop . . . 54
Structural Member . . . 79
groups . . . 79
orientation . . . 79
Studio lighting . . . 57
Sub-assembly . . . 91
from STEP . . . 21
Subtract Bodies . . . 58
Surface analysis . . . 103
Surface extrude . . . 62
Surface finish . . . 99
Surfaces . . . 60
Sweep . . . 45
guide curve . . . 45
model edge path . . . 45
orientation . . . 45
scale . . . 45
twist . . . 45
Swept Surface . . . 60
Symmetric mate . . . 89

T

Tab and Slot . . . 74
Table
drawing . . . 99
Table Driven Pattern . . . 50

Tabs 20
Tapped hole 48
Teardrop hole 84
Tees 80
Text 33
Theme 16
Thicken 64
Thickness Analysis 103
Thin feature 44
Threaded insert 83
Threads 49
  cosmetic 50
Title block 95
Tolerance stack-up 102
Tool wheel 28
Toolbars
  customising 15
Top-down design 91
Tour 16
Transition
  round to rectangle 80
Transition / Reducer 72
  axis line 72
  flanges 72
  spigots 72
Transparency 24
Trim 31
Trim Surface 64
Trim/Extend 80
Tutorial 41

U

Unbend 76
Undercut Check 103
Undo history 69
Units 18
Universal joint mate 88
Untrim Surface 64
Up axis 23
Update Now 14
Updates 14

V

Vent 53
View properties 97
Visibility 24

W

Weld Bead 80
Weld symbol 99
Weldments 79
Where Used 19
Width mate 88
Worst case 102
Wrap 50

Z

Z up 23
Zebra Stripes 103
Zoom 23